

Industries

1. Which of the following is the largest metal trading centre?

- (a) Johannesburg (b) New York
(c) London (d) Singapore

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

London Metal Exchange (LME) is the world's largest industrial-trading metal and cost-risk management center. The exchange also offers contracts on ferrous metals and precious metals.

The Transport

1. Match List- I with List- II and select the correct answer using the codes given below:

List- I

List II

- | | |
|---|---|
| <p>(A) European trans-Continental railway</p> <p>(B) Trans-Andean railway</p> <p>(C) Trans-Siberian railway</p> <p>(D) Orient Express</p> | <p>(1) Paris to Istanbul</p> <p>(2) Leningrad to Vladivostok</p> <p>(3) Leningrad to Volgograd</p> <p>(4) Buenos Aires to Valparaiso</p> <p>(5) Paris to Warsaw</p> |
|---|---|

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 5 | 4 | 2 | 1 |
| (b) | 1 | 4 | 3 | 2 |
| (c) | 5 | 1 | 2 | 3 |
| (d) | 1 | 2 | 3 | 4 |

47th B.P.S.C. (Pre) 2005

Ans. (a)

European trans-continental railway travel from Paris to Warsaw. Trans-Andean railway is the most important rail route of South America. It travels from Valparaiso to Buenos Aires of Argentina. The Trans-Siberian railway is the longest (about 9300 km.) railway in the world. It has connected Moscow with Vladivostok.

Note: The main track of Trans-Siberian Railway extends from Vladivostok to Moscow. By using a connecting service it can be travelled to Saint Petersburg in West. Orient Express is most important rail route of Europe. It runs across eight countries of Europe. It connects Paris & Le Harare to Istanbul.

The Ports and Harbours

1. Which one of the following is not a sea-port city?

- (a) Tokyo (b) Canberra
(c) New York (d) London

40th B.P.S.C. (Pre) 1995

Ans. (b)

Canberra, the capital of Australia is not surrounded by Sea. Tokyo - Capital of Japan and prominent seaport situated on Honshu island. New York - Sea-port city of the USA located on the banks of Hudson River in New York City. London - Port situated on the banks of Thames. Hence Canberra is not a seaport city. Thus, option (b) is the correct answer.

2. Who wrote Periplus of the Erythraean Sea?

- (a) Ctesias (b) Pliny
(c) Ptolemy (d) Strabo
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (e)

The Periplus of the Erythraean Sea or Periplus of the Red Sea is a Greco-Roman periplus written in Koine Greek that describes navigation and trading opportunities from Roman Egyptian ports like Berenice Troglodytica along the Coast of Red Sea and others along the Horn of Africa, the Sindh region of Pakistan, along with southwestern regions of India. The text has been ascribed to different dates between the first and the third century. A mid-first century date is now the most commonly accepted. While the author is unknown, it is clearly a first-hand description by someone familiar with the area and is nearly unique in providing accurate insights into what the ancient European world knew about the lands around the Indian Ocean.

3. Which one of the following is known as the 'Coffee Port' of the world?

- (a) Sao Paulo (b) Santos
(c) Rio de Janeiro (d) Buenos Aires
(e) None of the above / More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (b)

The Port of Santos is located in Brazil. It was established in 1546 for exporting coffee at large scale. Now it is known as Coffee Port due to highest coffee export in the world.

General Mental Ability

1. The value of

$$\frac{18}{(5+\sqrt{7})} + \frac{4}{(\sqrt{7}+\sqrt{3})} + \frac{1}{(\sqrt{3}+\sqrt{2})} + \frac{1}{(\sqrt{2}+1)}$$

- (a) 4 (b) 3
(c) 5 (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 1-5}-2024

Ans. (a)

$$= \frac{18}{(5+\sqrt{7})} + \frac{4}{(\sqrt{7}+\sqrt{3})} + \frac{1}{(\sqrt{3}+\sqrt{2})} + \frac{1}{(\sqrt{2}+1)}$$

We know that-

$$= \frac{18(5-\sqrt{7})}{(5+\sqrt{7})(5-\sqrt{7})} + \frac{4(\sqrt{7}-\sqrt{3})}{(\sqrt{7}+\sqrt{3})(\sqrt{7}-\sqrt{3})} +$$

$$\frac{(\sqrt{3}-\sqrt{2})}{(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2})} + \frac{\sqrt{2}-1}{(\sqrt{2}+1)(\sqrt{2}-1)}$$

$$= \frac{18(5-\sqrt{7})}{[5^2-(\sqrt{7})^2]} + \frac{4(\sqrt{7}-\sqrt{3})}{[(\sqrt{7})^2-(\sqrt{3})^2]} + \frac{\sqrt{3}-\sqrt{2}}{(\sqrt{3})^2-(\sqrt{2})^2}$$

$$+ \frac{\sqrt{2}-1}{[(\sqrt{2})^2-1^2]}$$

$$[\because (a-b)(a+b) = a^2 - b^2]$$

$$= \frac{18(5-\sqrt{7})}{25-7} + \frac{4(\sqrt{7}-\sqrt{3})}{7-3} + \frac{\sqrt{3}-\sqrt{2}}{3-2} + \frac{\sqrt{2}-1}{2-1}$$

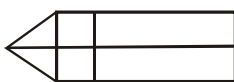
$$= \frac{18(5-\sqrt{7})}{18} + \frac{4(\sqrt{7}-\sqrt{3})}{4} + \frac{(\sqrt{3}-\sqrt{2})}{1} + \frac{\sqrt{2}-1}{1}$$

$$= 5-\sqrt{7} + \sqrt{7}-\sqrt{3} + \sqrt{3}-\sqrt{2} + \sqrt{2}-1$$

$$= 5-1$$

$$= 4$$

2. How many rectangles are there in the figure given below?



- (a) 9 (b) 12

(c) 8

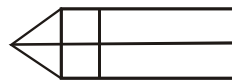
(d) More than one of the above

(e) None of the above

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Ans. (a)

The given figure-



Rectangle formed from (1) figure = 4

Rectangle formed from (2) figure = 4

Rectangle formed from (4) figure = 1

Total rectangle = 4+4+1 = 9

3. A player makes 7 complete revolutions of a circular path to complete a race of 2200 metres. The radius of the circular path is :

$$\left(\pi = \frac{22}{7} \right)$$

(a) 42 metres

(b) 45 metres

(c) 50 metres

(d) More than one of the above

(e) None of the above

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Ans. (c)

A player makes 7 complete revolutions of a circular path to complete a race of 2200 meter.

$\therefore$ 7 rounds = 2200 meters

$$1 \text{ round} = \frac{2200}{7} \text{ meters}$$

We know that-

1 round = circumference of the path then,

$$\frac{2200}{7} \text{ meters} = 2\pi r \quad (\because \text{where } r \text{ is the radius of path})$$

$$\frac{2200}{7} = 2 \times \frac{22}{7} \times r$$

$$r = \frac{100}{2}$$

$$r = 50 \text{ meters}$$

- 4 If the capacity of a cylindrical tank is 1848 m^3 and the diameter of its base is 14 m, the depth of the tank is :

$$\left(\pi = \frac{22}{7} \right)$$

- (a) 8 m (b) 12 m
(c) 16 m (d) More than one of the above
(e) None of the above

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Ans. (b)

Given that–

Volume of the tank is 1848 m^3 and diameter of its base is 14 m also, we know that–

Volume of Cylinder = $\pi r^2 h$

($\because$ where is the depth of the cylinder)

$$1848 = \frac{22}{7} \times 7 \times 7 \times h \quad (\because 2r = 14 \Rightarrow r = 7)$$

$$1848 = 22 \times 7 \times h$$

$$h = \frac{1848}{22 \times 7}$$

$$h = 12 \text{ meters}$$

- 5 If $\frac{1}{8}$ th of a number is 30, what will be 62% of that number?

- (a) 181.3 (b) 178.24
(c) 148.8 (d) More than one of the above
(e) None of the above

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Ans. (c)

Let, the number is x.

According to the question–

$$\frac{1}{8} \times x = 30$$

$$x = 240$$

$$62\% \text{ of } x = 240 \times \frac{62}{100} = \frac{14880}{100}$$

$$= \frac{1488}{10}$$

$$62\% \text{ of } x = 148.8$$

- 6 The salary of an officer is increased by 25%. By what percent should the new salary be decreased to restore the original salary?

- (a) 25 (b) 22.5

- (c) 20 (d) More than one of the above
(e) None of the above

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Ans. (c)

Let officer salary was 100 rupees.

Now, after the increment of 25%

$$= 100 \times \frac{125}{100}$$

$$= 125$$

To restore the original salary, the required percentage decrement

$$= 125 - 100$$

$$= 25$$

$$\text{decreased percentage} = \frac{25}{125} \times 100$$

$$= 20\%$$

- 7 If $x + \frac{1}{x} = 5$, what is the value of $x^2 + \frac{1}{x^4}$?

- (a) 525 (b) 527
(c) 529 (d) More than one of the above
(e) None of the above

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Ans. (b)

Given that–

$$x + \frac{1}{x} = 5$$

Squaring on both sides–

$$x^2 + \frac{1}{x^2} + 2 = 25$$

$$x^2 + \frac{1}{x^2} = 23$$

again, squaring on both sides–

$$x^4 + \frac{1}{x^4} + 2 = 529$$

$$x^4 + \frac{1}{x^4} = 527$$

- 8 A bag contains 5-rupees, 2-rupees and 1-rupee coins in the ratio 2 : 3 : 4. The total value of all the coins is ₹2,000. How many coins of 2-rupee are there in the bag?

- (a) 200 (b) 250
(c) 400 (d) More than one of the above
(e) None of the above

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A bag contains 5 rupees, 2 rupees and 1 rupee coins in a ratio 2 : 3 : 4.

Then,

Denomination	–	5	2	1
Ratio	–	2x	3x	4x
Total	–	10x	6x	4x
		= 20x		

According to the question–

$$20x = 2000$$

$$x = 100$$

$$\text{then 2 rupees coins will be } \frac{600}{2} = 300$$

Hence, option (e) will be correct.

9. The interior angle of a regular polygon exceeds its exterior angle by 108° . How many sides does the polygon have?

- (a) 10 (b) 9
(c) 8 (d) More than one of the above
(e) None of the above

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Ans. (a)

We know that–

$$\text{Interior angle} + \text{Exterior angle} = 180^\circ \quad \dots\dots\dots(i)$$

According to the question–

$$\text{Interior angle} = 108^\circ + \text{exterior angle}$$

$$\text{Interior angle} - \text{exterior angle} = 108^\circ \quad \dots\dots\dots(ii)$$

from equation (i) & (ii) □

$$2 \times \text{Interior angle} = 180^\circ + 108^\circ$$

$$2 \times \text{Interior angle} = 288^\circ$$

$$\text{Interior angle} = 144^\circ$$

From equation (i)–

$$\text{Exterior angle} = 180^\circ - 144^\circ$$

$$\text{Exterior angle} = 36^\circ$$

also we know that–

$$\text{exterior angle} = \frac{360^\circ}{x} \quad \{\because \text{where } x \text{ is side}\}$$

$$36^\circ = \frac{360^\circ}{x}$$

$$x = \frac{360^\circ}{36^\circ}$$

$$\boxed{x = 10}$$

10. Which of the following is the value of $(x+1/x)^2$?
- (a) $x^2 + 1/x^2$ (b) $x^2 - 1/x^2$
(c) $x^2 + 1/x^2 + 1$ (d) More than one of the above
(e) None of the above

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Ans. (e)

$$\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 \times x \times \frac{1}{x}$$

$$= x^2 + \frac{1}{x^2} + 2$$

Hence, option (e) is the correct answer.

11. An article is at 10% more than the CP. If discount of 10% is allowed then which of the following is right?

- (a) 1% gain
(b) 1% loss
(c) no gain no loss
(d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (b)

Let, the CP of an article = Rs. 100.

$$\therefore \text{Marked price of an article} = \text{Rs. } 100 + \frac{100 \times 10}{100}$$

$$= \text{Rs. } 110$$

Discount % = 10%

$$\therefore \text{Selling Price of an article} = 110 - 110 \times 10\%$$

$$= \text{Rs. } 99$$

$$\therefore \text{CP} > \text{SP}$$

$$\therefore \text{Loss \%} = \frac{\text{CP} - \text{SP}}{\text{CP}} \times 100 = \frac{100 - 99}{100} \times 100 = 1\%$$

Hence, option (b) is the correct answer.

12. By what number should 81 be divided to get a perfect cube?

- (a) 3
(b) 6
(c) 7
(d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (a)

$$81 = 3 \times 3 \times 3 \times 3$$

$$81 = 3^3 \times 3$$

Hence, 81 should be divided by 3 to get a perfect cube.

13. The difference between two whole numbers is 66. The ratio of the two numbers is 2:5. The two numbers are :

- (a) 60 and 6 (b) 100 and 33
(c) 110 and 44 (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Let, the two whole numbers be a and b.

According to question–

$$a \sim b = 66 \quad \dots\dots\dots(i)$$

$$\text{and } \frac{a}{b} = \frac{2}{5}$$

$$a = \frac{2b}{5} \quad \dots\dots\dots(ii)$$

$$\frac{2b}{5} \sim b = 66$$

$$\frac{3b}{5} = 66$$

$$\boxed{b = 110} \text{ and } \boxed{a = 44}$$

Hence, the two whole numbers are 44 and 110.

14. In a parallelogram ABCD, angle A and angle B are in the ratio 1:2. Find the angle A.

- (a) 30° (b) 45°
 (c) 60° (d) More than one of the above
 (e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (c)

Let, the angles A and B of parallelogram ABCD are x and 2x. As we know, the sum of adjacent angles of a parallelogram is 180° .

$$\angle A + \angle B = 180^\circ$$

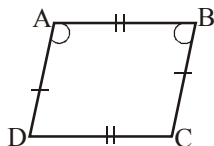
$$x + 2x = 180^\circ$$

$$3x = 180^\circ$$

$$\boxed{x = 60^\circ}$$

$$\boxed{2x = 120^\circ}$$

Hence, the value of angle A is 60° .



15. The height of a cylinder whose radius is 7 cm and the total surface area is 968 cm^2 is :

- (a) 15 cm (b) 17 cm
 (c) 19 cm (d) More than one of the above
 (e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (a)

Given that,

Radius of a cylinder = 7 cm

Total surface area of cylinder = 968 cm^2

Let, the height of a cylinder = x cm.

According to the question–

$$\text{Total surface area of cylinder} = 2\pi rh + 2\pi r^2$$

$$968 = 2\pi r(h+r)$$

$$968 = 2 \times \frac{22}{7} \times 7(x+7)$$

$$22 = x + 7$$

$$\boxed{x = 15 \text{ cm}}$$

Hence, the height of a cylinder is 15 cm.

16. If $(-3)^{m+1} \times (-3)^5 = (-3)^7$, then the value of m is :

- (a) 5
 (b) 7
 (c) 1
 (d) More than one of the above
 (e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (c)

$$(-3)^{m+1} \times (-3)^5 = (-3)^7$$

$$\therefore (-3)^{(m+1)+5} = (-3)^7$$

$$\therefore (m+1)+5 = 7$$

$$\boxed{m = 1}$$

Hence, the value of m is 1.

17. If x and y are inversely proportional, then:

- (a) $y/x = \text{constant}$
 (b) $xy = \text{constant}$
 (c) $x/y = \text{constant}$
 (d) More than one of the above
 (e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (b)

$$x \propto \frac{1}{y}$$

$$x = \frac{c}{y} \quad \{\text{where } C = \text{constant}\}$$

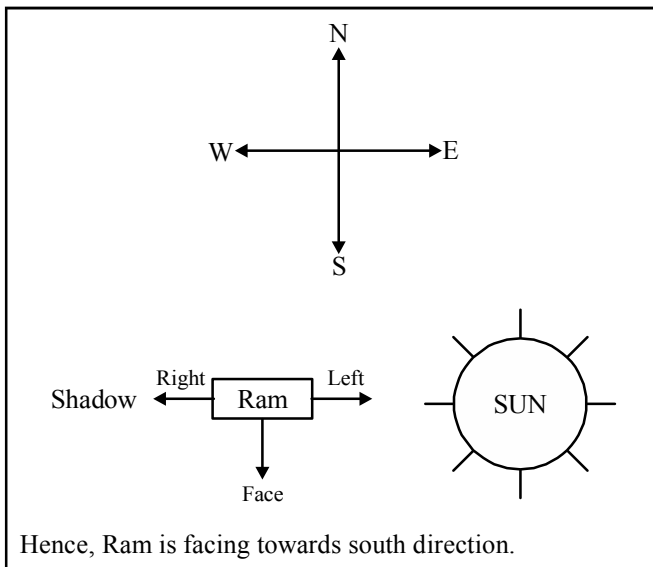
$$\boxed{xy = \text{constant}}$$

Hence, option (b) is the correct answer.

18. Early morning, Ram was standing on the ground in front of an effigy. The shadow of the effigy was appearing towards the right side of Ram. In which direction is Ram facing?
- (a) East
(b) West
(c) South
(d) None of the above

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Ans. (c)



19. If the 3rd day of a month is Friday, which day will be the 6th day before 27th of this month?
- (a) Tuesday (b) Wednesday
(c) Friday (d) None of the above

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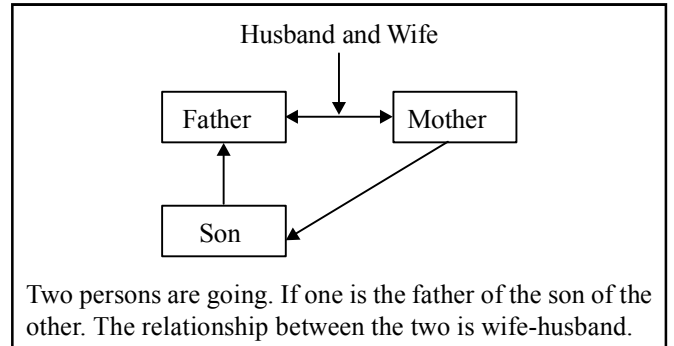
Ans. (a)

Given that,
 3rd Day of a month = Friday
 6th Day before 27th = 21st
 $\therefore$ 21st Day of a month = 3rd + 18 {18 = 21-3}
 $= 3^{\text{rd}} + 21 - 3$
 $= \text{Friday} - 3$
 $= \text{Tuesday}$

20. Two persons are going. One is the father of the son of the other. What is the relationship between the two?
- (a) Mother–Son (b) Father–Son
(c) Wife–Husband (d) None of the above

BPSC Agriculture Department-2024

Ans. (c)



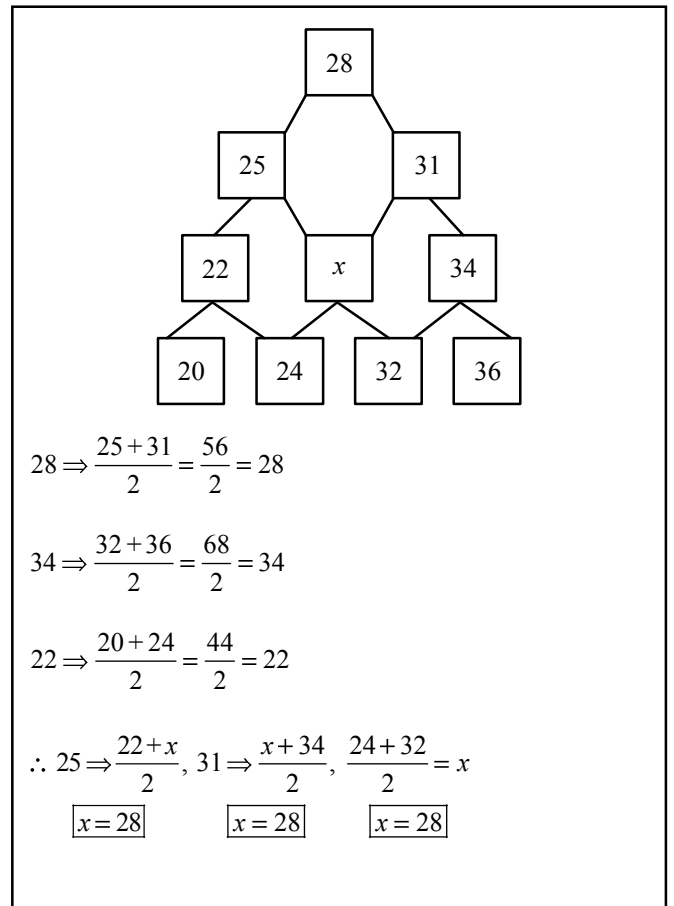
21. In the following table, obtain the missing number x :

		28	
	25		31
	22	x	34
20	24	32	36

- (a) 28 (b) 30
(c) 32 (d) None of the above

BPSC Agriculture Department-2024

Ans. (a)



22. In a certain code language, 'GAME' is written as '\$ ÷ * %' and 'BEAD' is written as '# % ÷ ×'. How will the word 'MADE' be written in that code language?

- (a) \$ ÷ × %
- (b) * ÷ \$ %
- (c) * ÷ × %
- (d) # ÷ × %

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Ans. (c)

In the given coding language, there is certain code for certain word—

Just like—

G — \$

A — ÷

M — *

E — %

In the same way—

B — #

E — %

A — ÷

D — ×

M — *

A — ÷

D — ×

E — %

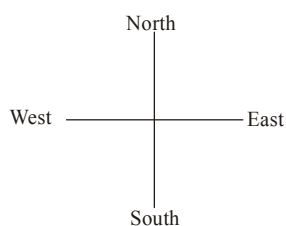
Hence, option (c) will be correct option.

23. Q travels towards East. M travels towards North. S and T travel in opposite directions. T travels towards right of Q. Which of the following is definitely true?

- (a) M and S travel in the opposite directions
- (b) S travels towards West
- (c) T travels towards North
- (d) M and S travel in the same direction

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Ans. (d)



From the above arrangement, it is clear that M & S are travelling in the same direction. Hence, option (d) will be the correct option.

24. A man 'Ramesh', who owns a plot of land of 100 square yards, increases his plot of land by acquiring 10% more from his neighbor 'Suresh', who also owns 100 square yards land. After 2 years, he sells back 10% of the total plot to the neighbor. Which of the following is correct?

- (a) Ramesh's land is more than Suresh
- (b) Suresh's land is more than Ramesh

- (c) Both are equal
- (d) None of the above

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Ans. (b)

According to the question—

Ramesh → 100 sq. yard

Suresh → 100 sq. yard

After the acquisition of 10% of land from his neighbour Suresh by 10% →

$$\text{Ramesh} \rightarrow 100 + 100 \times \frac{10}{100} = 110 \text{ sq. yard}$$

$$\text{Suresh} \rightarrow 100 - 100 \times \frac{10}{100} = 90 \text{ sq. yard}$$

After 2 years Ramesh sells back 10% of this total land to his neighbour—

$$\text{Ramesh} \rightarrow 110 - 110 \times \frac{10}{100} = 99 \text{ sq. yard}$$

$$\text{Suresh} \rightarrow 90 + 110 \times \frac{10}{100} = 101 \text{ sq. yard}$$

Hence, Suresh land is more than Ramesh land, which is mention in the option (b).

25. Find the odd pair among the following options.

- (a) Millinery : Hats
- (b) Brewery : Alcohol
- (c) Stationery : Paper
- (d) Snobbery : Shoes

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Ans. (d)

In the given question the option (d) is odd among other options because—

Millinery means women's Hat and other related goods and Brewery mean a company that makes beer or a place where beer is made & the third option Stationery means the things needed for writing where as Snobbery means behaviour and opinions that are typical of a snob: that means it is not related to shoes.

26. What should come in place of question mark (?) in the following number series?

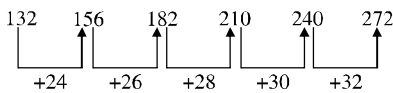
132 156 ? 210 240 272

- (a) 196
- (b) 182
- (c) 199
- (d) 204

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Ans. (b)

The given number series—



Hence, option (b) will be the right answer.

27. Select the missing number from the given alternatives :

44	49	37
52	?	41
58	35	53

(a) 56

(b) 77

(c) 66

(d) 63

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Ans. (b)

44	49	37	—R ₁
52	? = 77	41	—R ₂
58	35	53	—R ₃

from R₁ → (44 - 37) × 7 = 49

from R₃ → (58 - 53) × 7 = 35

Similarly R₂ → (52 - 41) × 7 = 11 × 7 = 77

Hence, option (b) will be the right answer.

28. Read the given statement and conclusions carefully. Decide which of the given conclusions logically follow(s) from the statement.

Statement : In a one-day cricket match, the total runs made by a team were 200, of which 160 runs were made by spinners.

Conclusions :

1. 80% of the team consisted of spinners.

2. The opening batsmen were spinners.

Select the correct answer.

(a) Only 1 follows the statement

(b) Only 2 follows the statement

(c) Both 1 and 2 follow the statement

(d) Neither 1 nor 2 follows the statement

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Ans. (d)

Statement : Total run = 200 runs

Runs made by Spinners = 160

Conclusion : (1) 80% of team is consisted of spinners—

This conclusion is wrong, here 80% of runs made by spinners but it does not clarify that the team 80% player is spinners.

(2) The opening batsman were spinners—

This conclusion is also wrong because, there is nothing about this conclusion is mentioned in the statement.

Therefore, neither conclusion (1) nor conclusion (2) follows the statement.

Hence, option (d) will be the right answer.

29. In a code language, if GREAT is written as 718222620 and MONK is written as 13121411, then how will VIGOROUS be written in the same language?

(a) 22187121812619

(b) 21177121811619

(c) 22187131813620

(d) 21187111711620

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Ans. (a)

Here, the vowels in the word is coded with its opposite place value.

Just like—

Place value	Code	Place value	Code
G—7	7	M—13	13
R—18	18	O—15	opposite 12
E—5	opposite 22	N—14	14
A—1	opposite 26	K—11	11
T—20	20		

In the same way—

Place value	Code
V—22	22
I—9	opposite 18
G—7	7
O—15	opposite 12
R—18	18
O—15	opposite 12
U—21	opposite 6
S—19	19

Hence, the given word "Vigorous" is written as 22187121812619 which is in option 'a'.

30. Find the odd one in the following groups :

Q,W,Z,B B,H,K,M W,C,G,J M,S,V,X

(a) Q,W,Z,B

(b) M,S,V,X

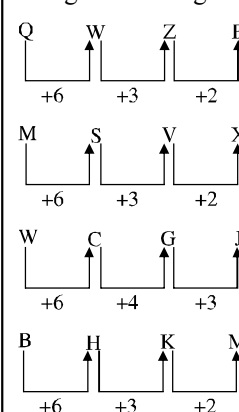
(c) W,C,G,J

(d) B,H,K,M

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Ans. (c)

The given letter groups—



Hence, 'W, C, G, J' is different from the other that is given in

31. Four branches of a company are located at M, N, O and P. M is in the North of N at a distance of 4 km; P is in the South of O at a distance of 2 km; N is in the Southeast of O by 1 km. What is the distance between M and P in km?

(a) 5.34 (b) 6.74
(c) 28.5 (d) None of the above

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Ans. (a)

Four branches of the company— M, N, O, P

In $\triangle MPA \Rightarrow$

$$MP^2 = \left(\frac{1}{\sqrt{2}}\right)^2 + \left[4 + \left(2 - \frac{1}{\sqrt{2}}\right)\right]^2$$

$$MP^2 = \frac{1}{2} + 16 + \left(2 - \frac{1}{\sqrt{2}}\right)^2 + 8\left(2 - \frac{1}{\sqrt{2}}\right)$$

$$= \frac{1}{2} + 16 + 4 + \frac{1}{2} - 2 \times 2 \times \frac{1}{\sqrt{2}} + 16 - \frac{8}{\sqrt{2}}$$

$$= 21 - \frac{4}{\sqrt{2}} + 16 - \frac{8}{\sqrt{2}}$$

$$= 37 - \frac{12}{\sqrt{2}}$$

$$= 37 - 6\sqrt{2}$$

$$= 37 - 6 \times 1.414$$

$$= 37 - 8.484$$

$$= 28.516$$

$MP = \sqrt{28.516} \approx 5.34 \text{ km}$

32. If $\sqrt{1 + \frac{27}{169}} = \left(1 + \frac{x}{13}\right)$, then the value of x is

(a) 1 (b) $-\frac{14}{13}$
(c) -27 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (d)

$$\sqrt{1 + \frac{27}{169}} = \left(1 + \frac{x}{13}\right)$$

$$\sqrt{\frac{169 + 27}{169}} = 1 + \frac{x}{13}$$

$$\sqrt{\frac{196}{169}} = 1 + \frac{x}{13}$$

$$\frac{14}{13} = 1 + \frac{x}{13} \quad \left| \frac{-14}{13} = 1 + \frac{x}{13} \right.$$

$$14 = 13 + x \quad | \quad -14 = 13 + x$$

$x = 1 \quad | \quad x = 27$

33. The radius of a solid sphere of some metal is 3 cm. How long a wire of diameter 4 mm can be drawn from it?

(a) 8.8m (b) 9m
(c) 9.1m (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Radius of a sphere = 3 cm
and we are supposed to make a wire of diameter 4 mm. So the length of the wire will be as follows—
Let the length of the wire is h, radius of sphere is 'r' & radius of cylinder is R
Volume of solid sphere = volume of cylinder
($\because$ Wire is in shape of cylinder)
 $r = 3 \text{ cm}$
 $R = 2 \text{ mm} \Rightarrow 0.2 \text{ cm}$

$$\frac{4}{3} \pi r^3 = \pi R^2 h$$

$$\frac{4}{3} \times 3 \times 3 \times 3 = 0.2 \times 0.2 \times h$$

$$36 = 0.2 \times 0.2 \times h$$

$$h = \frac{36}{0.04}$$

$$h = 900 \text{ cm}$$

$h = 9 \text{ meter}$

34. On selling 30 chairs, a furniture seller gets a loss equal to selling price of 6 chairs. His loss percent is

- (a) 16 (b) $16\frac{1}{3}$
 (c) $16\frac{2}{3}$ (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

On selling 30 chairs, a furniture seller get a loss equal to selling price of 6 chairs.

Now loss = cost price – selling price

According to question–

6 selling price (SP) = 30 cost price (CP) – 30 selling price (SP)

6 SP + 30 SP = 30 CP

$$\frac{SP}{CP} = \frac{30}{36}$$

$$L\% = \frac{CP - SP}{CP} \times 100 = \frac{36 - 30}{36} \times 100$$

$$L\% = \frac{6}{36} \times 100 = \frac{100}{6}$$

$$L\% = 16\frac{2}{3}\%$$

35. The product of the digits of a two-digit number is 36. If 45 is subtracted from the number, then the digits interchange mutually. The number is

- (a) 84 (b) 73
 (c) 49 (d) More than one of the above
 (e) None of the above

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Ans. (e)

Let the digit = $10x+y$

According to the question

$$x \times y = 36 \quad \dots\dots\dots(i)$$

and

$$(10x+y) - 45 = 10y+x$$

$$(10x+y) - (10y+x) = 45$$

$$9x - 9y = 45$$

$$x - y = 5 \quad \dots\dots\dots(ii)$$

from (i) & (ii)

$$x = 9 \text{ \& } y = 4$$

so the initial number is = 94

36. If $a + b + c = 0$, then what is the value of $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$?

- (a) 1 (b) 2

- (c) 3 (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

Given that–

$$a + b + c = 0$$

$$\text{so } a^3 + b^3 + c^3 = 3abc \quad \dots\dots\dots(i)$$

Now

$$\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$$

$$\frac{a^3}{abc} + \frac{b^3}{bca} + \frac{c^3}{cab}$$

$$\frac{a^3 + b^3 + c^3}{abc}$$

from equation (i)

$$\frac{3abc}{abc}$$

$$\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab} = 3$$

37. The solution of $4^{x+1} + 4^{1-x} = 10$ is

- (a) $-\frac{1}{2}$ (b) $\frac{1}{2}$
 (c) 0 (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

$$4^{x+1} + 4^{1-x} = 10$$

$$4^x \times 4 + 4^1 \times 4^{-x} = 10$$

$$4^x \times 4 + 4 \times \frac{1}{4^x} = 10$$

$$4^x + \frac{1}{4^x} = \frac{5}{2}$$

Let $4^x = P$ then–

$$P + \frac{1}{P} = \frac{5}{2} \quad \dots\dots\dots(i)$$

$$\left(P - \frac{1}{P}\right)^2 = \left(P + \frac{1}{P}\right)^2 - 4$$

$$= \frac{25}{4} - 4$$

$$= \frac{9}{4}$$

$$\left(P - \frac{1}{P}\right) = \pm \frac{3}{2} \quad \dots\dots\dots(ii)$$

from the equation (i) & (ii)

$$2P = 4$$

$$P = 2$$

Now

$$4^x = 2$$

$$4^x = \sqrt{4} = 4^x = 4^{1/2}$$

$$\text{so } \boxed{x = \frac{1}{2}}$$

38. On dividing $x^3 - 3x^2 + x + 2$ by a polynomial $g(x)$, the quotient and remainder are $(x^2 - x + 1)$ and $(-2x + 4)$ respectively.

Then $g(x)$ is

- (a) $x - 2$ (b) $x^2 + x + 1$
 (c) $x^2 - 1$ (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

Dividend = Divisor $\times$ Quotient + Remainder

Now

$$x^3 - 3x^2 + x + 2 = g(x) \times (x^2 - x + 1) + (-2x + 4)$$

$$g(x) \times (x^2 - x + 1) = x^3 - 3x^2 + x + 2 + 2x - 4$$

$$= x^3 - 3x^2 + 3x - 2$$

$$g(x) \times (x^2 - x + 1) = x^3 - 3x^2 + 3x - 2$$

$$g(x) = \frac{x^3 - 3x^2 + 3x - 2}{x^2 - x + 1}$$

$$g(x) = \frac{(x - 2)(x^2 - x + 1)}{(x^2 - x + 1)}$$

$$\boxed{g(x) = (x - 2)}$$

39. A toy is in the form of a cone mounted on a hemisphere of diameter 7 cm. The total height of the toy is 14.5 cm.

Then the volume of the toy is $\left(\pi = \frac{22}{7}\right)$

- (a) 231 cm^3
 (b) 331 cm^3
 (c) 131 cm^3
 (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift I

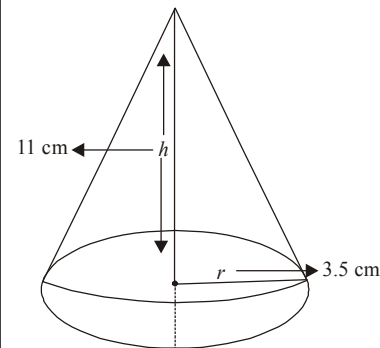
Ans. (a)

Given that–

Diameter of Hemisphere = 7 cm

Total height of the toy = 14.5 cm

Now



Total volume of the toy = Volume of the Hemisphere + Volume of the cone.

$$\text{Total volume} = \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h$$

$$= \frac{\pi}{3} r^2 (2r + h)$$

$$= \frac{22}{7 \times 3} \times \frac{7}{2} \times \frac{7}{2} \left(2 \times \frac{7}{2} + 11 \right)$$

$$= \frac{22}{3} \times \frac{7}{4} \times 18$$

$$= 11 \times 21$$

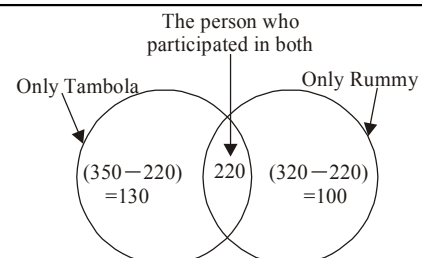
$$\boxed{\text{Total volume} = 231 \text{ cm}^3}$$

40. In a club, all the members participate either in Tambola or in Rummy. 320 participate in Rummy, 350 participate in Tambola and 220 participate in both. How many members does the club have?

- (a) 440 (b) 445
 (c) 450 (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)



Total member = only Tambola + only Rummy + Participant of both
 $= 130 + 100 + 220$

$$\boxed{\text{Total member} = 450}$$

41. The sum of the reciprocals of A's ages (in years) 2 years ago and 2 years from now is $\frac{2}{3}$. The present age of A is

- (a) 5 years (b) 7 years
(c) 4 years (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

Let A's present age be x .

The sum of the reciprocals of A's age 2 years ago ($x-2$) and 2 years from now ($x+2$) is given by—

$$\frac{1}{(x-2)} + \frac{1}{(x+2)}$$

According to the question—

$$\frac{1}{x-2} + \frac{1}{x+2} = \frac{2}{3}$$

$$\frac{x+2+x-2}{x^2-4} = \frac{2}{3}$$

$$\frac{2x}{x^2-4} = \frac{2}{3}$$

$$3x = x^2 - 4$$

$$x^2 - 3x - 4 = 0$$

$$x^2 - 4x + x - 4 = 0$$

$$x(x-4) + 1(x-4) = 0$$

$$x = 4, -1$$

Hence, the answer will be 4

42. For what value of k , the system of equations $x - ky = 2$ and $3x + 2y = -5$ has a unique solution?

- (a) $-\frac{2}{3}$ (b) 1
(c) 2 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

The given equation—

$$x - ky = 2 \quad \dots\dots\dots(i)$$

$$3x + 2y = -5 \quad \dots\dots\dots(ii)$$

If the given equation have unique solution, then—

$$\frac{1}{3} \neq \frac{-k}{2}$$

$$-3k \neq 2$$

$$k \neq \frac{-2}{3}$$

42. The radius of a circle is 14 cm and the area of a sector is 102.7 cm^2 . The central angle of the sector is

- (a) 30° (b) 60°
(c) 45° (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

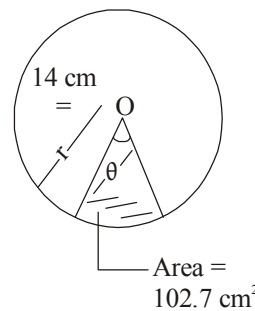
We know that—

$$\text{Area of sector} = \pi r^2 \frac{\theta}{360}$$

$$102.7 = \frac{22}{7} \times 14 \times 14 \times \frac{\theta}{360}$$

$$\theta = \frac{102.7 \times 45}{11 \times 7}$$

$$\theta \approx 60^\circ$$



44. What is the compound interest of ₹ 15,625 for 9 months when the rate of interest is 16% per annum and the interest is payable quarterly?

- (a) ₹ 1,961 (b) ₹ 1,951
(c) ₹ 2,051 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

The given sum (P) = 15625

(t) Time = 9 month

rate of interest (r) = 16%

if the interest is payable quarterly than the interest rate for a quarter—

$$12 \text{ m} \rightarrow 16\%$$

$$3 \text{ m} \rightarrow 4\%$$

and No. of quarter in 9 month = 3

We know that—

$$A = P \left(1 + \frac{r}{100} \right)^t$$

$$A = 15625 \times \left(1 + \frac{4}{100} \right)^3$$

$$= 15625 \times \frac{26}{25} \times \frac{26}{25} \times \frac{26}{25}$$

$$A = 17576$$

Now total interest = $A - P$

$$= 17576 - 15625$$

$$\text{Total Interest} = ₹ 1951$$

45. If the product of the roots of the equation $3kx^2 - 25kx + k + 8 = 0$ is 3, then k is

- (a) 1 (b) 4
(c) 3 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

The given equation–

$$3kx^2 - 25kx + (k + 8) = 0 \quad \dots\dots\dots(i)$$

if the equation is in the form of the equation given below & their roots is, then–

$$ax^2 + bx + c = 0 \quad \dots\dots\dots(ii)$$

then

$$\alpha + \beta = \frac{-b}{a}$$

$$\& \alpha \cdot \beta = \frac{c}{a}$$

on comparing equation (i) & (ii)

$$a = 3k, b = -25k, c = (k + 8)$$

Product of roots is given i.e. 3

$$\text{so } \frac{k + 8}{3k} = 3$$

$$k + 8 = 9k$$

$$8 = 8k$$

$$\boxed{k = 1}$$

46. The value of $\log_{\sqrt{2}}(32)$ is

- (a) $\frac{5}{2}$ (b) 5
(c) 10 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

Let

$$\log_{\sqrt{2}}(32) = x$$

We can also write this log function as–

$$32 = (\sqrt{2})^x$$

$$32 = [(2)^{1/2}]^x$$

$$2^5 = (2)^{x/2}$$

on comparing both side–

$$\frac{x}{2} = 5$$

$$\boxed{x = 10}$$

47. The altitude of a right-angled triangle is 6 cm less than its base. If the hypotenuse is $\sqrt{218}$ cm, the other two sides (in cm) are

- (a) 5 and 11 (b) 9 and 15
(c) 10 and 16 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (e)

Let base of right angle triangle = x

so Altitude of the right angle triangle = $(x - 6)$

According to the question–

by pythagoras theorem–

$$(\sqrt{218})^2 = (x)^2 + (x - 6)^2$$

$$218 = x^2 + x^2 + 36 - 12x$$

$$218 = 2x^2 + 36 - 12x$$

$$109 = x^2 + 18 - 6x$$

$$x^2 + 18 - 6x - 109 = 0$$

$$x^2 - 6x - 91 = 0$$

$$x^2 - (13 - 7)x - 91 = 0$$

$$x^2 - 13x + 7x - 91 = 0$$

$$x(x - 13) + 7(x - 13) = 0$$

$$(x + 7)(x - 13) = 0$$

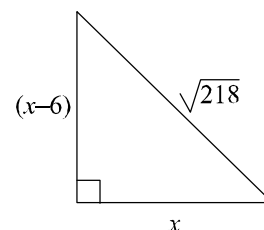
$$x = -7, 13$$

Length cannot be in negative so we take the positive value so that–

$$x = 13$$

$$\boxed{\text{Base} = 13}$$

$$\boxed{\text{Altitude} = 13 - 6 = 7}$$



48. Arrange the following fractions in ascending order :

$$\frac{13}{18}, \frac{29}{36}, \frac{41}{45}, \frac{51}{60}$$

$$(a) \frac{13}{18} < \frac{29}{36} < \frac{51}{60} < \frac{41}{45} \quad (b) \frac{13}{18} < \frac{51}{60} < \frac{29}{36} < \frac{41}{45}$$

$$(c) \frac{51}{60} < \frac{41}{45} < \frac{29}{36} < \frac{13}{18} \quad (d) \text{More than one of the above}$$

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

$$\frac{13}{18}, \frac{29}{36}, \frac{41}{45}, \frac{51}{60}$$

Now,

$$\begin{array}{cccc} \frac{13}{18}, & \frac{29}{36}, & \frac{41}{45}, & \frac{51}{60} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \frac{130}{180}, & \frac{145}{180}, & \frac{164}{180}, & \frac{153}{180} \end{array}$$

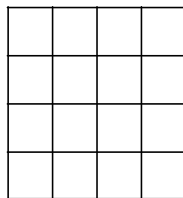
In this case, when denominator is equal than the highest valued numerator will be greater so the ascending order–

$$\frac{130}{180} < \frac{145}{180} < \frac{153}{180} < \frac{164}{180}$$

$$\frac{13}{18} < \frac{29}{36} < \frac{51}{60} < \frac{41}{45}$$

Hence, option (a) will be the right answer.

49. How many squares are there in the given figure?



- (a) 25 (b) 30
(c) 32 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

When the row and column are equal then–

				$-1^2 \rightarrow 1$
				$-2^2 \rightarrow 4$
				$-3^2 \rightarrow 9$
				$-4^2 \rightarrow 16$
				30

Hence, the total number of squares in the given figure will be 30.

50. A man fills a basket with eggs in such a way that the number of eggs added on each successive day is the same as the number already present in the basket. This way the basket gets completely filled in 24 days. After how many days, the basket was one-fourth full?

- (a) 6 (b) 12
(c) 22 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

If a man fills a basket with eggs in such a way that the number of eggs added on each successive day is the same as the number already present in the basket, then it goes like this

$$1 \ 2 \ 4 \ 8 \ 16 \ 32 \ 64 \dots\dots\dots 2^{23}$$

if the basket gets completely filled in 24 days, it means the basket was half full on the 23rd day. Therefore, it was one-fourth full on the 22nd day.

51. The sum of two numbers is 480. If the bigger number is decreased by 2% and the smaller number is increased by 10%, then the numbers obtained are equal. The larger number is approximately

(a) $\frac{3320}{13}$

(b) $\frac{3300}{13}$

(c) $\frac{3120}{13}$

(d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Let the bigger number is x & other is y

According to the question–

$$x + y = 480 \dots\dots\dots(i)$$

Now bigger is decreased by 2% and smaller number is increased by 10% then–

$$\frac{98x}{100} = \frac{110y}{100}$$

$$\frac{x}{y} = \frac{110}{98} = \frac{55}{49}$$

from equation (i)

$$x + \frac{49x}{55} = 480$$

$$104x = 480 \times 55$$

$$x = \frac{480 \times 55}{104}$$

$$x = \frac{60 \times 55}{13}$$

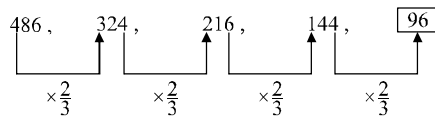
$$x = \frac{3300}{13}$$

52. The next number in the finite sequence 486, 324, 216, 144, ... is

- (a) 96 (b) 76
(c) 36 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

The Given Series–



Hence, option (a) will be the right answer.

53. Which number should replace ‘?’ in the following series?

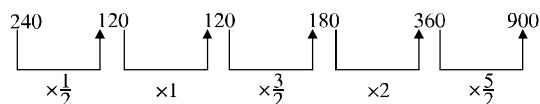
240, 120, ?, 180, 360, 900

- (a) 120 (b) 130
(c) 150 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

The given series–



Hence, option (a) will be the right answer.

54. The term in the blank

_____ : FICTION :: GRIND : BMDIY is

- (a) LOIZOUT (b) KNHYNTS
(c) IMGXMSR (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Like as–

B $\xrightarrow{+5}$ G
M $\xrightarrow{+5}$ R
D $\xrightarrow{+5}$ I
I $\xrightarrow{+5}$ N
Y $\xrightarrow{+5}$ D

Similarly–

F $\xrightarrow{+5}$ K
I $\xrightarrow{+5}$ N
C $\xrightarrow{+5}$ H
T $\xrightarrow{+5}$ Y
I $\xrightarrow{+5}$ N
O $\xrightarrow{+5}$ T
N $\xrightarrow{+5}$ S

Hence, option (b) will be the right answer.

55. There are 5 routes to go from Prayagraj to Patna and 4 routes to go from Patna to Kolkata. How many ways are possible for going from Prayagraj to Kolkata via Patna?

- (a) 45 (b) 20
(c) 54 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (B)

Since there are 5 routes from Prayagraj to Patna and 4 routes from Patna to Kolkata, the total number of ways to go from Prayagraj to Kolkata via Patna is the product of the two route options.

= 5 routes from Prayagraj to Patna $\times$ 4 routes from Patna to Kolkata

right answer.

$$= 5 \times 4$$

$$= 20$$

Hence, there are 20 possible ways which is given in the option (b).

56. Select the most suitable statement.

- (a) Every circle is an ellipse.
(b) No circle is hyperbola.
(c) Every square is a rectangle.
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (d)

From option (a) – Every circle is an ellipse is correct because, the formal definition of an ellipse is the set of all points such that the sum of the distance between those points and two fixed points is constant.

From option (c) – "Every square is a rectangle" is correct while every rectangle is not correct.

Hence, option (d) will be the right answer.

57. The word CODE is coded as 15161991 and STEP is coded as 18769122. Then SPOTE is coded as

- (a) 1822761691 (b) 1822167691
(c) 1816227691 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

The given code for the words–

CODE $\rightarrow$ 15 16 19 91

STEP $\rightarrow$ 18 76 91 22

so the code for the word 'SPOTE' is–

= S P O T E $\rightarrow$ 18 22 16 76 91

Hence, option (b) will be the right answer of the given question.

58. The area of a circle is $40\pi^2$. The area of a square with sides equal to radius of the circle is

- (a) 4π (b) $4\pi^2$
(c) 40π (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

Let the radius of the circle is a then the side of square will also be ' a '–

According to the question–

$$\pi a^2 = 40\pi^2$$

$$a^2 = 40\pi$$

($\therefore$ Area of square of sides $a = a^2$)

59. Select the odd word in the group :

Book, Paper, Pencil, Pen, Eraser

- (a) Book (b) Pencil
(c) Eraser (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

In the given words-'Book' is odd one because "Paper, Pencil, Pen, Eraser, is used to write while book is used to "Study". Therefore, option (a) will be the right answer.

60. If in English alphabet, A = 1, D = 2, P = 4, then which of the following denotes 3?

- (a) H (b) I
(c) J (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Place value	Code
A = 1	(1) ²
D = 4	(2) ²
P = 16	(4) ²
∴ I = 9	(3) ²

Therefore, option (b) will be the right answer.

61. Two numbers are in the ratio 21:10. If their HCF is 11, then the sum of the numbers is

- (a) 341 (b) 141
(c) 241 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

Ratio of two numbers are – 21 : 10
HCF = 11
Let the numbers are – 21x & 10x
so the number will be – 21x × 11 = 231x
— 10x × 11 = 110x
The sum of numbers will be
= 231x + 110x
= 341x
Therefor sum of numbers will be either 341 or multiples of 341.

62. A profit of 10% is made when a smartphone is sold at P rupees and there is 4% loss when the smartphone is sold at Q rupees. Then P : Q is

- (a) 55 : 48 (b) 45 : 34
(c) 110 : 96 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (d)

Let the price of smartphone = 100 ₹

When it is sold at 10% profit

$$P = 100 \times \frac{110}{100} = 110 \text{ ₹}$$

When it is sold at 4% loss

$$Q = 100 \times \frac{96}{100} = 96 \text{ ₹}$$

$$\text{Now } \frac{P}{Q} = \frac{110}{96} = \frac{55}{48}$$

Hence, option (d) will be the right answer.

63. In how many different ways can the letters of the word EDUCATION be arranged so that the vowels come together?

- (a) 7200 (b) 14400
(c) 12100 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

The given word–

E D U C A T I O N

Vowels – E, U, A, I, O

Consonants – D, C, T, N

If vowels considered as one group then total ways to arrange the consonants and group = 5! = 120

Also the group of vowels can be arranged in – 5!

ways ⇒ 5! ⇒ 120

Total Number of arrangement–

= 5! × 5!

= 120 × 120

= 14400

Hence, option (b) will be the right answer.

64. If $\left(\frac{3}{4}\right)^{-6} \times \left(\frac{4}{3}\right)^{-8} = \left(\frac{3}{4}\right)^n$ then the value of n is–

- (a) 1 (b) 4
(c) 3 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (e)

$$\left(\frac{3}{4}\right)^{-6} \times \left(\frac{4}{3}\right)^{-8} = \left(\frac{3}{4}\right)^n$$

$$\left(\frac{3}{4}\right)^{-6} \times \left(\frac{3}{4}\right)^8 = \left(\frac{3}{4}\right)^n$$

$$\left(\frac{3}{4}\right)^{-6+8} = \left(\frac{3}{4}\right)^n$$

$$\left(\frac{3}{4}\right)^2 = \left(\frac{3}{4}\right)^n$$

$$n = 2$$

Hence, option (e) will be the right answer.

65. The simplified form of $(4^{-1} + 8^{-1}) \div \frac{1}{\left(\frac{3}{2}\right)^{-2}}$ is-

- (a) $\frac{1}{6}$ (b) $\frac{2}{3}$
 (c) $\frac{1}{2}$ (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

$$(4^{-1} + 8^{-1}) \div \frac{1}{\left(\frac{3}{2}\right)^{-2}}$$

$$= \left(\frac{1}{4} + \frac{1}{8}\right) \div \left(\frac{3}{2}\right)^2$$

$$= \frac{(2+1)}{8} \div \frac{9}{4}$$

$$= \frac{3}{8} \times \frac{4}{9} = \frac{1}{6}$$

66. If $A : B = \frac{1}{2} : \frac{3}{8}$, $B : C = \frac{1}{3} : \frac{5}{9}$ and $C : D = \frac{5}{6} : \frac{3}{4}$, then the ratio $A : B : C : D$ is-

- (a) 6 : 8 : 9 : 10 (b) 8 : 6 : 10 : 9
 (c) 4 : 6 : 8 : 10 (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

$$A : B = \frac{1}{2} : \frac{3}{8} \Rightarrow 4 : 3$$

$$B : C = \frac{1}{3} : \frac{5}{9} \Rightarrow 3 : 5$$

$$C : D = \frac{5}{6} : \frac{3}{4} \Rightarrow 10 : 9$$

Now

$$A : B : C = 4 : 3 : 5$$

$$C : D = 10 : 9$$

Now

$$A : B : C : D = 8 : 6 : 10 : 9$$

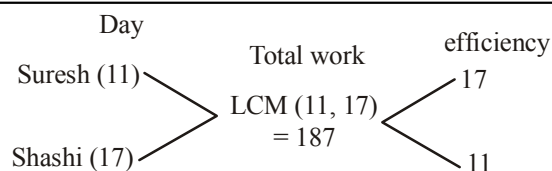
Hence, option (b) will be the right answer.

67. Suresh and Shashi complete a work in 11 days and 17 days respectively. They received a remuneration of ₹5,600 after completing the work jointly. Their shares in the remuneration are respectively-

- (a) ₹ 3600 and ₹ 2000 (b) ₹ 3400 and ₹ 2200
 (c) ₹ 3500 and ₹ 2100 (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)



Total remuneration will be distributed as per the ratio of their efficiency-

$$(17 + 11) \rightarrow 5600$$

$$1 \rightarrow 200$$

$$\text{Suresh} = 17 \times 200$$

$$= 3400$$

$$\text{Shashi} = 11 \times 200 = 2200$$

68. A's weight is 25% of B's weight and 40% of C's weight. What percentage of C's weight is B's weight?

- (a) 160 (b) 180
 (c) 50 (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

According to the question-

A's weight = 25% of B's Weight = 40% of C's weight

Now

$$25\% B = 40\% \text{ of } C$$

$$\frac{B}{C} = \frac{40}{25}$$

$$\frac{B}{C} = \frac{8}{5}$$

Percentage of B's weight of C's weight

$$= \frac{8}{5} \times 100 = 160\%$$

69. The diameter of a copper sphere is 6 cm. The sphere is melted and is drawn into a long wire of uniform circular cross-section. If the length of the wire is 36 cm, then its radius is-

- (a) 0.5 cm (b) 2 cm
 (c) 1.5 cm (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift II

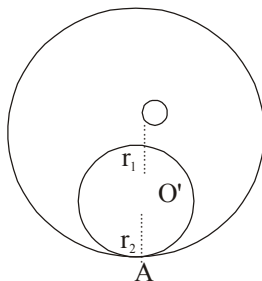
Let the radius of sphere = R
 radius of long wire = r
 length of the wire = h
 According to the question-
 $2R = 6\text{ cm}$
 $R = 3\text{ cm}$
 $h = 36\text{ cm}$
 Volume of the sphere = volume of the wire
 $\frac{4}{3} \times \pi \times 3 \times 3 \times 3 = \pi \times r^2 \times 36$
 $r = 1\text{ cm}$
 Hence, option (e) will be the right answer.

70. Two circles touch internally. The sum of their areas is $116\pi\text{ cm}^2$ and the distance between their centres is 6 cm. The radii of the circles are respectively—
 (a) 16 cm and 10 cm (b) 10 cm and 4 cm
 (c) 12 cm and 6 cm (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

According to the question-
 $(r_1 - r_2) = 6\text{ cm}$ (i)
 $\pi r_1^2 + r_1^2 = 116\pi$
 $r_1^2 + r_2^2 = 116$ (ii)
 from equation (i)–
 $(r_1 - r_2)^2 = 6^2$
 $r_1^2 + r_2^2 - 2r_1r_2 = 36$
 $116 - 2r_1r_2 = 36$
 $2r_1r_2 = 80$
 Now
 $(r_1 + r_2)^2 = r_1^2 + r_2^2 + 2r_1r_2$
 $(r_1 + r_2)^2 = 116 + 80$
 $(r_1 + r_2)^2 = 196$
 $r_1 + r_2 = 14$ (iii)
 from (i) and (iii) –
 $2r_1 = 20\text{ cm}$
 $r_1 = 10\text{ cm}$
 again
 $2r_2 = 8\text{ cm}$
 $r_2 = 4\text{ cm}$
 Hence, option (b) will be the right answer.



71. A fast train takes 3 hours less than a slow train for a journey of 600 km. If the speed of the slow train is 10 km/hr less than that of the fast train, then the speeds of the two trains are respectively
 (a) 50 km/hr and 40 km/hr
 (b) 30 km/hr and 20 km/hr

- (c) 40 km/hr and 30 km/hr
 (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

Let the speed of the high speed train = x
 then slow train speed = $(x - 10)$
 According to question-
 $\frac{600}{(x - 10)} - \frac{600}{x} = 3$
 $\frac{x - x + 10}{x(x - 10)} = \frac{3}{600} = \frac{1}{200}$

$2000 = x(x - 10)$
 $x^2 - 10x - 2000 = 0$
 $x^2 - (50 - 40)x - 2000 = 0$
 $x^2 - 50x + 40x - 2000 = 0$
 $x(x - 50) + 40(x - 50) = 0$
 $(x - 50)(x + 40) = 0$
 $x = 50\text{ km}$
 Hence, High speed train speed is 50 km/h and slow speed train speed is $= 50 - 10 = 40\text{ km/h}$

72. The value of x satisfying the equations

$$\frac{1}{x} + \frac{1}{y} = 8, \frac{1}{y} + \frac{1}{z} = 12, \frac{1}{z} + \frac{1}{x} = 10 \text{ is}$$

- (a) 3 (b) $\frac{1}{3}$
 (c) $3\frac{1}{3}$ (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

$\frac{1}{x} + \frac{1}{y} = 8$ (i)
 $\frac{1}{y} + \frac{1}{z} = 12$ (ii)
 $\frac{1}{z} + \frac{1}{x} = 10$ (iii)
 From equation (ii) – (i) –
 $\frac{1}{y} + \frac{1}{z} - \frac{1}{x} - \frac{1}{y} = 12 - 8$
 $\frac{1}{z} - \frac{1}{x} = 4$ (iv)
 from equation (iii) and (iv) –
 $\frac{1}{-} + \frac{1}{-} + \frac{1}{-} - \frac{1}{-} = 4 + 10$

$$\frac{2}{z} = 14$$

$$z = \frac{1}{7}$$

from equation (iii)

$$\frac{1}{z} + \frac{1}{x} = 10$$

$$7 + \frac{1}{x} = 10$$

$$\frac{1}{x} = 3$$

$$\Rightarrow x = \frac{1}{3}$$

Hence, option (b) will be the right answer.

73. There is 50% increase in an amount in 5 years at simple interest. What will be the compound interest of ₹12,000 after 3 years at the same rate?

- (a) ₹ 3972 (b) ₹ 6240
(c) ₹ 3120 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

Let the principle amount (P) = 100

after 5 year at simple interest the amount will be = $100 \times \frac{150}{100}$

= 150

Now

simple interest = 150 - 100 = 50

$$50 = \frac{100 \times 5 \times r}{100}$$

r = 10%

now, at the same rate compound interest of ₹12000

so-

$$A = 12000 \times \frac{110}{100} \times \frac{110}{100} \times \frac{110}{100}$$

$$A = 12 \times 11 \times 11 \times 11$$

$$A = 12 \times 1331$$

Now compound Interest = A - P

$$= 15972 - 12000$$

Compound Interest = 3972

74. $37\frac{1}{2}\%$ of the candidates in an examination were girls.

75% of the boys and $62\frac{1}{2}\%$ of the girls passed, and 342 girls failed. The number of boys failed was

- (a) 370 (b) 380

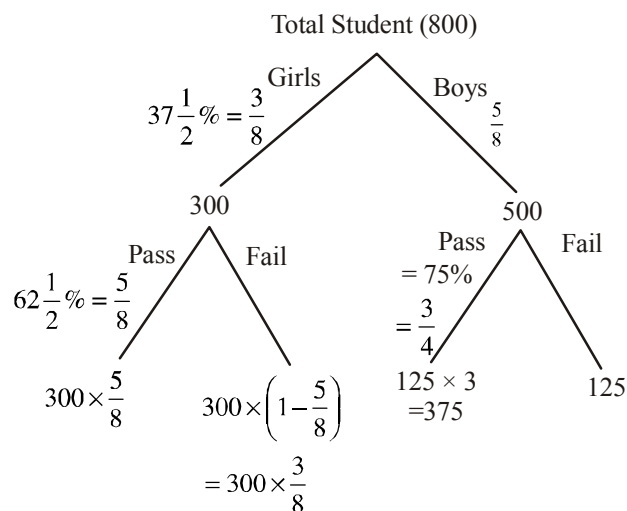
- (c) 360 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

$$37\frac{1}{2}\% = \frac{3}{8}$$

Let Total student in the class = 800



Now failed boys-

$$125 \rightarrow 125 \times 342 \times \frac{8}{3 \times 300}$$

$$\rightarrow 5 \times 342 \times \frac{2}{9}$$

$$\rightarrow 5 \times 38 \times 2$$

Number of boys failed = 380 boys

75. Two possible rational numbers between $-\frac{2}{3}$ and $\frac{1}{2}$ are

- (a) $\frac{2}{6}, \frac{3}{5}$ (b) $-\frac{2}{6}, \frac{2}{6}$
(c) $-\frac{1}{6}, \frac{4}{6}$ (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

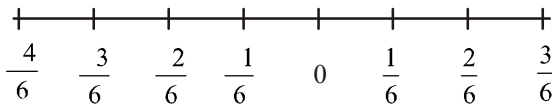
Ans. (b)

In order to determine two possible number

b/w $-\frac{2}{3}$ and $\frac{1}{2}$ have to draw number line-

$$\frac{-2}{3} \rightarrow \frac{-4}{6}$$

$$\frac{1}{2} \rightarrow \frac{3}{6}$$



There are the possible rational numbers between, $-\frac{2}{3}$ and $\frac{1}{2}$

Hence option (b) will be the right answer.

76. A father is five times as old as his son, but after 15 years, he will remain only twice as old as his son. What is the age of the father?
- (a) 20 years (b) 25 years
(c) 30 years (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

Let the age of son = x

then, the age of father = 5x

According to the question-

$$(5x + 15) = 2 \times (x + 15)$$

$$5x + 15 = 2x + 30$$

$$3x = 30 - 15$$

$$3x = 15$$

$$x = 5$$

Now the age of Father = $5 \times 5 = 25$ years

77. Neelu buys lemons at the rate of 2 lemons in one rupee and sells them at the rate of 5 lemons in three rupees. Her profit is
- (a) 15% (b) 18%
(c) 20% (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

Cost price of 2 lemons (cp) = ₹1

Selling price of 5 lemons (sp) = ₹3

Now

$$2cp = ₹1 \quad \dots(i)$$

$$5sp = ₹3 \quad \dots(ii)$$

multiply by 3 on the (i) equation-

$$6cp = ₹3 \quad \dots(iii)$$

from equation (iii) and (ii)-

$$6cp = 5sp$$

$$\frac{sp}{cp} = \frac{6}{5}$$

$$P\% = \frac{sp - cp}{cp} \times 100$$

$$P\% = \frac{6 - 5}{5} \times 100$$

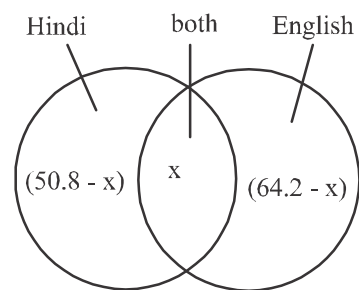
$$P\% = 20\%$$

78. In a school, every student offers either Hindi or English or both. 50.8% offer Hindi and 64.2% offer English. If the total number of students is 500, how many students offer both Hindi and English?

- (a) 60 (b) 65
(c) 75 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)



According to question

$$(50.8\% - x) + x + (64.2\% - x) = 100\%$$

$$50.8\% - x + x + 64.2\% - x = 100\%$$

$$115.0\% - x = 100\%$$

$$x = 15\%$$

Now the number of students who offer both Hindi and English

$$= 500 \times \frac{15}{100} = 75 \text{ Student}$$

79. What is the value of the following?

$$x^{a(b-c)} \cdot x^{b(c-a)} \cdot x^{c(a-b)}$$

- (a) 0 (b) 1
(c) x (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

$$x^{a(b-c)} \cdot x^{b(c-a)} \cdot x^{c(a-b)}$$

$$= x^{ab-ac+bc-ba+ca-cb}$$

$$= x^0$$

$$= 1$$

80. How many 4-digit numbers can be formed using the digits 0, 1, 2, 3, 4 without repetition?

- (a) 120
(b) 5^4
(c) 4^5
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

We have 5-digit

0, 1, 2, 3, 4

we are supposed to form 4-digit number using these digits with repetition

(i) If we chose First digit we have 5 options

(ii) If we chose second digit we have 4 options

(iii) If we chose Third digit we have 3 options

(iv) If we chose fourth digit we have 2 options

Now the total number of four digit no.

$$= 5 \times 4 \times 3 \times 2$$

$$= 120$$

81. If the word HEIGHT is coded as 96108921 and LOOSE is coded as 131616206, then the code for MOBILE, in the same language, is

- (a) 1416310136 (b) 1416411136
(c) 1416312136 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

The given coded word-like as-

Place value Code
H ——— 8 ——— +1 ———> 9

E ——— 5 ——— +1 ———> 6

I ——— 9 ——— +1 ———> 10

G ——— 7 ——— +1 ———> 8

H ——— 8 ——— +1 ———> 9

T ——— 20 ——— +1 ———> 21

Place value Code
L ——— 12 ——— +1 ———> 13

O ——— 15 ——— +1 ———> 16

O ——— 15 ——— +1 ———> 16

S ——— 19 ——— +1 ———> 20

E ——— 5 ——— +1 ———> 6

Similarly-

Place value code
M ——— 13 ——— +1 ———> 14

O ——— 15 ——— +1 ———> 16

B ——— 2 ——— +1 ———> 3

I ——— 9 ——— +1 ———> 10

L ——— 12 ——— +1 ———> 13

E ——— 5 ——— +1 ———> 6

Now 'MOBILE' is coded as '14 16 3 10 13 6'

82. Arrange the following words in logical order :

1. Associate Professor 2. Professor

3. Assistant Professor

Select the correct answer using the codes given below.

- (a) 2, 3, 1 (b) 2, 1, 3
(c) 1, 2, 3 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

In the academic hierarchy, the logical order is typically as follows— From lower to higher.

Assistant Professor Associate Professor Professor
3 1 2

So the correct order will be "2 1 3" from higher to lower

83. A person divides some items among four children. The

first child gets $\frac{1}{2}$ of the total items and the second child

gets $\frac{1}{4}$ of the first child. The third child gets $\frac{3}{4}$ of the first child. The fraction of the items the fourth child gets is

- (a) $\frac{3}{4}$ of the total items
(b) $\frac{3}{8}$ of the total items
(c) $\frac{1}{4}$ of the total items
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (e)

Let the total item = 1

$$\text{First child} = 1 \times \frac{1}{2} = \frac{1}{2}$$

$$\text{Second child} = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$$

$$\text{Third child} = \frac{3}{4} \times \frac{1}{2} = \frac{3}{8}$$

Therefore, Fourth child get Nothing

Remaining

$$= 1 - \left(\frac{1}{2} + \frac{1}{8} + \frac{3}{8} \right)$$

$$= 1 - \left(\frac{1}{2} + \frac{1}{2} \right)$$

$$= 1 - 1$$

84. If the area of a square is equal to the area of an equilateral triangle, the ratio of the side of the square and the side of the equilateral triangle is

- (a) 1 : 1 (b) 2 : 1
(c) $\sqrt{3} : 2$ (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (e)

Let the side of a square is 'a' and side of the equilateral triangle is 'b'

Now we know that-

Area of square = a^2

Area of a equilateral triangle = $\frac{\sqrt{3}}{4} b^2$

According to the question-

$$a^2 = \frac{\sqrt{3}}{4} b^2$$

$$\frac{a^2}{b^2} = \frac{\sqrt{3}}{4}$$

$$\frac{a}{b} = \frac{\sqrt[4]{3}}{2}$$

85. Find the wrong number in the series given below :

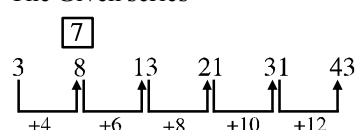
3, 8, 13, 21, 31, 43

- (a) 21 (b) 31
(c) 8 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

The Given series-



Therefore, In the place 8, there should be 7 for the correct sequencing of the series.

86. Find the wrong number from the table given below :

2	4	1	5
6	12	3	15
18	36	9	45
54	108	27	100

- (a) 9 (b) 36
(c) 100 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

The Given table-

2 } $\times 3$	4 } $\times 3$	1 } $\times 3$	5 } $\times 3$
6 } $\times 3$	12 } $\times 3$	3 } $\times 3$	15 } $\times 3$
18 } $\times 3$	36 } $\times 3$	9 } $\times 3$	45 } $\times 3$
54 } $\times 3$	108 } $\times 3$	27 } $\times 3$	100 } $\times 3$ 135

In the Given table- 100 is incorrect, in the place of 100 there should be 135.

87. Let $a = 10$ cm, $b = 3$ cm and $c = 6$ cm. Which of the following is true?

- (a) a , b and c are the sides of a triangle.
(b) a , b and c form right-angled triangle.
(c) a , b and c form isosceles triangle.
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

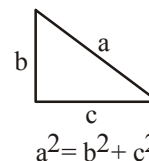
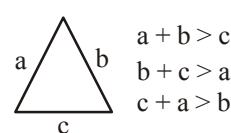
Ans. (e)

Given that-

$a = 10$, $b = 3$, $c = 6$

from the Given parameter no triangle can be formed because as per the triangle properties-

for right angled triangle



Here

$$10 + 3 > 6$$

but

$$3 + 6 \not> 10$$

Hence, No triangle can be formed

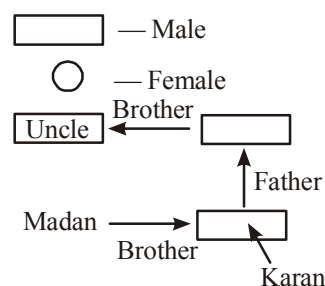
88. Madan's brother's uncle is Karan's father's brother. What can be the relationship between Madan and Karan?

- (a) Madan is Karan's uncle
(b) Madan is Karan's brother
(c) Madan is Karan's son
(d) More than one of the above
(e) None of the above

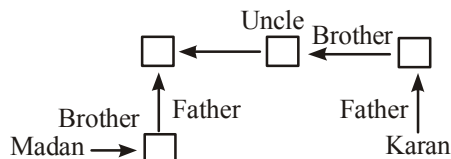
B.P.S.C. PRT 2023 Shift II

Ans. (e)

Case - 1



Madan's Gender is not defined he may be brother of Karan
Case - 2



Here, Madan & Karan both gender is not defined So we cannot define relation.

89. Find the missing letters in the series given below :

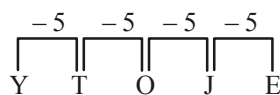
Y, T, O, _____, _____

- (a) J, E (b) J, D
 (c) K, F (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift II

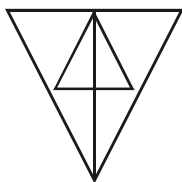
Ans. (a)

The Given Series-



Hence, the missing Letter will be J & E.

90. How many triangles are there in the figure given below?



- (a) 13 (b) 11
 (c) 15 (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)



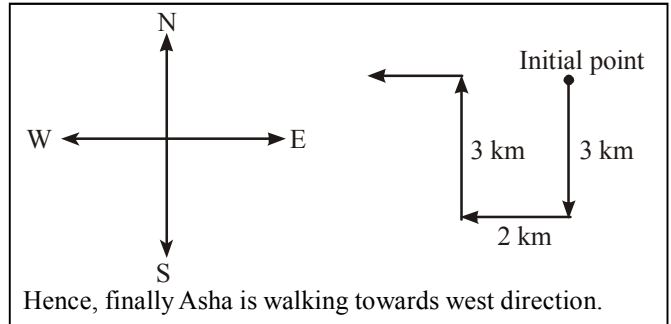
Triangle from single Image-6
 Triangle from double Image-2
 Triangle triple Image-2
 Triangle, Comprising all part of the picture-1
 Total triangle = 6 + 2 + 2 + 1 = 11

91. Asha walks 3 km southward and then turns right and walks 2 km. She again turns right and walks 3 km, and then turns towards her left and starts walking straight. In which direction is she walking now?

- (a) North (b) West
 (c) East (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)



Hence, finally Asha is walking towards west direction.

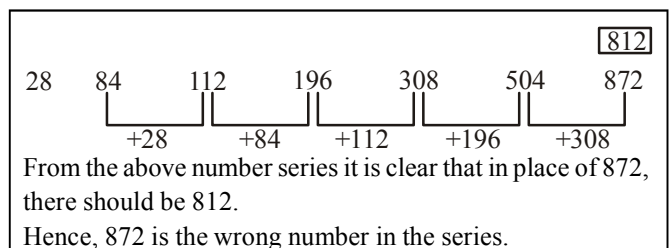
92. Find the wrong number in the series given below :

28, 84, 112, 196, 308, 504, 872

- (a) 112 (b) 308
 (c) 872 (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)



From the above number series it is clear that in place of 872, there should be 812.

Hence, 872 is the wrong number in the series.

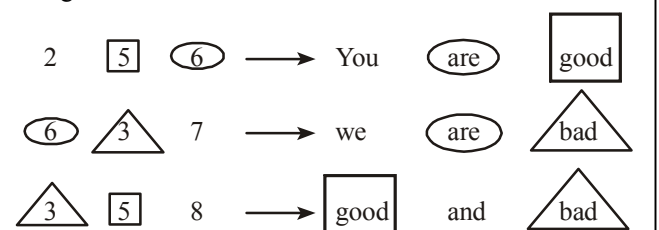
93. In a certain code, '256' means 'you are good'; '637' means 'we are bad' and '358' means 'good and bad'. Which of the following digits represents 'and' in that code?

- (a) 2 (b) 5
 (c) 8 (d) More than one of the above
 (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

The given code-



from the above arrangement it is clear in that 'and' is coded as '8' in the question.

94. Six roads lead to a country. They may be indicated by the letters X, Y, Z and the digits 1, 2, 3. When there is a storm, Y is blocked. When there are floods, X, 1 and 2 will be affected. When road 1 is blocked, Z is also blocked. At a time, when there are floods and a storm also blows, which road can be used?

- (c) Z (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

Roads	Affected or Blocked	Unblocked
X	✓	✗
Y	✓	✗
Z	✓	✗
1	✓	✗
2	✓	✗
3		✓

Therefore, In the Given Situation, road 3 can be used,

95. In a certain code, DELHI is written as CCIDD. How is BOMBAY written in that code?

- (a) AJMTVT (b) MJXVSU
(c) AMJXVS (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

Just Like-	Similarly
D $\xrightarrow{-1}$ C	B $\xrightarrow{-1}$ A
E $\xrightarrow{-2}$ C	O $\xrightarrow{-2}$ M
L $\xrightarrow{-3}$ I	M $\xrightarrow{-3}$ J
H $\xrightarrow{-4}$ D	B $\xrightarrow{-4}$ X
I $\xrightarrow{-5}$ D	A $\xrightarrow{-5}$ V
	Y $\xrightarrow{-6}$ S

Here, option (c) will be the correct option.

96. One of the two digits of a two-digit number is three times the other digit. If you inter- change the digits of this two-digit number and add the resulting number to the original number, you get 88. What is the original number?

- (a) 62 (b) 31
(c) 64 (d) 33

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Ans. (a)

Let the Unit digit of the number is 'x', then other number will be '3x'
Now,
$3x \times 10 + x = 31x$
after interchanging, the number-
$10x + 3x = 13x$
Now

$$31x + 13x = 88$$

$$44x = 88$$

$$x = 2$$

Now the Original Number is- $31 \times 2 = 62$

Hence, Option (a) will be the right answer.

97. Two candles are of different lengths and thicknesses. The short and the long ones can burn respectively for 3.5 hours and 5 hours. After burning for 2 hours, the lengths of the candles become equal in length. What fraction of the long candle's height was the short candle initially?

- (a) $\frac{5}{7}$ (b) $\frac{3}{5}$
(c) $\frac{4}{5}$ (d) $\frac{2}{7}$

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Ans. (a)

Let the lengths are l_1 and l_2 and thickness are x_1 and x_2

$$l_1 = 3.5 \text{ hours and } l_2 = 5 \text{ hours}$$

According to the question-

$$l_1 - 2 \times \frac{l_1}{3.5} = l_2 - 2 \times \frac{l_2}{5}$$

$$\frac{3.5l_1 - 2l_1}{3.5} = \frac{5l_2 - 2l_2}{5}$$

$$\frac{1.5l_1}{3.5} = \frac{3l_2}{5}$$

$$\frac{l_1}{l_2} = \frac{2.1}{1.5}$$

$$\frac{l_1}{l_2} = \frac{7}{5} \text{ or } \boxed{\frac{l_2}{l_1} = \frac{5}{7}}$$

Hence, option (a) will be the right answer.

98. In a survey of 500 students of a college, it was found that 49% liked watching football, 53% liked watching hockey and 62% liked watching basketball. Also, 27% liked watching football and hockey both, 29% liked watching basketball and hockey both and 28% liked watching football and basketball both. 5% liked watching none of these games. How many students like watching all the three games?

- (a) 10% (b) 15%
(c) 12% (d) 20%

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Ans. (b)

n(F) = Percentage of Student who like watching football = 49%

n(H) = Percentage of Student who like watching hockey = 53%

n(B) = Percentage of Student who like watching basketball

$$n(F \cap H) = 27\%, n(B \cap H) = 29\%, n(F \cap B) = 28\%$$

Since, 5% like watching none of the given games so,

$$n(F \cap H \cap B) = 95\%$$

Now

$$n(F \cap H \cap B) = n(F) + n(H) + n(B) - n(F \cap H) - n(B \cap H) - n(F \cap B) + n(F \cap H \cap B)$$

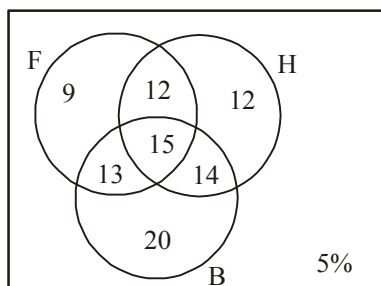
$$95\% = 49\% + 53\% + 62\% - 27\% - 29\% - 28\% + n(F \cap H \cap B)$$

$$95\% = 164\% - 84\% + n(F \cap H \cap B)$$

$$95\% = 80\% + n(F \cap H \cap B)$$

$$n(F \cap H \cap B) = 15\%$$

Now-



From the above venn diagram-

Number of Students who like watching all the three games = 15%

99. Mother was asked how many gifts she had in the bag. She replied that there were all dolls but six, all cars but six, and all books but six. How many gifts had she in all?

- (a) 18 (b) 27
(c) 36 (d) 9

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Ans. (d)

Let the total number of toys be x

Then Number of dolls = x - 6

Number of cars = x - 6

Number of books = x - 6

According to the question-

$$(x - 6) + (x - 6) + (x - 6) = x$$

$$3x - 18 = x$$

$$2x = 18$$

$$x = 9$$

Hence, option (d) will be the right answer.

100. In a dairy, there are 60 cows and buffalos. The number of cows is twice that of buffalos. Buffalo X ranked seventeenth in terms of milk delivered. If there are 9 cows ahead of buffalo X, how many buffalos are after in rank in terms of milk delivered?

(c) 13

(d) 10

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Ans. (b)

Let the number of buffalo be x, then no. of cows will be 2x

Now

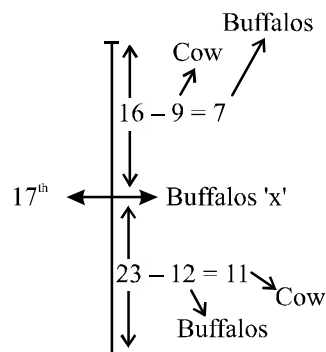
$$2x + x = 60$$

$$3x = 60$$

$$x = 20$$

Cows- 40

Buffalo - 20



There are total 20 buffalos and 7 of them are above buffalos 'x' then after 'x' there will be 12 buffalos

101. If you write down all the numbers from 1 to 100, then how many times do you write 3?

- (a) 17 (b) 21
(c) 20 (d) 18

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Ans. (c)

According to the question-

1 → 9 → 1 time

10 → 20 → 1 time

21 → 30 → 2 times

31 → 40 → 10 times

41 → 50 → 1 time

51 → 60 → 1 time

61 → 70 → 1 time

71 → 80 → 1 time

81 → 90 → 1 time

91 → 100 → 1 time

$$\text{Total} = 20$$

Therefore, from 1 to 100, 3 comes, 20 times

102. If a cube of 25 cm side is divided into 125 smaller cubes of equal volume, then the side of smaller cube thus formed will be

- (a) 3 cm (b) 5 cm
(c) 6 cm (d) 2 cm

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Ans. (b)

Let us assume that the sides of the smaller cube is x-
According to the question-

$$25 \times 25 \times 25 = 125x^3$$

$$x^3 = 125$$

$$x = 5 \text{ cm}$$

Hence, option (b) will be the right answer.

103. A, B, C, D, E and F, not necessarily in that order, are sitting at a round table. A is between D and F, C is opposite to D; and D and E are not on neighboring chairs. Which one of the following pairs must be sitting on neighboring chairs?

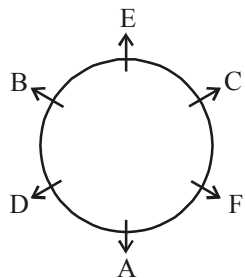
- (a) C and E (b) B and F
(c) A and C (d) A and B

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Ans. (a)

There are six person A, B, C, D, E and F sitting around a table with certain condition-

On following that condition the sitting arrangement as follows-



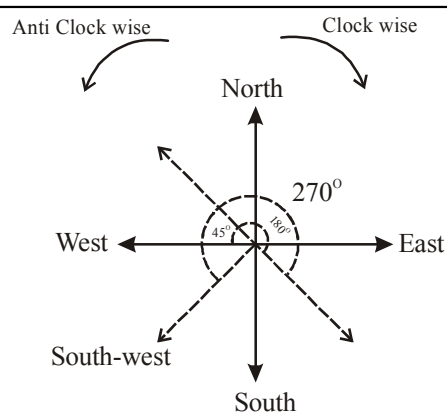
Here, C and E must be sitting on neighboring chairs

104. A man is facing West. He turns 45° in the clockwise direction and then another 180° in the same direction and then 270° in the anticlockwise direction. Which direction is he facing now?

- (a) North-West
(b) West
(c) South-West
(d) South

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Ans. (c)



Hence, option (c) will be the right answer.

105. If a, b, c, d, e and f are six consecutive even numbers, then their average is

- (a) $a + 5$ (b) $6(a + 5)$
(c) $(abcdef)/6$ (d) $a + 4$

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Ans. (a)

Given that-

a, b, c, d, e and f are six consecutive even numbers-

$$b = a + 2$$

$$c = a + 4$$

$$d = a + 6$$

$$e = a + 8$$

$$f = a + 10$$

Now-

$$\text{Average} = \frac{a + b + c + d + e + f}{6}$$

$$= \frac{a + (a + 2) + (a + 4) + (a + 6) + (a + 8) + (a + 10)}{6}$$

$$= \frac{6a + 30}{6}$$

$$\text{Average} = (a + 5)$$

Hence, option (a) will be the right answer.

106. The question given below has a problem and two statements I and II. Decide if the information given in the statements is sufficient for answering the problem:

K, R, S and T are four players in Indian cricket team.

Who is the oldest among them?

I. The total age of K and T together is more than that of S.

II. The total age of R and K together is less than that of S.

- (a) Data in statement II alone is sufficient
(b) Data in both statements together is sufficient
(c) Data in both statements together is not sufficient
(d) Data in statement I alone is sufficient

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Ans. (c)

As per the Given statement in the question-

$$K + T > S$$

$$R + K < S$$

This implies that-

$$R < T$$

$$\text{and } R, K < S$$

Therefore either T or S is the oldest, but there is no relation between ages of T and S

Hence, data in both statement together is not sufficient.

107. An accurate clock shows 8 O'clock in the morning. Through how many degrees will the hour hand rotate when the clock shows 2 O'clock in the afternoon?

- (c) 180 degrees (d) 150 degrees

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Ans. (c)

Given that – hour hand of the Clock moves from 8 O'clock to 2 O' Clock –

Time Gap between 8 O' Clock and 2' O' Clock = 6 hours

Now, We know that–

hour hand moves 360° in 12 hours

12 hours = 360°

1 hours = 30°

6 hours = 180°

Therefore, hour hand moves 180° from 8 O'clock to 2 O'clock

108. If $(x-1)$ is a factor of the equation $x^3(p)3x^2+3x(q)1=0$, then signs in place of (p) and (q) are

- (a) $-, +$ (b) $+, +$
(c) $+, -$ (d) $-, -$

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Ans. (d)

Given that $(x-1)$ is a factor of the equation

$x^3(p)3x^2+3x(q)1=0$

then $x-1=0$

$x=1$

$1(p)3+3(q)1=0$

From the option (d)-

L.H.S.

$1-3+3-1$

$-2+2=0$

R.H.S.

Hence, option (d) will be the right answer.

109. Mr. Prasad invested an amount of ₹13,900 divided in two different schemes A and B at the simple interest rate of 14% p.a. and 11% p.a. respectively. If the total amount of simple interest earned in 2 years be ₹3,508, what was the amount invested in scheme B?

- (a) ₹ 6800 (b) ₹ 7100
(c) ₹ 6400 (d) ₹ 6600

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Ans. (c)

Given that-

$A+B=13900$ (1)

and

$$\frac{A \times 14\% \times 2}{100} + \frac{B \times 11\% \times 2}{100} = 3508$$

$28A+22B=3508 \times 100$

$14A+11B=175400$ (ii)

Multiplying by 14 in equation (i) & subtract equation (ii) from it-

$14A+14B=13900 \times 14$

$$14A+14B=194600$$

$$14A+11B=175400$$

$$\underline{\hspace{1cm}}$$

$$3B=19200$$

$$B=\text{₹ } 6400$$

Hence, Option (c) will be the right answer

110. If the national day of a country was celebrated on the 4th Saturday of a month, then the date of celebration was (it is given that the first day of that month is Tuesday)

- (a) 25th (b) 26th
(c) 27th (d) 24th

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Ans. (B)

Let us assume a month that have 30 day's

If the 1st day of the month is "Tuesday" then the 1st Saturday of this month will be on '5'

Now

IInd Saturday-12

IIIrd Saturday- 19

IVth Saturday- 26

Hence, Option (b) will be the right answer.

111. Find the quadratic equation whose roots are the reciprocals of the roots of

$$2x^2+5x+3.$$

- (a) $3x^2-5x-2=0$ (b) $5x^2+3x+2=0$
(c) $3x^2+5x+2=0$ (d) None of the above

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Ans. (c)

The Given equation–

$$2x^2+5x+3=0$$

We know that–

$$ax^2+bx+c=0 \text{ whose roots are } \alpha \text{ \& \& } \beta$$

then

$$\alpha+\beta=\frac{-b}{a}, \alpha\beta=\frac{c}{a}$$

similarly–

$$\alpha+\beta=\frac{-5}{2}$$

$$\alpha\beta=\frac{3}{2}$$

reciprocal of $\alpha, \beta, \frac{1}{\alpha}$ and $\frac{1}{\beta}$ is

Now–

$$\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}=\frac{\frac{-5}{2}}{\frac{3}{2}}=\frac{-5}{3}$$

and—

$$\frac{1}{\alpha\beta} = \frac{2}{3}$$

Now the new equation—

$$x^2 - \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)x + \left(\frac{1}{\alpha} \times \frac{1}{\beta}\right) = 0$$

$$x^2 - \left(\frac{-5}{3}\right)x + \frac{2}{3} = 0$$

$$x^2 + \frac{5}{3}x + \frac{2}{3} = 0$$

$$3x^2 + 5x + 2 = 0$$

Hence, Option (c) will be the right answer.

- 112. Suppose that in a town, 800 people are selected randomly. 280 go to work by car only, 220 go to work by bicycle only and 140 use both ways-sometimes go with a car and sometimes with a bicycle. How many people go by neither car nor bicycle?**

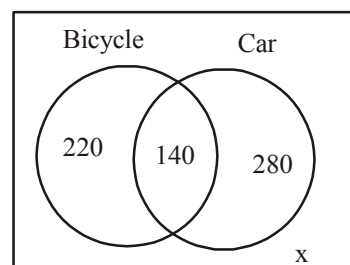
- (a) 500 (b) 160
(c) 60 (d) 140

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Ans. (b)

Total People = 800

Let x person does not go by neither car nor bicycle-



$$220 + 140 + 280 + x = 800$$

$$640 + x = 800$$

$$x = 160$$

Hence, option (b) will be the right answer.

- 113. A worker is paid ₹ 20 for a full day work. He works 1, 1/3, 2/3, 1/8, 3/4 days in a week. What is the total amount paid for that worker?**

- (a) 58 (b) 57.50
(c) 56.50 (d) 57

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Ans. (b)

Given that-

Worker's full day salary = ₹20

Total days work in a week

$$= 1 + \frac{1}{3} + \frac{2}{3} + \frac{1}{8} + \frac{3}{4}$$

$$= 1 + 1 + \frac{1}{8} + \frac{6}{8}$$

$$= 2 + \frac{1}{8} + \frac{6}{8}$$

$$= \frac{23}{8} \text{ day}$$

$$\text{Total amount paid} = \frac{23}{8} \times 20 = \frac{115}{2} = ₹57.50$$

Hence, option (b) will be the right answer.

- 114. The odometer of a car reads 2000 km at the start of a trip and 2400 km at the end of the trip. If the trip took 8 hours, the average speed of the car will be**

- (a) 40 km/h (b) 75 km/h
(c) 50 km/h (d) 60 km/h

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Ans. (c)

Reading of odometer at the start of trip = 2000 km

Reading of odometer at the end of trip = 2400 km

Distance covered = 2400 - 2000 = 400 km

Time taken for the trip = 8 hours.

Now

$$\begin{aligned} \text{Average speed of car} &= \frac{400}{8} \text{ km/hr} \\ &= 50 \text{ km/hr} \end{aligned}$$

- 115. The average marks of 80 students are given to be 60. It was found later that marks of two students were 45 and 31 which were misread as 85 and 71. The corrected mean mark will be**

- (a) 59 (b) 53
(c) 61 (d) 49

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Ans. (a)

The average marks of 80 students = 60

Total marks of 80 students = 60 × 80 = 4800

It was found that 45 & 31 misread as 85 and 71. Now the correct total marks of 80 students

$$= 4800 + (45 + 31) - (85 + 71)$$

$$= 4800 + 76 - 156$$

$$= 4720$$

$$\text{Now the correct average of 80 students} = \frac{4720}{80} = 59$$

Hence, option (a) will be the right answer.

- 116. If out of given six consecutive natural numbers, the sum of first three consecutive natural numbers is 27, what is the sum of rest last three consecutive natural numbers?**

(c) 36

(d) 96

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Ans. (c)

Let us consider the six consecutive natural number-
 $x, (x+1), (x+2), (x+3), (x+4), (x+5)$

According to the question-

$$x + (x+1) + (x+2) = 27$$

$$3x + 3 = 27$$

$$x + 1 = 9$$

$$x = 8$$

Now, the sum of last three consecutive natural number-

$$= (x+3) + (x+4) + (x+5)$$

$$= 3x + 12$$

$$= 3 \times 8 + 12$$

$$= 36$$

Hence, option (c) will be the right answer.

117. If $3^{2x-1} + 3^{2x+1} = 270$ then x is equal to

(a) 5

(b) 3

(c) 2

(d) 4

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Ans. (c)

$$3^{2x-1} + 3^{2x+1} = 270$$

$$3^{2x} \times 3^{-1} + 3^{2x} \times 3^1 = 270$$

$$3^{2x} \left(\frac{1}{3} + 3 \right) = 270$$

$$3^{2x} \times \frac{10}{3} = 270$$

$$3^{2x} = 3 \times 3 \times 3 \times 3$$

$$3^{2x} = 3^4$$

$$2x = 4$$

$$x = 2$$

Hence, option (c) will be the right answer.

118. A train reaches at its destination on time if its speed is 40 km per hour from its initial station. If this train runs with a speed of 35 km per hour, it reaches its destination late by 15 minutes. Accordingly what is the distance between its initial station and destination station?

(a) 60 km

(b) 70 km

(c) 80 km

(d) 35 km

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Ans. (b)

Let the Distance is x
we know that-

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Now, According to the question-

$$\frac{x}{35} - \frac{x}{40} = \frac{15}{60}$$

$$x \left(\frac{1}{35} - \frac{1}{40} \right) = \frac{15}{60}$$

$$x \times \frac{5}{40 \times 35} = \frac{1}{4}$$

$$x = \frac{40 \times 35}{20}$$

$$x = 70 \text{ km}$$

Hence, option (b) will be the right answer.

119. If two numbers x and y are less than 50% and 20% respectively with any third number, the percentage x of y is equal to

(a) 62.5

(b) 65

(c) 66

(d) 62

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Ans. (a)

Let the third number is 'z'

$$\text{Then } x = z \times \frac{50}{100} \Rightarrow \frac{z}{2}$$

$$y = z \times \frac{20}{100} \Rightarrow \frac{2z}{5}$$

$$\text{Now percentage x of y} = \frac{\frac{z}{2}}{\frac{2z}{5}} \times 100 = 62.5\%$$

Hence, option (a) will be the right answer.

120. A shopkeeper fixes selling price of its items by 10% increase on its cost. On this fixed selling price, a discount of 10% is given. What is the percentage loss or profit on sale?

(a) 1% profit

(b) 10% loss

(c) 10% profit

(d) 1% loss

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Ans. (d)

Let the cost price of item = 100

$$\text{then market price} = 100 \times \frac{110}{100} = 110$$

$$\text{After 10\% discount, the selling price} = 110 \times \frac{90}{100} = 99$$

Now

$$\text{Loss (L\%)} = \frac{\text{CP} - \text{SP}}{\text{CP}} \times 100$$

$$= \frac{100 - 99}{100} \times 100$$

$$\text{L\%} = 1\%$$

121. Two numbers are such that their sum, difference and product are in the ratio 9:5:28. The difference of their square is

- (a) 170 (b) 280
(c) 370 (d) 180

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Ans. (d)

Let two numbers are x and y according to the question-
 $(x + y) : (x - y) : xy = 9 : 5 : 28$
 $x + y = 9k$ (i)
 $x - y = 5k$ (ii)
 from equation (i) and (ii)-
 $2x = 14k$
 $x = 7k$
 also
 $y = 2k$
 Now
 $xy = 28k$
 $14K^2 = 28k$
 $k = 2$
 then the numbers are-
 $x = 14, y = 4$
 According to the question-
 $x^2 - y^2 = 196 - 16$
 $x^2 - y^2 = 180$

122. The cost price of an item is ₹800. At the time of its sale, a discount of 10% is given on its fixed selling price. In spite of this discount, a profit of 12.5% is earned. Accordingly, what is the fixed selling price of this item?

- (a) ₹1200 (b) ₹1100
(c) ₹1000 (d) ₹1300

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Ans. (c)

Given that-
 cost price of article = 800
 Discount = 10%
 Profit = 12.5%
 We know that-

$$\frac{MP}{CP} = \frac{100 \pm \text{Profit or loss}}{100 - \text{discount\%}}$$

$$\frac{MP}{800} = \frac{100 + 12.5}{100 - 10}$$

$$\frac{MP}{800} = \frac{112.5}{90}$$

$$MP = \frac{112.5 \times 80}{9}$$

$$MP = ₹1000$$

 Hence, option (c) will be the right answer.

123. If the sum of square of two numbers and difference of the squares of these numbers are 58 and 40 respectively, then the sum of these numbers is

- (a) +10 (b) -10
(c) 0 (d) Both (a) and (b)

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (d)

Let the numbers are x & y
 According to the question-
 $x^2 + y^2 = 58$ (i)
 $x^2 - y^2 = 40$ (ii)
 On adding both the equation-
 $2x^2 = 98$
 $x^2 = 49$
 $x = \pm 7$
 from equation (i)-
 $49 + y^2 = 58$
 $y^2 = 9$
 $y = \pm 3$
 then sum of these numbers are +10 and - 10 which is given in the option (d).

124. How many words, no matter if they are meaningless, can be formed by the letters of the word 'DIARY'?

- (a) 24 (b) 25
(c) 10 (d) More than one of the above
(e) None of the above

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Ans. (e)

No. of letters in word 'DIARY' = 5
 According to questions
 No. of words that can be formed by the letters of the word 'DIARY' = 5!
 $= 5 \times 4 \times 3 \times 2 \times 1$
 $= 120$

125. Select the missing number from the given alternatives:

44	49	37
52	?	41
58	35	53

- (a) 66 (b) 56
(c) 77 (d) More than one of the above
(e) None of the above

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44	49	37
52	?	41
58	35	53

In row I $\rightarrow 44 - 37 = 7 \Rightarrow 7 \times 7 = 49$

In row III $\rightarrow 58 - 53 = 5 \Rightarrow 5 \times 7 = 35$

Similarly:

In row II $\rightarrow 52 - 41 = 11 \Rightarrow 11 \times 7 = 77$

126. Find the missing number in the given series following the same pattern :

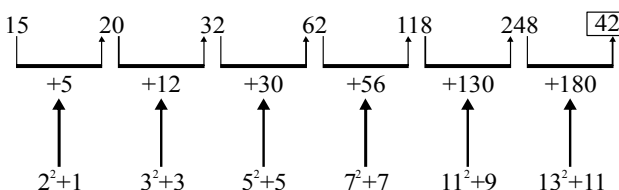
15, 20, 32, 62, 118, 248, ?

- (a) 428 (b) 322
(c) 368 (d) More than one of the above
(e) None of the above

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Ans. (a)

15, 20, 32, 62, 118, 248, ?



Pattern $\rightarrow$ (Prime no.)² + (odd no.) in increasing order.

So the correct answer is option (a) 428.

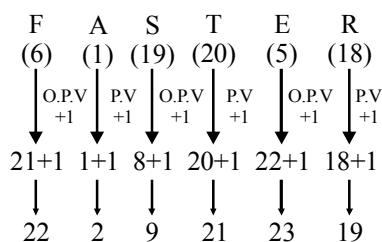
127. In a code language, if FASTER is written as 2229212319 and MONK is written as 15161412, then how will GUIDE be written in the same language?

- (a) 192019423 (b) 212219523
(c) 222119522 (d) More than one of the above
(e) None of the above

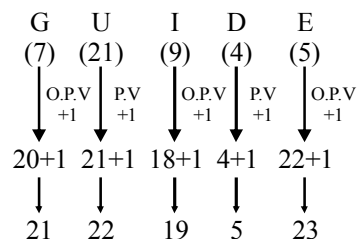
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Ans. (b)

As



Similarly-



Hence, GUIDE will be written as 212219523.

[Note: O.P.V. $\rightarrow$ Opposite Positional Value
P.V. $\rightarrow$ Positional Value.]

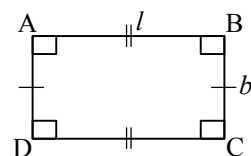
128. A rectangle has an area 30 cm² and perimeter 26 cm. Its sides (in cm) are

- (a) 10, 3 (b) 5, 6
(c) 2, 15 (d) More than one of the above
(e) None of the above

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Ans. (a)

Given that,



Area of a rectangle = 30 cm²

Perimeter of rectangle = 26 cm

Let l and b are the length and breadth of rectangle

$lb = 30$ cm²(i)

$2(l+b) = 26$ cm

$l+b = 13$ cm(ii)

From eqⁿ (i) and eqⁿ (ii)

$$l = 10 \text{ cm}$$

and

$$b = 3 \text{ cm}$$

129. A shopkeeper offers 10% discount on an item with marked price Rs. 400. If he charges 10% GST, then the final price of the item is

- (a) Rs. 380 (b) Rs. 400
(c) Rs. 396 (d) More than one of the above
(e) None of the above

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Ans. (c)

Given that

Marked price (M.P.) of an item = Rs. 400

$$\text{Price of item after 10\% discount} = \text{Rs. } 400 \times \frac{\text{Rs. } 400 \times 10}{100}$$

$$= \text{Rs. } 400 - \text{Rs. } 40$$

$$= \text{Rs. } 360$$

$$\text{GST on item} = \frac{360 \times 10}{100} = 36$$

$$\text{Total payable price of item} = \text{Rs. } 360 + \text{Rs. } 36$$

$$= \text{Rs. } 396$$

130. Arrange the following words in logical and meaningful order :

1. Vice President
 2. President
 3. Speaker
 4. Prime Minister
 5. Members of the Parliament
- (a) 5, 1, 2, 3, 4 (b) 4, 2, 1, 3, 5
 (c) 2, 1, 4, 3, 5 (d) More than one of the above
 (e) None of the above

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Ans. (c)

The logical and meaningful order is →
 President > Vice President > Prime Minister > Speaker >
 Member of Parliament
 (2) (1) (4) (3)
 (5)
 Hence, the correct answer is option (c).

131. Reshma donates one-fourth of her property to a charity organization and divides the remaining property equally among her three children. The part of property each child gets is

- (a) half (b) one-fourth
 (c) two-thirds (d) More than one of the above
 (e) None of the above

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Ans. (b)

Let the property of Reshma is x .
 Part of property donated to charity organisation = $\frac{x}{4}$
 Remaining property = $x - \frac{x}{4} = \frac{3x}{4}$
 Now, Part of property one child = $\frac{\frac{3x}{4}}{3} = \frac{3x}{4 \times 3} = \frac{x}{4}$
 Hence, the part of property each child gets is one fourth.

132. Which of the following statements is false?

- (a) Every square is a rhombus.
 (b) Every square is a rectangle.
 (c) Every rhombus is a parallelogram.
 (d) More than one of the above
 (e) None of the above

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Ans. (e)

The correct answer is option (e).
 Every square is a rectangle and also rhombus.
 Every rhombus is a parallelogram. So all the statement are true.

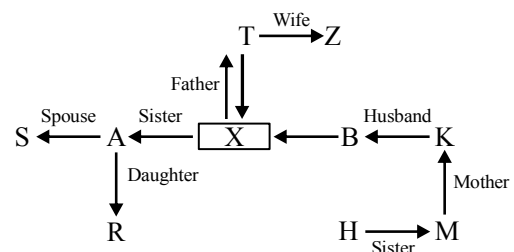
133. In a family of some persons, B says that R is the daughter of my sister A, who is the only daughter of T. X is the child of T and Z, who is the grandmother of H. K is the mother of M, who is the only sister of H. X is unmarried. If S is the spouse of A, how is K related to S?

- (a) Brother-in-law
 (b) Sister-in-law
 (c) Wife of brother-in-law
 (d) More than one of the above
 (e) None of the above

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Ans. (c)

According to question



Hence, K is the wife of B, who is brother-in-law of S.
 So option (c) is the correct answer i.e. K is the wife of brother-in-law of S.

134. The missing number in the sequence 2, 3, 6, 18, ?, 1944 is

- (a) 154 (b) 180
 (c) 108 (d) 450
 (e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (c)

The Given series-

2 3 6 18 108 1944
 2×3 6×3 18×6 108×18

Hence, option (c) will be the right answer.

135. At 4:30 AM, what is the angle formed between the hour hand and the minute hand?

- (a) 60° (b) 45°
 (c) 30° (d) 90°
 (e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Formula for finding angle between hour hand and minute hand

$$\theta = \left[\frac{11}{2} \times \text{minute} - 30 \times \text{hour} \right]$$

At 4 : 30 am

$$\theta = \left[\frac{11}{2} \times 30 - 30 \times 4 \right]$$

$$= [11 \times 15 - 120]$$

$$= [165^\circ - 120^\circ]$$

$$= 45^\circ$$

136. If ${}^nC_9 = {}^nC_8$, then the value of n is

- (a) 1 (b) 17
(c) 72 (d) 36
(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (b)

$${}^nC_9 = {}^nC_8$$

$$\frac{n!}{9!(n-9)!} = \frac{n!}{8!(n-8)!} \quad \left[\because {}^nC_r = \frac{n!}{r!(n-r)!} \right]$$

$$\frac{1}{9 \times 8!(n-9)!} = \frac{1}{8! \times (n-8) \times (n-9)!}$$

$$(n-8) = 9$$

$$n = 17$$

137. If the average of 8 observations is 16 and that of 4 observations is 64, then the average of $(8 + 4) = 12$ observations is

- (a) 4/8 (b) 8/12
(c) 32 (d) 4
(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (c)

Average of 8 observations is 16 and average of 4 observations is 64, then the average of 12 observation will be-

$$= \frac{8 \times 16 + 4 \times 64}{(8 + 4)}$$

$$= \frac{128 + 256}{12}$$

$$= \frac{384}{12}$$

$$= 32$$

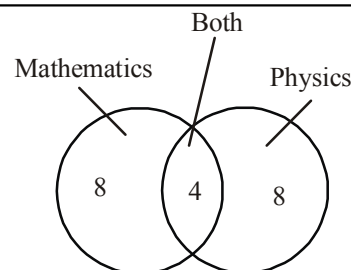
Hence, option (c) will be the right answer.

138. In a school, there are 20 teachers who teach Mathematics or Physics. Of these, 12 teach Mathematics and 4 teach both Mathematics and Physics. How many teachers teach Physics?

- (a) 4 (b) 8
(c) 12 (d) 16
(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (c)



Total teacher = 20

only maths teacher = 8

only physics teacher = 8

and teacher who teach both subject = 4

Now the physics teacher = 12

139. The 101st term of the arithmetic progression 5, 11, 17, ... is

- (a) 605 (b) 505
(c) 405 (d) 305
(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (a)

The Given Arithmetic progression-

5, 11, 17,

$a = 5$

$d = 6$

$T_n = a + (n - 1)d$

For 101st term-

$T_{101} = 5 + (101 - 1) \times 6$

$= 5 + 600$

$T_{101} = 605$

140. The two roots of the quadratic equation

$$2x^2 + \sqrt{6}x - 6 = 0 \text{ are}$$

- (a) $\sqrt{6}$ and $\sqrt{6}/2$
(b) $-\sqrt{6}$ and $\sqrt{6}$
(c) $-\sqrt{6}$ and $\sqrt{6}/2$
(d) 6 and $\sqrt{6}$
(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

The given quadratic equation–

$$2x^2 + \sqrt{6}x - 6 = 0$$

$$2x^2 + (2\sqrt{6} - \sqrt{6})x - 6 = 0$$

$$2x^2 + 2\sqrt{6}x - \sqrt{6}x - 6 = 0$$

$$2x(x + \sqrt{6}) - \sqrt{6}(x + \sqrt{6}) = 0$$

$$(2x - \sqrt{6})(x + \sqrt{6}) = 0$$

$$x = \frac{\sqrt{6}}{2}, -\sqrt{6}$$

Hence, option (c) will be the right answer.

141. Given that

$$2x + 3y = 10$$

$$3x - y = 4$$

then x and y are respectively

(a) -2 and 2 (b) 2 and -2

(c) 2 and -3 (d) 2 and 2

(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (d)

$$2x + 3y = 10 \quad \dots\dots(i)$$

$$3x - y = 4 \quad \dots\dots(ii)$$

multiply by 3 in equation (ii)

$$9x - 3y = 12 \quad \dots\dots(iii)$$

Adding equation (i) and (iii)

$$11x = 22$$

$$x = 2$$

from equation (i)-

$$2 \times 2 + 3y = 10$$

$$3y = 10 - 4$$

$$3y = 6$$

$$y = 2$$

Hence, option (d) will be the right answer.

142. The Value of

$$\frac{(35 + 27)^2 + (35 - 27)^2}{(35)^2 + (27)^2}$$

(a) 4 (b) 8

(c) 62 (d) 2

(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (d)

$$\frac{(35 + 27)^2 + (35 - 27)^2}{(35)^2 + (27)^2}$$

$$= \frac{(35)^2 + (27)^2 + 2 \times 35 \times 27 + (35)^2 + (27)^2 - 2 \times 35 \times 27}{(35)^2 + (27)^2}$$

$$= \frac{2[(35)^2 + (27)^2]}{[(35)^2 + (27)^2]}$$

$$= 2$$

Hence, option (d) will be the right answer.

143. The population of a village is 16500. If it increases annually at the rate of 10%, then what will be its population after two years?

(a) 19965 (b) 19925

(c) 19800 (d) 19250

(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (a)

Present population = 16500

It increase by 10% annually then after two year-

$$= 16500 \times \frac{110}{100} \times \frac{110}{100}$$

$$= 165 \times 121$$

$$= 19965$$

Hence, option (a) will be the right answer.

144. Sarita purchases lemons at the rate of 2 lemons in 1 rupee and sells them at the rate of 5 lemons in 3 rupees. Her profit percentage is

(a) 12.5 (b) 18

(c) 15 (d) 20

(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (d)

Cost price of 2 lemons = 1 rupee

So 2 CP = 1 ... (i)

Now, selling price of 5 lemons = 3 rupees

So 5 SP = 3 ... (ii)

from equation (i) & (ii)

2 CP = 1 ... (i)

5 SP = 3 ... (ii)

multiply equation (i) by 3

6 CP = 3 ... (iii)

from (ii) & (iii)

5 SP = 6 CP ($\because$ RHS is same)

$$\frac{SP}{CP} = \frac{6}{5}$$

We know that

$$P\% = \frac{SP - CP}{CP} \times 100$$

Now

$$\frac{SP - CP}{CP} \times 100 = \frac{6 - 5}{5} \times 100 \Rightarrow 20\%$$

145. The parallel sides of a trapezium are 12 cm and 8 cm respectively and the distance between them is 7 cm. Its area is

- (a) 70 cm² (b) 90 cm²
 (c) 120 cm² (d) 140 cm²
 (e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (a)

Let ABCD is a trapezium-

According to the area formula of trapezium =

$$\text{Area} = \frac{1}{2} \times \text{distance} \times (\text{addition of parallel sides})$$

$$= \frac{1}{2} \times 7 \times (12 + 8)$$

$$= \frac{1}{2} \times 7 \times 20$$

$$\text{Area} = 70 \text{ cm}^2$$

146. A and B can finish a work in 11 and 17 days respectively. They worked together and received remuneration of ₹ 5,600 after finishing the work. Their shares in the remuneration are respectively

- (a) ₹ 3,600, ₹ 2,000 (b) ₹ 3,400, ₹ 2,200
 (c) ₹ 3,500, ₹ 2,100 (d) ₹ 3,300, ₹ 2,300
 (e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

A takes 11 days to finish work and B takes 17 days to complete the same work so-

Efficiency

A (11) ——— 17 ——— B

Total work (LCM of 11, 17) = 187

B (17) ——— 11 ——— A

As we know that total remuneration of 5600 will be distributed between them in the ratio of their efficiency.

Now,

(17 + 11) → 5600

28 → 5600

1	→	200
17	→	3400
11	→	2200

The share of A & B will be ₹ 3400 & ₹ 2200 respectively

147. The average weight of 60 students of a class is 50 kg. By including the class teacher, the average weight increases by 500 gm. The weight of the class teacher is

- (a) 79 kg (b) 80 kg
 (c) 80.5 kg (d) 81 kg
 (e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (c)

Average weight of 60 student = 50 kg

After including teachers weight the average weight of the class increased by 500 gm

Now the weight of the Class teacher

$$= 50 + 61 \times 0.5 \text{ kg}$$

$$= 50 + 61 \times \frac{5}{10}$$

$$= 50 + 30.5$$

weight of Class teacher = 80.5 kg

148. The missing number in the sequence 60, 180, 90, 270, ?, 405 is

- (a) 675 (b) 540
 (c) 360 (d) 135
 (e) None of the above / More than one of the above

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Ans. (d)

60 180 90 270 135 405

×3 ÷2 ×3 ÷2 ×3

Hence the missing number will be '135'

149. A clock is started at 12:00 noon. By 20 minutes past 04:00, the hour hand will turn through

- (a) 125° (b) 130°
 (c) 135° (d) 140°
 (e) None of the above / More than one of the above

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Ans. (b)

Time between 12 : 00 noon & 20 minute past 4 : 00 = 4 hours 20 min.

$$= \left(4 + \frac{1}{3}\right) \text{ hours} \Rightarrow \frac{13}{3} \text{ hours}$$

Hour hand angle in 12 hours = 360°

$$\frac{13}{3} \text{ hours} = \frac{360}{12} \times \frac{13}{3} \Rightarrow 130^\circ$$

150. Given that

$$113x + 97y = 517$$

$$97x + 113y = 533$$

then x and y are respectively

- (a) 3 and 2 (b) 2 and 3
(c) 3 and 4 (d) 4 and 3
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

The Given equation-

$$113x + 97y = 517 \quad \text{--- (i)}$$

$$97x + 113y = 533 \quad \text{--- (ii)}$$

multiply (i) by 113 & (ii) by 97, then we get-

$$12769x + 10961y = 58421 \quad \text{--- (iii)}$$

$$9409x + 10961y = 51701 \quad \text{--- (iv)}$$

substract (iii) & (iv)

$$3360x = 6720$$

$$x = \frac{6720}{3360} = 2$$

put the value of x in the equation (i)

$$113 \times 2 + 97y = 517$$

$$226 + 97y = 517$$

$$97y = 517 - 226$$

$$97y = 291$$

$$y = 3$$

Hence, the value of x & y will be 2 & 3 respectively.

151. The population of a town is 3528000. If it increases annually at the rate of 5%, then what will be its population after 3 years?

- (a) 4084102 (b) 3984102
(c) 4048102 (d) 4148102
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (e)

Given that the population of the town
= 3528000

population increases annually by 5% per annum so-

$$\text{Increased population} = 3528000 \times \frac{105}{100} \times \frac{105}{100} \times \frac{105}{100}$$

$$= 3528 \times \frac{21 \times 21 \times 21}{8}$$

$$= 21 \times 21 \times 21 \times 441$$

$$= 441 \times 441 \times 21$$

$$= 4084101$$

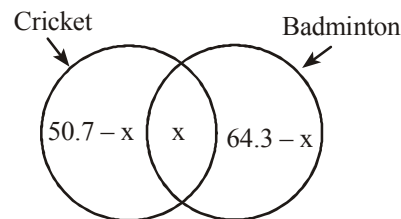
152. In a school, every student plays either cricket or badminton or both. If 50.7% play cricket, 64.3% play badminton and the total number of students is 600, how many students play both cricket and badminton?

- (c) 90 (d) 120
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (c)

Total student = 600



let x student play both the games

So,

$$50.7 - x + x + 64.3 - x = 100$$

$$115 - x = 100$$

$$x = 115 - 100$$

$$x = 15\%$$

$$\begin{aligned} \text{Total student who play both the games} &= 600 \times \frac{15}{100} \\ &= 90 \text{ students} \end{aligned}$$

153. The ratio of wine and water in a mixture of 35 litres is 4 : 1. How many litres of water should be added to it so that the ratio of wine and water becomes 2 : 1?

- (a) 5 litres (b) 6 liters
(c) 7 litres (d) 8 litres
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (c)

Total mixture of wine and water = 35 litres

The ratio of wine and water = 4 : 1

As we know that- water is being added to the mixture

So the quantity of wine will be same-

So-

Wine : water = 4 : 1 — Initial

$$\begin{array}{r|l} +1 & \text{addition of water} \\ \hline 2 : 1 & - \text{ Final} \\ 4 : 2 & \end{array}$$

1 Unit is being added, initial quantity of water is- 1 Unit

$$(4 + 1) \text{ Unit} = 35$$

$$1 \text{ Unit} = 7 \text{ litre}$$

Hence, 7 litre water is being added to the mixture.

154. Shashi loses 25% by selling oranges at the rate of ₹ 150 per dozen. At what rate should she sell them to get a profit of 20%?

- (c) ₹ 240 per dozen (d) ₹ 250 per dozen
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (c)

According to the question
75% of cost prize = 150 (since there is loss of 25%)
120% of cost price = $\left(\frac{150}{75} \times 120\right)$ Rs. (since profit is 20%
than 20% is profit) = 240 Rs.
Hence to make a profit of 20%, oranges should be sold at the rate of Rs. 240 per dozen

155. The lengths of the diagonals of a rhombus are 10 cm and 24 cm. Its perimeter is

- (a) 48 cm (b) 52 cm
(c) 56 cm (d) 60 cm
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (b)

We know that—
 $d_1^2 + d_2^2 = 4a^2$ (where a = side d_1, d_2 are diagonals of rhombus)
Side of rhombus = $\frac{1}{2} \times \sqrt{(\text{Diagonal}_1)^2 + (\text{Diagonal}_2)^2}$
 $= \frac{1}{2} \sqrt{(10)^2 + (24)^2} = \frac{1}{2} \sqrt{100 + 576}$
 $= \frac{1}{2} \sqrt{676} = \frac{1}{2} \times 26 = 13 \text{ cm}$
 $\therefore$ Perimeter of rhombus = $4 \times \text{side}$
 $= (4 \times 13) \text{ cm} = 52 \text{ cm}$

156. A train of length 140 m is running at a speed of 60 km/hr and a dog is running in the same direction parallel to the train at a speed of 18 km/hr. The train will cross the dog in

- (a) 10 seconds (b) 11 seconds
(c) 12 seconds (d) 121 seconds
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (c)

Relative speed = $(60 - 18) \text{ km/hr}$
 $= 42 \text{ km/hr}$
($\therefore$ both going in the same direction)
 $\therefore$ Time taken by Train to cross dog = $\frac{\text{Length of train}}{\text{Relative Speed}}$
 $= \frac{140 \text{ m}}{\left(42 \times \frac{5}{18}\right) \text{ m/s}}$

clock are together then $\theta = 0^\circ$

$$= \left(\frac{140 \times 18}{42 \times 5} \right)$$

$$= 12 \text{ second}$$

157. A clock is set right at 6 a.m. It gains 10 minutes in 24 hours. What will be the right time when the clock indicates 11 a.m. on the next day?

- (a) 10 a.m. (b) 48 minutes past 10
(c) 50 minutes past 10 (d) 54 minutes past 10
(e) None of the above/More than one of the above]

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (b)

24 hours = (24×60) minutes
Actual time of $(24 \times 60 + 10)$ minutes = (24×60) minutes
Actual time of 1450 minutes = 1440 minutes
Total time from 6 AM to 11 AM next day
 $= (24 \times 60)$ minutes + (5×60) minutes
 $= (1440 + 300)$ minutes = 1740 minutes
 $\therefore$ Actual time of 1450 minutes = 1440 minutes
 $\therefore$ Actual time of 1740 minutes = $\left(\frac{1440}{1450} \times 1740 \right)$ minutes
 $= 1728$ minutes
 $= (28 \times 60)$ minutes + 48 minutes
 $= 28 \text{ Hrs } 48 \text{ minutes}$
 $= 24 \text{ Hrs } + 4 \text{ Hrs } + 48 \text{ Minutes}$
On the next day at 11:00 am at the time when the hand of the clock shows the actual time will be = 6 AM + 4 Hrs + 48 minutes = 10:48 in the morning
Second Method
 $= (6 + 4)$ O'clock + 48 minutes
 $= 10:48 \text{ AM}$

158. At what time between 5 and 6 will the two hands of a watch coincide?

- (a) 26 minutes past 5
(b) 27 minutes past 5
(c) $27\frac{3}{11}$ minutes past 5
(d) 28 minutes past 5
(e) None of the above/More than one of the above

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Ans. (c)

The angle between the hour hand and the minute hand,
 $\theta = \left[30^\circ \text{H} - 11\frac{M}{2} \right]$
 $\text{H} = 5$ and $\theta = 0^\circ$

$$\therefore \theta = \left[30 \times 5 - \frac{11 \times M}{2} \right]$$

$$M = \frac{150 \times 2}{11} = \frac{300}{11}$$

$$M = 27 \frac{3}{11}$$

Therefore between 5 and 6 O'clock, both the clock hands will coincide at 5 O'clock 27.31 minutes.

159. Anil and Suman together can do a work in 12 days, in which, Anil alone can do in 20 days. If Suman alone has to do this work, she will take

- (a) 27 days (b) 28 days
(c) 29 days (d) 30 days
(e) None of the above/More than one of the above

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Ans. (d)

Let the total work = 60

Anil and Suman both can do in one day = $\frac{60}{12} = 5$ work

Work done by Anil in one day = $\frac{60}{20} = 3$

$\therefore$ Work done by Suman in one day = $(5 - 3)$ work = 2 work

$\therefore$ Time taken by Suman to complete Total work = $\frac{60}{2}$ Days
= 30 Days

160. In a class of 55 students, 34 like to play Cricket and 26 like to play Badminton. Also, each student likes to play at least one of the two games. How many students like to play both Cricket and Badminton?

- (a) 3 (b) 4
(c) 5 (d) 6
(e) None of the above/More than one of the above

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Ans. (c)

Number of students playing both the games
= $(34 + 26) - 55 = 60 - 55 = 5$

161. What are the two natural numbers whose product is 2400 and the sum of whose squares is 5200?

- (a) 120, 20 (b) 80, 30
(c) 75, 32 (d) 60, 40
(e) None of the above/More than one of the above

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Ans. (d)

Let both the numbers be x and y respectively.

According to question

$$x \times y = 2400$$

$$x^2 + y^2 = 5200$$

$$(x + y)^2 = x^2 + y^2 + 2xy$$

$$= 5200 + 2 \times 2400$$

$$= 5200 + 4800 = 10000$$

$$x + y = \sqrt{10000} = 100 \text{ ————— (i)}$$

$$\text{and } (x - y)^2 = x^2 + y^2 - 2xy$$

$$= 5200 - 4800$$

$$= 400$$

$$x - y = \sqrt{400} = 20 \text{ ————— (ii)}$$

From the equation (i) and (ii)

$$x = 60 \text{ and } y = 40$$

Hence both the numbers will be 60 and 40.

162. How many numbers between 100 and 500 are divisible by 4, 5 and 6?

- (a) 5 (b) 6
(c) 7 (d) 8
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (c)

$$\begin{array}{r|l} 2 & 4, 5, 6 \\ \hline 2 & 2, 5, 3 \\ \hline 3 & 1, 5, 3 \\ \hline 5 & 1, 5, 1 \\ \hline & 1, 1, 1 \end{array}$$

$$\therefore \text{LCM} = 2 \times 2 \times 3 \times 5 = 60$$

Now the number of multiples of 60 which lie between 100 and 500 = 120, 180, 240, 300, 360, 420 to 480. Hence these are the 7 numbers between 100 and 500 which are divisible by 4, 5 and 6.

163. In a mixture of 70 kg, the ratio of sand and cement is 4 :

1. How much sand should be added to the mixture so that the ratio of sand and cement in it becomes 6 : 1?

- (a) 24 kg (b) 28 kg
(c) 30 kg (d) 32 kg
(e) None of the above/More than one of the above

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Ans. (b)

Given that—

$$5 \text{ Unit} \rightarrow 70$$

$$\therefore 1 \text{ unit} = 14 \text{ kg.}$$

$$\therefore 6 \text{ unit} = 14 \times 6 = 84 \text{ kg.}$$

Hence the required amount of additional sand to the mixture

164. How many numbers between 200 and 600 are divisible by 4, 5 and 6?

- (a) 4 (b) 5
(c) 6 (d) 8
(e) None of the above/More than one of the above

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Ans. (c)

The number of such multiples of LCM of 4, 5 and 6, which lie between 200 and 600.

2	4, 5, 6
2	2, 5, 3
3	1, 5, 3
5	1, 5, 1
	1, 1, 1

$$\text{L.C.M.} = 2 \times 2 \times 3 \times 5 = 60$$

Now, such multiples of 60 which lie between 200 and 600. are 240, 300, 360, 420, 480 & 540

Thus total numbers are 6.

165. If n is any positive integer, then $(3^{4n} - 4^{3n})$ is always divisible by _____.

- (a) 7 (b) 17
(c) 112 (d) 145
(e) None of the above/More than one of the above

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Ans. (b)

Let, $n = 1, 2, 3, \dots$

If $n = 1$

$$\text{Then, } 3^{4n} - 4^{3n} = 3^4 - 4^3$$

$$\Rightarrow 81 - 64 = 17$$

If $n = 2$

$$\text{Then, } 3^8 - 4^6 = 3^4 \times 3^4 - 4^3 \times 4^3$$

$$= 81 \times 81 - 64 \times 64$$

$$= (81 + 64)(81 - 64)$$

$$= 145 \times 17$$

$\therefore$ If n is any positive integer then $3^{4n} - 4^{3n}$ is always divisible by 17.

166. The difference between the square of two numbers is 256000 and the sum of the numbers is 1000. The numbers are-

- (a) 600, 400
(b) 640, 360
(c) 628, 372
(d) 650, 350
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (c)

Let, two number are x and y respectively.

According to question-

$$x^2 - y^2 = 256000$$

$$(x + y)(x - y) = 256000 \quad \dots (i)$$

$$\text{and given that, } x + y = 1000 \quad \dots (ii)$$

Putting the value eqⁿ (ii) in eqⁿ (i)

$$1000 \times (x - y) = 256000$$

$$x - y = 256 \quad \dots (iii)$$

Adding eqⁿ (ii) and (iii)

$$2x = 1256$$

$$x = 628$$

Putting the value of x in eqⁿ (ii)

$$y = 1000 - 628$$

$$y = 372$$

So, the numbers are 628 and 372.

167. The value of $\log_{\sqrt{2}}(32)$ is.

- (a) $\frac{5}{2}$ (b) 5
(c) 10 (d) $\frac{1}{10}$
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (c)

$$\log_{\sqrt{2}}(32) = \log_{\sqrt{2}}(2)^5$$

$$= \log_{\sqrt{2}}(\sqrt{2} \cdot \sqrt{2})^5$$

$$= \log_{\sqrt{2}}(\sqrt{2})^{10}$$

$$= 10 \log_{\sqrt{2}}(\sqrt{2}) \quad (\because \log_x x = 1)$$

$$\text{Thus } \log_{\sqrt{2}}(32) = 10$$

168. By selling 45 lemons for Rs. 40, a man loses 20%. How many lemon should be sell for Rs. 24 to gain 20% in the transaction?

- (a) 16 (b) 18
(c) 20 (d) 22
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

Selling price of 45 lemons at loss of 20%. = Rs. 40

$$\text{Cost price of 45 lemons} = 40 \times \frac{100}{(100 - 20)}$$

$$= 40 \times \frac{100}{80} = \text{Rs. } 50$$

Selling price of 45 lemons at profit of 20%

$$= 50 \times \frac{120}{100} = \text{Rs. } 60$$

$$\text{No. of lemon for Rs. 1 at 20\% profit} = \frac{45}{60}$$

$$\text{No. of lemon for Rs. 24 at 20\% profit} = \frac{45}{60} \times 24 = 18$$

169. A 110m long train is travelling at a speed of 58 kmph, pass a passerby walking at 4 kmph in the same direction in-

- (a) 6 Seconds (b) $7\frac{1}{2}$ Seconds
(c) $7\frac{1}{3}$ Seconds (d) 8 Seconds
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (c)

$$\text{Relative speed of train} = (58 - 4) = 54 \text{ Km/hr.}$$

$$= \left(54 \times \frac{5}{18} \right) = 15 \text{ m/s}$$

$$\text{Length of train} = 110 \text{ m.}$$

The train will take time to cross passer by

$$= \frac{110}{15} \text{ s.}$$

$$= \frac{22}{3} \text{ s.} = 7\frac{1}{3} \text{ s.}$$

170. The perimeter of a rhombus is 52m and its shorter diagonal is 10m. The length of the longer diagonal is-

- (a) 12 m (b) 18 m
(c) 10 m (d) 24 m
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (d)

$$\text{Each sides of rhombus} = \frac{52}{4} = 13 \text{ m}$$

Let, the length of longer diagonal of rhombus is $2x$ m. Since half of both diagonal and one side form a right angled triangle, whose hypotenuse is the side of the rhombus.

So,

$$\left(\frac{10}{2} \right)^2 + \left(\frac{2x}{2} \right)^2 = (13)^2$$

$$(5)^2 + (x)^2 = 169$$

$$x^2 = 169 - 25$$

$$x = \sqrt{144} = 12 \text{ m}$$

So, the length of longer diagonal of rhombus

$$= (2 \times 12) \text{ m}$$

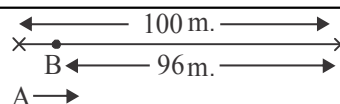
$$= 24 \text{ m}$$

171. In a 100m race, A runs at 8 kmph. If A gives B a start of 4m and still beats him by 15 seconds, what is the speed of B?

- (a) 5 kmph (b) 5.76 kmph
(c) 6 kmph (d) 6.34 kmph
(e) None of the above/More than one of the above

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Ans. (b)



$$\text{Speed of A} = \left(8 \times \frac{5}{18} \right) = \frac{20}{9} \text{ m/s.}$$

$$\text{Time taken by A to cover 100m} = \frac{100}{\frac{20}{9}} = 45 \text{ s.}$$

$$\text{Time taken by B to cover 96m} = (45 + 15) = 60 \text{ s.}$$

$$\text{Speed of B} = \frac{96 \text{ m}}{60 \text{ s}} = \left(\frac{96}{60} \times \frac{18}{5} \text{ km/hr} \right) = 5.76 \text{ km./hr}$$

172. A clock is set right at 8 a.m. The clock gains 10 minutes in 24 hours. What will be the true time when the clock indicates 1 p.m. on the following day?

- (a) 12 noon (b) 48 minutes past 12 noon
(c) 1 p m (d) 2 pm
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

From 8 am to 8 am next day, the clock will show 8:10 am.

$$24 \text{ hr} = (24 \times 60) \text{ minutes}$$

$$\text{Real time of } (24 \times 60 + 10) \text{ minute}$$

$$= (24 \times 60) \text{ minutes}$$

$$\text{Real time of 1450 minutes} = 1440 \text{ minutes}$$

$$\text{Real time of } (24 \times 60 + 5 \times 60) \text{ minutes}$$

$$= \left(\frac{1440}{1450} \times 1740 \right) \text{ minutes}$$

[$\therefore$ Total time from 8 am to 1 pm the next day.

$$= (24 \times 60) \text{ min.} + (5 \times 60) \text{ min.} = 1740]$$

$$= \frac{144}{145} \times 1740$$

$$= 144 \times 12 = 1728 \text{ min.}$$

$$= 28 \text{ hr.} + 48 \text{ min.}$$

$$= 24 \text{ hr.} + 4 \text{ hr.} + 48 \text{ min.}$$

So, on next day, at 1 o'clock in the afternoon, when the clock shows the hand, actual time will be.

$$= 8 \text{ am} + (4 \text{ hr} + 48 \text{ min})$$

173. The value of continued fraction.

$$2 - \frac{1}{2 - \frac{1}{2 - \frac{1}{\dots}}} \text{ is--}$$

- (a) 1 (b) -1
(c) 2 (d) 0
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (a)

$$2 - \frac{1}{2 - \frac{1}{2 - \frac{1}{\dots}}}$$

Let,

$$x = 2 - \frac{1}{2 - \frac{1}{2 - \frac{1}{\dots}}}$$

Now, $x = 2 - \frac{1}{x}$

$$x = \frac{2x-1}{x}$$

$$x^2 - 2x + 1 = 0$$

$$x^2 - x - x + 1 = 0$$

$$x(x-1) - 1(x-1) = 0$$

$$(x-1)^2 = 0$$

$$x = 1$$

So, the value of the continued fraction is 1.

174. Arrange the following in a logical sequence:

1. Patient 2. Diagnosis
3. Bill 4. Doctor
5. Treatment

- (a) 1, 4, 2, 3, 5 (b) 1, 4, 3, 2, 5
(c) 1, 4, 2, 5, 3 (d) 4, 1, 2, 3, 5
(e) None of the above / More than one of the above

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Ans. (c)

The logical sequence of the Given Word-

Patient → Doctor → Diagnosis → Treatment → Bill

1 4 2 5 3

Hence, The correct sequence is '14253' which is given in the option (c)

175. In a class, 72 students drink only tea, 50% of the students drink coffee, 25% of the students drink both tea and coffee and 5% of the students drink neither tea nor coffee. How many of the students drink only tea or only coffee?

- (a) 113 (b) 112

- (c) 114 (d) 128
(e) None of the above / More than one of the above

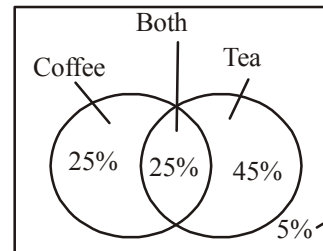
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Ans. (b)

Given that-

The Student, who drink only tea = 72

Now-



from the venn diagram-

$$45\% = 72$$

$$5\% = 8$$

one who drink only tea-

$$45\% = 72$$

one who drink only coffee-

$$25\% = 40$$

$$\text{Total student who drink either tea or coffee} = 72 + 40 = 112$$

176. Arrange the words given below in a meaningful sequence:

1. Rain 2. Monsoon
3. Rescue 4. Flood
5. Shelter 6. Relief

- (a) 1, 2, 3, 4, 5, 6 (b) 1, 2, 4, 5, 3, 6
(c) 2, 1, 4, 3, 5, 6 (d) 4, 1, 2, 3, 5, 6
(e) None of the above / More than one of the above

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Ans. (c)

Meaningful sequence of the Given Words-

Monsoon → Rain → Flood → Rescue → Shelter → Relief

2 1 4 3 5 6

Hence, the Correct sequence is "214356" which is given in the option (c).

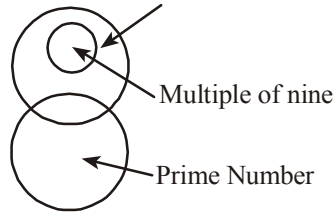
177. Given figure shows exactly correct relation in which of the following?



- (a) Europe, Asia, Africa
(b) Odd numbers, prime numbers, odd multiples of nine
(c) Blue, red, green
(d) Players, students, actors
(e) None of the above / More than one of the above

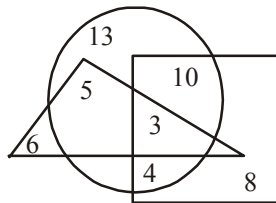
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The Given venn diagram
Odd Number



Hence, option (b) will be the right answer of the given question.

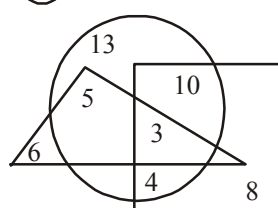
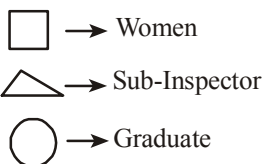
178. In the following diagram, square represents women, triangle represents sub-inspectors of police and circle represents graduates. Which numbered area represents women graduates sub-inspector of police?



- (a) 13 (b) 8
(c) 5 (d) 3
(e) None of the above / More than one of the above

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Ans. (d)



The area number, which represents women graduates sub-inspector of police = 3

179. When the reports of the annual profits of five companies L, M, N, O and P were submitted, it was found that no two companies had the same profit. M got more profit than O but less than P, whereas L got more profit than N but less than P. Which of the following companies got the highest profit?

- (a) P (b) L
(c) M (d) O
(e) None of the above / More than one of the above

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Ans. (a)

Let us analyze the given information
M got more profit than O but less than P
 $P > M > O$

L got more profit than N but less than P
 $P > L > N$

Here, we can say that, P got the highest profit
Therefore, option (a) will be the right answer.

180. Given below are two statements: "All men are vertebrates." "Some mammals are vertebrates."

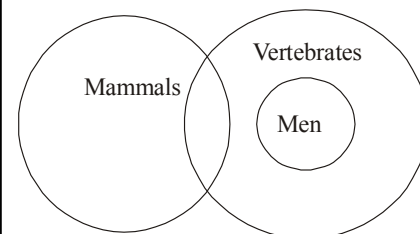
Which one of the following is correct?

- (a) All men are mammals.
(b) All mammals are men.
(c) Some vertebrates are mammals.
(d) All vertebrates are men.
(e) None of the above / More than one of the above

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Ans. (c)

Representing the Given statement through venn diagram-
Through Option-



- (a) All men are mammals. (×)
(b) All mammals are men. (×)
(c) Some vertebrates are mammals. (✓)
(d) All vertebrates are men. (×)
Hence, Option (c) will be the right answer.

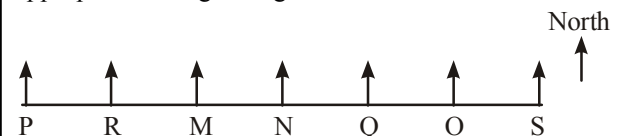
181. M, N, O, P, Q, R and S are seven persons sitting in a row for a photo session. O sits to the immediate left of S and R is to the immediate right of P. N has equal number of persons on either side of him in a row. M is to the immediate left of N and there are exactly three persons between M and S. How many persons are there between R and Q?

- (a) 2 (b) 3
(c) 4 (d) 1
(e) None of the above / More than one of the above

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Ans. (a)

From the given instruction in the question, we will draw a appropriate sitting arrangement-



Now- we can clearly see that, there are two person (M, N) sitting between R and Q
Hence, Option (a) will be the right answer of the given question.

182. If A+B means A is the son of B; A-B means A is the daughter of B; A×B means A is the father of B; A÷B means A is the mother of B, then which of the following means S is the son-in-law of P?

- (a) $P+Q÷R×S-T$ (b) $P×Q÷R-S+T$
(c) $P+Q×R-S÷T$ (d) $P×Q-R÷S×T$
(e) None of the above / More than one of the above

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Ans. (b)

The Given code-

$A+B \Rightarrow A$ is son of B

$A-B \Rightarrow A$ is daughter of B

$A \times B \Rightarrow A$ is father of B

$A \div B \Rightarrow A$ is mother of B

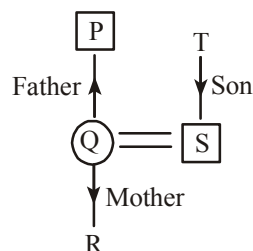
Now - We are required to show that S is the son-in-Law of P from the given option-

Now, from the option (b)-

$\square \rightarrow$ Male

$\bigcirc \rightarrow$ Female

$P \times Q \div R - S + T$



Hence, option (b) will be the right answer-

183. The number of the digits in the number 97415628, which remain unchanged while the digits are written in descending order, is

- (a) 1 (b) 2
(c) 3 (d) 4
(e) None of the above / More than one of the above

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Ans. (c)

The Number

9	7	4	1	5	6	2	8
9	8	7	6	5	4	2	1

right answer.

Hence, there are 3 number's which position does not change when we write the digits of the number in descending order- Therefore, option (c) will be the right Answer.

184. If you write down all the numbers from 1 to 100, then how many times do you write 3?

- (a) 11 (b) 18
(c) 20 (d) 21
(e) None of the above / More than one of the above

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Ans. (c)

If we write down all the number from 1 to 100 then the number of times we write 3 is 20.

3, 13, 23, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 43, 53, 63, 73, 83, 93. Hence '20' is correct answer.

185. The significant figure in the number 0.00386 is

- (a) 6 (b) 5
(c) 4 (d) 3
(e) None of the above / More than one of the above

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Ans. (d)

Leading zeroes to the Left of the First non-zero digit are not significant.

All non-zero digits are significant-

According to the rule, the given number 0.00386 has three significant Numbers (3, 8, 6)

Hence, option (d) will be the right answer

186. In a row of persons, the position of Kirti from the left side of the row is 27th and position of Kirti from the right side of the row is 34th, then the total number of persons in the row is

- (a) 60 (b) 61
(c) 62 (d) 59
(e) None of the above / More than one of the above

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Ans. (a)

Kirti's Position from Left (L) = 27th

Kirti's Position from Right (R) = 34th

We know that-

Total person in Row = Left + Right - 1

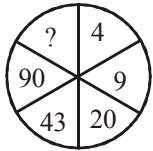
$T = 27 + 34 - 1$

$T = 61 - 1$

$T = 60$

There are 60 person in the Row-

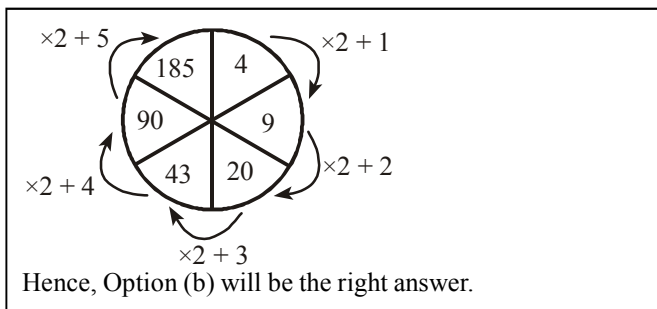
187. In the following figure, the missing number is



- (a) 126 (b) 185
(c) 239 (d) 145
(e) None of the above / More than one of the above

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Ans. (b)



188. If the number 12.464762 is rounded off to four places of decimal, then the correct number is

- (a) 12.4648 (b) 12.4647
(c) 12.4640 (d) 12.4649
(e) None of the above / More than one of the above

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Ans. (a)

To round off the number 12.464762 to four decimal places-
Then the round off number will be 12.4648
There fore, option (a) will be the right answer.

189. If A = 26, SUN = 27, then CAT = ?

- (a) 24 (b) 27
(c) 57 (d) 58
(e) None of the above / More than one of the above

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Ans. (c)

Letter	Positional value	Opposite
A →	1	→ 26
Similarly		
S → 19 →	8	C → 3 → 24
U → 21 →	6	A → 1 → 26
N → 14 →	13	T → 20 → 7
		27
		57
Now		
CAT = 57		

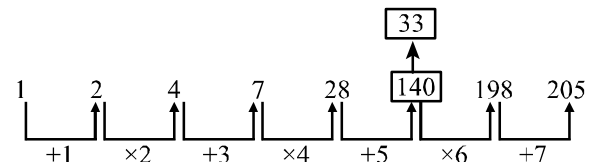
190. In the series 1, 2, 4, 7, 28, 140, 198, 205, a wrong number is

- (a) 198 (b) 4
(c) 205 (d) 140
(e) None of the above / More than one of the above

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Ans. (d)

The Given series-



In the Given series, 140 is wrong number, there should be 33 so we can make a meaningful number series,
Hence, Option (d) will be the right answer-

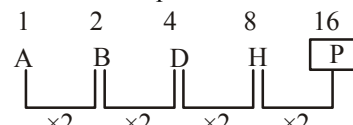
191. The next letter in the series, A, B, D, H will be

- (a) L (b) N
(c) R (d) P
(e) None of the above / More than one of the above

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Ans. (d)

The Given Alphabet series-



Hence, Option (d) will be the right answer-

192. The missing number in

27 : 51 :: 83 : ? is

- (a) 102 (b) 117
(c) 123 (d) 138
(e) None of the above / More than one of the above

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Ans. (c)

27 : 51 :: 83 : ?

Just as-

$$27 \rightarrow 5^2 + 2$$

$$51 \rightarrow 7^2 + 2$$

Similarly

$$83 \rightarrow 9^2 + 2$$

$$123 \rightarrow 11^2 + 2$$

Hence, Option (c) will be the right answer

193. Choose the odd one among the following.

- (a) 17 (b) 27
(c) 37 (d) 47
(e) None of the above / More than one of the above

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In the given option of the question- 27 is the odd one because except 27 all are Prime Numbers, Hence, Option (b) will be the right answer.

194. In a certain code language, if the word RECTANGLE is coded as TGEVCPING, then how will be the word RHOMBUS coded in that language?

- (a) TJQODWU (b) TJQNDWU
(c) TJOQDWV (d) TJQOEWU
(e) None of the above / More than one of the above

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Ans. (a)

Just as-	Similarly-
R $\xrightarrow{+2}$ T	R $\xrightarrow{+2}$ T
E $\xrightarrow{+2}$ G	H $\xrightarrow{+2}$ J
C $\xrightarrow{+2}$ E	O $\xrightarrow{+2}$ Q
T $\xrightarrow{+2}$ V	M $\xrightarrow{+2}$ O
A $\xrightarrow{+2}$ C	B $\xrightarrow{+2}$ D
N $\xrightarrow{+2}$ P	U $\xrightarrow{+2}$ W
G $\xrightarrow{+2}$ I	S $\xrightarrow{+2}$ U
L $\xrightarrow{+2}$ N	
E $\xrightarrow{+2}$ G	

Hence Option (a) will be the right answer.

195. In a certain code language, if CRICKET is coded as 3923564, ROCKET is coded as 913564 and KETTLE is coded as 564406, then in that language LITTLE is coded as

- (a) 244060 (b) 024406
(c) 020446 (d) 200446
(e) None of the above / More than one of the above

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Ans. (b)

Here, we compare the word's code with the given unknown word "LITTLE"

C — 3	R — 9	K — 5	L — 0
R — 9	O — 1	E — 6	I — 2
I — 2	C — 3	T — 4	T — 4
C — 3	K — 5	T — 4	T — 4
K — 5	E — 6	L — 0	L — 0
E — 6	T — 4	E — 6	E — 6
T — 4			

Hence, Option (b) will be the right answer.

196. Choose the odd one among the following.

- (a) SORE (b) SOTLU
(c) NORGAE (d) MEJNIAS
(e) None of the above / More than one of the above

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Ans. (c)

From the Given Option-

SORE $\rightarrow$ ROSE

SOTLU $\rightarrow$ LOTUS

MEJNIAS $\rightarrow$ JASMINE

But the word given in the option (c) i.e. NORGAE cannot make a meaningful word of flower

Hence, option (c) will be the right answer.

197. Complete the following series :

$6 + \sqrt{216}, 7 + \sqrt{343}, 8 + \sqrt{512}, 9 + \sqrt{729}, \dots$

- (a) $10 + \sqrt{100}$ (b) $10 + \sqrt{1000}$
(c) $100 + \sqrt{1000}$ (d) $100 + \sqrt{10}$
(e) None of the above / More than one of the above

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Ans. (b)

The given series,

$6 + \sqrt{216}, 7 + \sqrt{343}, 8 + \sqrt{512}, 9 + \sqrt{729}, ?$

Now,

$6 + \sqrt{6^3}, 7 + \sqrt{7^3}, 8 + \sqrt{8^3}, 9 + \sqrt{9^3}, 10 + \sqrt{10^3}$

$6 + \sqrt{216}, 7 + \sqrt{343}, 8 + \sqrt{512}, 9 + \sqrt{729}, 10 + \sqrt{1000}$

Hence, Option (b) will be the right answer.

198. If 1st January, 1992 is Tuesday, then on which day of the week will be 1st January, 1993 fall?

- (a) Thursday (b) Wednesday
(c) Friday (d) Saturday
(e) None of the above / More than one of the above

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Ans. (a)

1 January, 1992

Since, 1992 is perfectly divided by 4, so it is a leap year that has 366 days

Now-

$\frac{366}{7} = \text{remainder} - 2$

Thus, the day on 1st January 1993 is— Tuesday +2
= Thursday

199. When the clock shows 3 hours 14 minutes, what is the angle between the hands of the clock?

- (a) 10° (b) 12°
 (c) 13° (d) 16°
 (e) None of the above / More than one of the above

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Ans. (c)

Clock shows 3 hours 14 minutes-
 then angle between hour hand and minute hand

$$\theta = 30h - \frac{11}{2}m$$

$$\theta = 30 \times 3 - \frac{11}{2} \times 14$$

$$\theta = 90 - 11 \times 7$$

$$\theta = 90^\circ - 77^\circ$$

$$\theta = 13^\circ$$

Hence, Option (c) will be the right answer

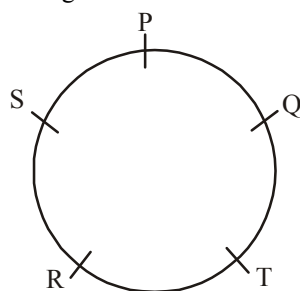
200. P, Q, R, S and T sit around a table. P sits two seats to the left of R and Q sits two seats to the right of R. If S is not sitting next to Q, who is sitting between Q and S?

- (a) P (b) R
 (c) T (d) Both R and P
 (e) None of the above / More than one of the above

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Ans. (a)

According to the question the appropriate sitting arrangement-



Hence, we can see that P is sitting between Q & S.

201. P, Q, R, S and T are five speakers who have to speak on a particular day, not necessarily in the same order. R is neither the first nor the last speaker. There are three speakers after S and three speakers ahead of T. If P speaks after Q, then who is the last speaker to speak?

- (a) S (b) T
 (c) P (d) R
 (e) None of the above / More than one of the above

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Ans. (c)

According to the question the given arrangement are as following-

1	Q
2	S
3	R
4	T
5	P

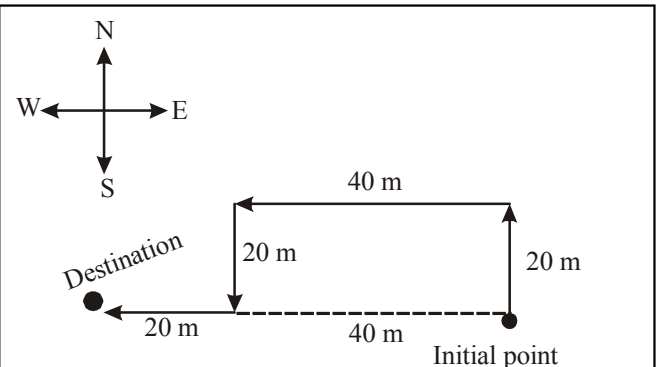
Here, we can see that, the last speaker is 'P'
 Hence, Option (c) will be the right answer.

202. Vinod walks 20 metres towards north. He then turns left and walks 40 metres. He again turns left and walks 20 metres. Further, he moves 20 metres after turning to the right. How far is he from his original position?

- (a) 20 meters (b) 60 meters
 (c) 30 meters (d) 50 meters
 (e) None of the above / More than one of the above

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Ans. (b)



Hence, we can see that, the distance between Destination and Initial point is 60 meter, which is given in the option (b)

203. The series

$$1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots$$

- (a) geometric series
 (b) harmonic series
 (c) exponential series
 (d) p-series
 (e) None of the above / More than one of the above

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Ans. (c)

The Given Series-

$$1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots$$

This series is an exponential series

This series is an expansion of e^x

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

Hence, option (c) will be the right answer.

204. If $\cos \theta = \sin \theta$, then the value of θ is

- (a) π (b) $\frac{\pi}{12}$
 (c) $\frac{\pi}{4}$ (d) $\frac{2\pi}{3}$
 (e) None of the above / More than one of the above

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Ans. (c)

$$\begin{aligned}\cos \theta &= \sin \theta \\ \cos \theta &= \cos(90^\circ - \theta) \\ \theta &= 90^\circ - \theta \\ 2\theta &= 90^\circ \text{ or } \frac{\pi}{2}\end{aligned}$$

$$\theta = \frac{\pi}{4} \text{ or } 45^\circ$$

Hence, option (c) will be the right answer-

205. If $\frac{{}^nC_2}{{}^{n-1}C_2} = \frac{2}{1}, n \geq 1$ then the value of n is

- (a) 2
 (b) 4
 (c) 1
 (d) 3
 (e) None of the above / More than one of the above

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Ans. (b)

$$\frac{{}^nC_2}{{}^{n-1}C_2} = \frac{2}{1}$$

$$\frac{\frac{n!}{2!(n-2)!}}{\frac{(n-1)!}{2!(n-1-2)!}} = \frac{2}{1} \quad \left[\because {}^nC_r = \frac{n!}{r!(n-r)!} \right]$$

$$\frac{\frac{n(n-1)!}{2!(n-2)!}}{\frac{(n-1)!}{2!(n-3)!}} = \frac{2}{1}$$

$$\frac{n(n-3)!}{(n-2)!} = \frac{2}{1}$$

$$\frac{n(n-3)!}{(n-2)(n-3)!} = \frac{2}{1}$$

$$n = 2(n-2)$$

$$n = 2n - 4$$

$$n = 4$$

Hence, option (b) will be the right answer.

206. $\frac{5}{3}x^3 + \frac{5}{3}x - \frac{10}{3} = 0$, then one of the value of x is

- (a) $\frac{5}{3}$ (b) $\frac{10}{3}$
 (c) $\frac{3}{5}$ (d) 1
 (e) None of the above / More than one of the above

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Ans. (d)

$$\frac{5}{3}x^3 + \frac{5}{3}x - \frac{10}{3} = 0$$

$$5x^3 + 5x - 10 = 0$$

$$x^3 + x - 2 = 0$$

Let putting the value of $x = 1$

$$2 - 2 = 0$$

$$0 = 0$$

Therefore-

$$x^2(x-1) + x(x-1) + 2(x-1) = 0$$

$$(x-1)(x^2 + x + 2) = 0$$

Here, one of the value of x is 1

Hence, option (d) will be the right answer.

207. If $f: N \rightarrow N$ is defined by $f(x) = 3x + 7$ and $g: N \rightarrow N$ is defined by $g(x) = 5x + 1$, then $(g \circ f)(1)$ is

- (a) 25 (b) 51
 (c) 57 (d) 26
 (e) None of the above / More than one of the above

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Ans. (b)

To find $(g \circ f)(1)$, you need to substitute the value of $f(1)$ into $g(x)$

Now,

$$f(x) = 3x + 7$$

$$f(1) = 10$$

Now, substitute this value into $g(x)$:

$$g(x) = 5x + 1$$

$$g[f(1)] = g(10) = 5 \times 10 + 1$$

$$(g \circ f)(1) = 51$$

208. If $\log\left(\frac{a}{b}\right) + \log\left(\frac{b}{a}\right) = \log(a + b)$ then,

- (a) $a + b = 0$ (b) $a + b = 1$
 (c) $a = b$ (d) $ab = 0$
 (e) None of the above / More than one of the above

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Ans. (b)

$$\log\left(\frac{a}{b}\right) + \log\left(\frac{b}{a}\right) = \log(a + b)$$

$$\log\left(\frac{a}{b} \times \frac{b}{a}\right) = \log(a + b) \quad [\because \log m + \log n = \log mn]$$

$$\log 1 = \log(a + b)$$

$$a + b = 1$$

Hence, option (b) will be the right answer.

209. Find the number which when multiplied by 15 is increased by 196.

- (a) 12 (b) 14
 (c) 17 (d) 19
 (e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (b)

Let the number is 'x'

According to the question

$$x \times 15 = x + 196$$

$$15x - x = 196$$

$$14x = 196$$

$$x = 14$$

Hence option (b) will be the right answer.

210. The solution of the equations

$$2x - 3y = 23$$

$$4x + 3y = 43$$

is

(a) $x = 11; y = -\frac{1}{3}$

(b) $x = 23; y = 43$

(c) $x = 43; y = 23$

(d) $x = 11; y = 11$

(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (a)

$$2x - 3y = 23 \quad \text{-----(i)}$$

$$4x + 3y = 43 \quad \text{-----(ii)}$$

Adding the equation (i) and (ii)

$$6x = 66$$

$$x = 11$$

Putting the value of x in the equation (i)

$$2 \times 11 - 3y = 23$$

$$22 - 3y = 23$$

$$3y = -1$$

$$y = -\frac{1}{3}$$

Hence, option (a) will be the right answer.

211. The ratio of two number is 3 : 4 and their HCF is 4.

Their LCM is

- (a) 16 (b) 12
 (c) 24 (d) 48
 (e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (d)

The ratio of Two numbers are 3 : 4

Let the number-

3x and 4x

$$\text{HCF}(x) = 4$$

Now

$$\text{LCM} = \text{HCF} \times 3 \times 4$$

$$= 4 \times 3 \times 4$$

$$\text{LCM} = 48$$

Hence, LCM will be 48

212. The sum of the first 8 prime numbers is

- (a) 75 (b) 77
 (c) 60 (d) 58
 (e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (b)

First 8 prime number are- 2, 3, 5, 7, 11, 13, 17, 19

According to the question-

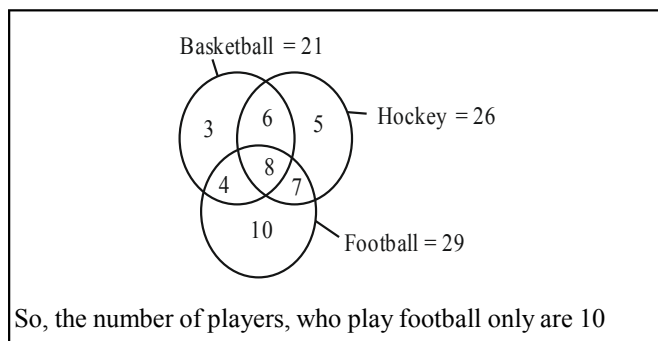
$$2 + 3 + 5 + 7 + 11 + 13 + 17 + 19 = 77$$

213. In a group of athletics teams in a school, 21 are in the basketball team, 26 in the hockey team and 29 in the football team. If 14 play hockey and basketball 12 play football and basketball, 15 play hockey and football, 8 play all the three games, then how many play football only?

- (a) 10 (b) 29
(c) 21 (d) 18
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)



214. Mohan can do a bit of work in 25 days which can be completed by Sohan in 20 days. Both together labour for 5 days and afterward Mohan leaves off. How long will Sohan take to complete the remaining work?

- (a) 20 days (b) 11 days
(c) 14 days (d) 21 days
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans.-(b)

Let, total work = 100 (LCM of 25 and 20)

Mohan, (25) > Total Mark < 4
Sohan, (20) > 100 < 5

5 days work of Mohan and Sohan = $(4 + 5) \times 5$
= 45

Remaining work = $100 - 45 = 55$

Time taken by Sohan to complete remaining work = $\frac{55}{5}$
= 11 days

215. A clock is started at 12:00 noon. By 10 minutes past 5:00, the hour hand has turned through.

- (a) 135° (b) 145°
(c) 155° (d) 165°
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

∴ Angle formed by hour hand in 1 min = $\frac{1}{2}^\circ$
Angle formed by hour hand in 5 hr 10 min or 310 min is
 $= 310 \times \frac{1}{2}^\circ = 155^\circ$
So, by 10 minutes past 5:00, the hour hand turned through 155°

216. Which one of the following cannot be the square of a natural numbers?

- (a) 26569 (b) 143642
(c) 30976 (d) 28561
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

The unit digit of square of any natural number can never be 2, 3, 7 and 8, So the number given in option (b) 143642 cannot be the square number.

217. Given that-

$$217x + 131y = 913$$

$$131x + 217y = 827$$

Then x and y are respectively

- (a) 5 and 7 (b) 3 and 2
(c) -5 and -7 (d) 2 and 5
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

$217x + 131y = 913$ (i)
 $131x + 217y = 827$ (ii)
Add eqⁿ (ii) and eqⁿ (i)
 $348x + 348y = 1740$
 $x + y = 5$ (iii)
Sub. eqⁿ (ii) from eqⁿ (i)
 $86x - 86y = 86$
 $x - y = 1$ (iv)
Add eqⁿ (iii) and (iv)
 $2x = 6$
 $x = 3$
Put the value of x in eqⁿ (iv)
 $3 - y = 1$
 $y = 2$
So the value of x and y are 3 and 2 respectively.

218. $\frac{(598 + 479)^2 - (598 - 479)^2}{598 \times 479} = ?$

- (a) 4 (b) 10
(c) 132 (d) 8
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

$$\frac{(598 + 479)^2 - (598 - 479)^2}{598 \times 479}$$

$$= \frac{4 \times 598 \times 479}{598 \times 479} \quad \{ \because (a + b)^2 - (a - b)^2 = 4ab \}$$

$$= 4$$

219. The population of a town is 176400. If it increases annually at the rate of 5%, then what will be its population after two years?

- (a) 194481 (b) 296841
(c) 394481 (d) 396841
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

$\therefore$ Population of a town after two years

$$= 176400 \times \frac{105}{100} \times \frac{105}{100}$$

$$= 194481$$

220. The missing number in the sequence- 4, 18, 48, 100, ?, 294, 448 is.

- (a) 94 (b) 164
(c) 180 (d) 192
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

The sequence is—

$$\begin{array}{ccccccc} 4 & 18 & 48 & 100 & \boxed{180} & 294 & 448 \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ (2)^3 - (2)^2 & (3)^3 - (3)^2 & (4)^3 - (4)^2 & (5)^3 - (5)^2 & (6)^3 - (6)^2 & (7)^3 - (7)^2 & (8)^3 - (8)^2 \end{array}$$

$\therefore ? = 180$

221. If ${}^{2n}C_3 : {}^nC_2 = 44 : 3$, then the value of n is—

- (a) 1 (b) 6
(c) 11 (d) 4
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

$$\frac{{}^{2n}C_3}{{}^nC_2} = \frac{44}{3}$$

$$\frac{\frac{2n!}{3!(2n-3)!}}{\frac{n!}{2!(n-2)!}} = \frac{44}{3}$$

$$\frac{2n \times (2n-1)(2n-2)}{3 \times 2 \times 1} = \frac{44}{3}$$

$$\frac{2n \times (2n-1)(2n-2)}{2 \times 1} = \frac{44}{3}$$

(b) 2 : 1

$$\frac{2n \times 2 \times (2n-1)(n-1)}{3n(n-1)} = \frac{44}{3}$$

$$\frac{4(2n-1)}{3} = \frac{44}{3}$$

$$2n-1 = 11$$

$$2n = 12$$

$$\therefore n = 6$$

222. If the average of m numbers is n^2 and that of n numbers is m^2 , then the average of $m+n$ numbers is—

- (a) $\frac{n}{m}$ (b) $\frac{m}{n}$
(c) $m n$ (d) $m - n$
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

$$\text{Average} = \frac{mn^2 + nm^2}{(m+n)}$$

$$= \frac{mn(n+m)}{(m+n)}$$

$$= m n$$

223. The number of ways in which 12 identical pens can be distributed between two students if each student is to get at least two pens, is—

- (a) 8 (b) 9
(c) 10 (d) 11
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

$\therefore$ Each students is to get at least two pens, if

Pen received by first student	Pen received by second student
2	10
3	9
4	8
5	7
6	6
7	5
8	4
9	3
10	2

So, total ways = 9

224. If the radii of circles A and B are in the ratio of 1.5:1, then the area of the circles A and B will be in the ratio of—

- (c) 2.25 : 1
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

Let the radius of circles A and B are $1.5x$ and x respectively.

Area of circle = $\pi \times (\text{radius})^2$

$\therefore$ Area of circle A : Area of circle B

$$= \pi (1.5x)^2 : \pi (x)^2$$

$$= 2.25x^2 : x^2$$

$$= 2.25 : 1$$

Note - If the ratio of radius or circumference of two circles are $x : y$ then the ratio of their areas will be $x^2 : y^2$

Ratio of area of circle A and B = $(1.5)^2 : (1)^2 = 2.25 : 1 = 9 : 4$

225. One tap can fill a water tank in 3 hours and another tap can empty it in 4 hours. If the tank is one third full and both the taps are opened together, then the time taken by taps to fill the tank will be-

- (a) 8 hours (b) 9 hours
(c) 10 hours (d) 11 hours
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (a)

Let the capacity of the tank is 12 liters.

$$1/3 \text{ capacity of tank} = 12 \times \frac{1}{3} = 4 \text{ l.}$$

Remaining capacity of tank = $12 - 4 = 8 \text{ l.}$

$$\text{Tank filled by first tap in 1 hr.} = \frac{12}{3} = 4 \text{ l.}$$

$$\text{and Second tap will empty in 1 hr.} = \frac{12}{4} = 3 \text{ l.}$$

$$\text{Both tap will fill the tank in 1 hr.} = 4 - 3 = 1 \text{ l.}$$

$$\text{Time taken by taps to fill remaining 8 ltr.} = \frac{8}{1} = 8 \text{ hr.}$$

226. The next term in the sequence 1, 3, 9, 15, 25, 35, 49,will be?

- (a) 80 (b) 64
(c) 81 (d) 63
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

The given number series is as follows -

$$\begin{array}{ccccccc} 1 & 3 & 9 & 15 & 25 & 35 & 49 & ? \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1^2 & (2)^2-1 & 3^2 & (4)^2-1 & 5^2 & (6)^2-1 & 7^2 & (8)^2-1 \end{array}$$

$$\begin{aligned} \text{Next term} &= (8)^2 - 1 \\ &= 64 - 1 \end{aligned}$$

227. If $x^2 - y^2 = 7$ and $x - y = 1$, then the length of a diagonal of a rectangle with length and width respectively x cm and y cm will be -

- (a) 5 cm (b) 6 cm
(c) 7 cm (d) 8 cm
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (a)

Given that,

$$x^2 - y^2 = 7 \text{ and}$$

$$x - y = 1 \quad \dots\dots\dots(i)$$

$$(x - y)(x + y) = 7$$

$$1(x + y) = 7$$

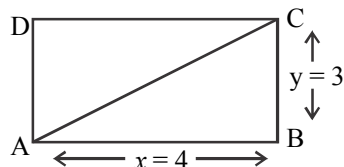
$$x + y = 7 \quad \dots\dots\dots(ii)$$

On Adding eqⁿ (i) and (ii)

$$2x = 8$$

$$x = 4$$

and $y = 4 - 1 = 3$ [by putting the value of x in eqⁿ (i)]



$$\begin{aligned} \text{Length of diagonal of rectangle ABCD} &= \sqrt{(AB)^2 + (BC)^2} \\ &= \sqrt{(4)^2 + (3)^2} \\ &= \sqrt{16 + 9} = \sqrt{25} = 5 \text{ cm} \end{aligned}$$

228. In a class of 80 students, 60% students play carrom, 45% play chess and 10% student neither play carrom nor chess. What is the number of student who play chess only?

- (a) 36 (b) 24
(c) 12 (d) 8
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

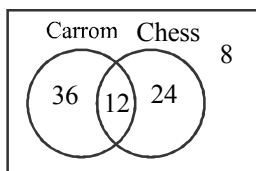
Ans. (b)

$$\text{No. of students play carrom in class} = 80 \times \frac{60}{100} = 48$$

$$\begin{aligned} \text{and no. of students play chess in class} &= 80 \times \frac{45}{100} \\ &= 36 \end{aligned}$$

$$\text{Students who neither play carrom nor chess} = 80 \times \frac{10}{100} = 8$$

$$\begin{aligned} \text{No. of such students play atleast one game.} \\ &= 80 - 8 = 72 \end{aligned}$$



No. of students who play both game = $(48 + 36) - 72$
 $= 84 - 72 = 12$

No. of students who play only chess = $36 - 12 = 24$

229. If $x^3 - y^3 = \frac{117}{8}$ and $x - y = \frac{3}{2}$, then the value of $x^2 + xy + y^2$ will be-

- (a) 1 (b) $\frac{5}{8}$
 (c) $\frac{39}{4}$ (d) $\frac{39}{8}$

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

Given that,

$$x^3 - y^3 = \frac{117}{8} \text{ and } x - y = \frac{3}{2}$$

$$x^3 - y^3 = (x - y)(x^2 + xy + y^2) \quad (\text{Formula})$$

$$\frac{117}{8} = \frac{3}{2}(x^2 + xy + y^2)$$

$$\therefore (x^2 + xy + y^2) = \frac{117 \times 2}{8 \times 3} \Rightarrow \frac{39}{4}$$

230. If $(5)^{x^2+2x+7} = (125)^{2x+1}$ then the value of x is-

- (a) 5 (b) 4
 (c) 3 (d) 2

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

$$(5)^{x^2+2x+7} = (125)^{2x+1}$$

$$(5)^{x^2+2x+7} = (5^3)^{2x+1} \Rightarrow 5^{(x^2+2x+7)} = (5)^{6x+3}$$

Bases of both sides are equal, so the powers will also equal.

$$x^2 + 2x + 7 = 6x + 3$$

$$x^2 + 2x - 6x + 7 - 3 = 0$$

$$x^2 - 4x + 4 = 0$$

$$(x - 2)^2 = 0$$

$$x - 2 = 0$$

$$x = 2$$

231. If $S = \sum_{n=1}^{15} \left(n + \frac{1}{3} \right)$ then the value of S is-

- (a) 125 (b) $120 + \frac{1}{3}$

- (c) $135 + \frac{1}{3}$ (d) 130

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (a)

$$S = \sum_{n=1}^{15} \left(n + \frac{1}{3} \right)$$

$$= \left(1 + \frac{1}{3} \right) + \left(2 + \frac{1}{3} \right) + \dots + \left(15 + \frac{1}{3} \right)$$

$$= \frac{4}{3} + \frac{7}{3} + \dots + \frac{46}{3}$$

$$= \frac{4 + 7 + \dots + 46}{3}$$

$$\therefore S_n = \frac{n}{2} [2a + (n - 1)d] \quad \{ \text{where } a=4, d=3 \}$$

$$= \frac{15}{2} [2 \times 4 + (15 - 1)3] = \frac{15}{2} \times 50 = \frac{15 \times 25}{3}$$

$$= 5 \times 25 = 125$$

232. Two trains, each of length 150 metres, are moving in opposite directions with equal speed of 90 km per hour.

The time taken by the trains to cross each other will be.

- (a) 3 seconds (b) 4.5 seconds
 (c) 6 seconds (d) 9 seconds

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

$$\text{Relative speed} = (90 + 90) \text{ km./hr} = 180 \text{ km/hr}$$

$$= \left(180 \times \frac{5}{18} \right) \text{ m/s} = 50 \text{ m/s}$$

Distance = total length of both train

$$= (150 + 150) \text{ m} = 300 \text{ m}$$

Time taken by the trains to cross each other

$$= \frac{300}{50} = 6 \text{ second}$$

233. How many prime numbers are between 1 and 50?

- (a) 17 (b) 15
(c) 14 (d) 16
(e) None of the above

64th B.P.S.C. (Pre) 2018

Ans. (b)

Prime numbers between 1 and 50 are
2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43 and 47
There are 15 prime numbers between 1 and 50

234. A Number when divided by 342, gives a remainder 47. If the same numbers is divided by 19, what will be remainder?

- (a) 0 (b) 9
(c) 18 (d) 8
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) Exam. 2017

Ans. (b)

Let the number be x
Dividen = (Divisor $\times$ Quotient) + Remainder.
 $x = (342 \times \text{Quotient}) + 47$
Now dividing "dividen" by 19.
$$= \left(\frac{342 \times \text{Quotient}}{19} \right) + \frac{47}{19}$$
$$= (18 \times \text{Quotient}) + 2 \text{ Quotient} + 9 \text{ Remainder}$$

So, the remainder is 9.

235. A person who spends $66\frac{2}{3}\%$ of his income is able to

save Rs. 1,200 per month. His monthly expense (in Rs.) is-

- (a) 2,400 (b) 3,000
(c) 2,000 (d) 3,600
(e) 2,800

63rd B.P.S.C. (Pre) Exam. 2017

Ans. (a)

Monthly saving of a person = $100\% - 66\frac{2}{3}\%$
$$= \left(100 - \frac{200}{3} \right) \%$$
$$= \frac{100}{3} \%$$

$\therefore$ Monthly saving of person = $\frac{100}{3} \%$

According to questions, $\frac{100}{3} \% = \text{Rs. } 1200$

$\therefore$ Monthly Expense of person $\frac{200}{3} \%$
 $= \text{Rs. } 1200 \times 2 = 2400 \text{ Rs.}$

236. A man gains 20% by selling an article for a certain price. If he sells it at the double the price, the percentage of profit will be-

- (a) 140 (b) 200
(c) 160 (d) 160
(e) 120

63rd B.P.S.C. (Pre) Exam. 2017

Ans. (a)

Let the cost price of an article = Rs. 100
Selling price of article at 20% profit = Rs. 120
If the article is sold at double the price, the new selling price
 $= 2 \times 120 \Rightarrow 240 \text{ Rs.}$
$$\text{Now profit \%} = \left(\frac{240 - 100}{100} \times 100 \right) \%$$
$$= 140 \%$$

237. 10 women can complete a work in 7 days and 10 children take 14 days to complete the work. How many days will 5 women and 10 children take to complete the work?

- (a) 6 (b) 5
(c) 3 (d) 7
(e) 4

63rd B.P.S.C. (Pre) Exam. 2017

Ans. (d)

$\therefore$ Work of (10×7) women = work of (14×10) children

$\therefore$ Work of 1 women = work of $\left(\frac{14 \times 10}{10 \times 7} \right)$ children

$\therefore$ Work of 1 women = work of 2 children

$\therefore$ Work of 5 women and 10 children

= Work of $(5 \times 2 + 10)$ Children

= Work of 20 Children

$M_1 D_1 = M_2 D_2$

$14 \times 10 = 20 \times D_2$

$D_2 = \frac{140}{20} \Rightarrow 7 \text{ Day}$

238. How many numbers between 11 and 90 are divisible by 7?

- (a) 10 (b) 9

- (c) 13 (d) 12
(e) 11

63rd B.P.S.C. (Pre) Exam. 2017

Ans. (e)

Such numbers (between 11 and 90) which are divisible by 7 are as-

14, 21, 28, 84

The series of these numbers are arithmetic progression (AP) sequence.

∴ First term (a) = 14, d = 7

$$T_n (\text{n}^{\text{th}} \text{ term}) = 84$$

$$T_n = a + (n - 1) d$$

$$84 = 14 + (n - 1) 7$$

$$7(n - 1) = 84 - 14$$

$$n - 1 = \frac{70}{7} \Rightarrow 10$$

$$\therefore n = 10 + 1 \Rightarrow 11$$

239. A man can row 7.5 km per hours in still water. If in a river, running at 1.5 km per hour, it takes 50 minutes to row to a place and back then how far is the place?

- (a) 3 km (b) 4 km
(c) 2 km (d) 5 km
(e) 7 km

63rd B.P.S.C. (Pre) Exam. 2017

Ans. (a)

$$\text{Speed of boat in downstream} = \left(7\frac{1}{2} + 1\frac{1}{2}\right) \text{ km/hr}$$

$$= 9 \text{ km/hr}$$

$$\text{and speed of boat in upstream} = \left(7\frac{1}{2} - 1\frac{1}{2}\right) \text{ km/hr}$$

$$= 6 \text{ km/hr}$$

Let, man goes x km and return x km back.

$$\therefore \frac{x}{9} + \frac{x}{6} = \frac{50}{60} \text{ hr}$$

$$\frac{4x + 6x}{36} = \frac{5}{6}$$

$$\frac{2x + 3x}{18} = \frac{5}{6}$$

$$\frac{5x}{18} = \frac{5}{6} \Rightarrow 5x = 15$$

$$x = 3 \text{ km}$$

240. A sum of money invested at compound interest amounts to Rs. 4624 in 2 years and to Rs. 4913 in 3 years. The sum of money is-

- (a) Rs. 4,240 (b) Rs. 4,280
(c) Rs. 4,096 (d) Rs. 4,346
(e) Rs. 4,406

63rd B.P.S.C. (Pre) Exam. 2017

Ans. (c)

Let, the principle is Rs. P and rate is R%.

$$\text{Formula - } A = P \left(1 + \frac{R}{100}\right)^x$$

According to question,

$$P \left(1 + \frac{R}{100}\right)^2 = 4624 \dots \dots \dots (i)$$

$$P \left(1 + \frac{R}{100}\right)^3 = 4913 \dots \dots \dots (ii)$$

On dividing eqⁿ (ii) by eqⁿ (i)

$$1 + \frac{R}{100} = \frac{4913}{4624}$$

$$\frac{R}{100} = \frac{4913}{4624} - 1 = \frac{4913 - 4624}{4624}$$

$$R = \frac{289 \times 100}{4624} = \frac{25}{4}$$

Putting the value of R in eqⁿ (i)

$$P \left(1 + \frac{25}{4 \times 100}\right)^2 = 4624$$

$$P \times \frac{425}{400} \times \frac{425}{400} = 4624$$

$$P = \frac{4624 \times 400 \times 400}{425 \times 425} \Rightarrow 4096$$

241. A man buys Rs. 20 shares by paying 9% dividend. Man wants to have an interest of 12% on his money. The market value of each shares is:-

- (a) Rs.18 (b) Rs. 15
(c) Rs. 21 (d) Rs. 21
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) Exam. 2017

Ans. (b)

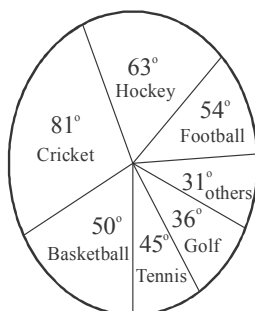
Let the market value of each share is Rs. x

$\therefore 12\% \text{ of } x = 9\% \text{ of } 20$

$$x \times \frac{12}{100} = 20 \times \frac{9}{100}$$

$$x = \frac{20 \times 9}{12} \Rightarrow 15 \text{ Rs.}$$

242. The following pie chart shows the expenditure of the country on various sports during the particular year. Study the chart carefully and answer the question.



How much percent less is spent on football than that of cricket?

- (a) $35\frac{1}{2}$ (b) 29
(c) $32\frac{1}{2}$ (d) 31
(e) $33\frac{1}{3}$

63rd B.P.S.C. (Pre) Exam. 2017

Ans. (e)

It is clear from the pie-chart that

Expenditure on football = 54°

and expenditure on cricket = 81°

$\therefore$ less, percent spent on football than that of cricket.

$$= \left(\frac{81^\circ - 54^\circ}{81^\circ} \times 100 \right) \%$$

$$= \left(\frac{27^\circ}{81^\circ} \times 100 \right) \%$$

$$= \frac{100}{3} \% \Rightarrow 33\frac{1}{3} \%$$

243. A invested Rs. 76,000 in a business. After some months, B joined him with Rs. 57,000. At the end of the year, the total profit was divided between them in the ratio 2:1 After how many months did B join?

- (a) 6 (b) 4

(c) 3

(d) 8

(e) 5

63rd B.P.S.C. (Pre) 2017

Ans. (b)

(Let B, invest money after x months)

	A	:	B
Investment $\rightarrow$	76000	:	57000
	4	:	3
	$\times$	:	$\times$
Time $\rightarrow$	12	:	$(12-x)$
Profit $\rightarrow$	2	:	1

$$\therefore \frac{4 \times 12}{3 \times (12-x)} = \frac{2}{1}$$

$$12-x=8$$

$$x = 12 - 8 \Rightarrow 4 \text{ Months}$$

So, the B join the business after 4 months.

244. How many times the digit 9 occurs in the numbers between 100 to 999?

- (a) 280 (b) 218
(c) 229 (d) 228
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (a)

Digit 9, occurs 20 times between 100 and 200.

$\therefore$ No. of times digit 9 occurs between 100 to 899.
 $= 20 \times 8 = 160$ times

digits 9 occurs between 900 to 990
 $= (11 \times 9 + 2)$ Times
 $= 101$ Times

digit 9 occurs between 991 to 999
 $= 9 \times 2 + 1$
 $= 19$ Times

So the digit 9 occurs between 100 to 999
 $= 160 + 101 + 19$
 $= 280$ Times

$\therefore$ So option (a) is the correct answer.

245. Find the value of expression. $\frac{4}{\sqrt[3]{9} - \sqrt[3]{3} + 1}$ is—

- (a) $3^{1/3} + 1$ (b) $3^{1/2} + 1$
(c) $3^{1/3} - 1$ (d) $3^{1/2} - 1$

48th to 52nd B.P.S.C. (Pre) 2008

Given that, expression $\frac{4}{\sqrt[3]{9} - \sqrt[3]{3} + 1}$

$$= \frac{3+1}{(3^2)^{\frac{1}{3}} - (3)^{\frac{1}{3}} + 1}$$

$$= \frac{(3^{\frac{1}{3}})^3 + (1^{\frac{1}{3}})^3}{(3^{\frac{1}{3}})^2 - (3)^{\frac{1}{3}} \times 1^{\frac{1}{3}} + (1^{\frac{1}{3}})^2}$$

$$= \frac{(3^{\frac{1}{3}} + 1^{\frac{1}{3}}) \left[(3^{\frac{1}{3}})^2 - (3)^{\frac{1}{3}} \times 1^{\frac{1}{3}} + (1^{\frac{1}{3}})^2 \right]}{(3^{\frac{1}{3}})^2 - (3)^{\frac{1}{3}} \times 1^{\frac{1}{3}} + (1^{\frac{1}{3}})^2}$$

$$[\because a^3 + b^3 = (a + b)(a^2 - ab + b^2)]$$

$$= 3^{\frac{1}{3}} + 1$$

246. The dearness-allowance of a person with basic salary Rs. 7700 is increased to 132% from 125% and the tax deduction on both is increased from 20% to 22%. He got salary increased by.

- (a) Rs. 74.00 (b) Rs. 77.00
(c) Rs.385.00 (d) Rs. 369.60
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (e)

Basic salary of person = Rs.7700

$$\text{Initial dearness-allowance} = 7700 \times \frac{125}{100} = \text{Rs. } 9625$$

$$\text{Increased dearness- allowance} = 7700 \times \frac{132}{100} \Rightarrow \text{Rs. } 10164$$

$$\text{Salary after 20\% tax deduction in basic salary and dearness allowance} = (7700 + 9625) \frac{80}{100}$$

$$= 17325 \times \frac{80}{100} = 13860 \text{ Rs.}$$

Salary after 22% tax deduction in basic salary and increased dearness allowance.

$$= (7700 + 10164) \frac{78}{100}$$

$$= 17864 \times \frac{78}{100} = 13933.92 \text{ Rs.}$$

$$\text{Increased salary of person} = 13933.92 - 13860 = \text{Rs. } 73.92$$

So, the correct answer is option (e)

247. A, B, C Invested 20 lack rupees in a business in the ratio 7:2:1. The total profit is 18%. A has to pay 30% tax and B has to pay 20% tax. A's net profit is what percent of B's net profit?

- (a) 118.8% (b) 180.0%
(c) 306.25% (d) 304.5%
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (e)

$$\text{Total profit of A, B, C,} = 2000000 \times \frac{18}{100} = \text{Rs. } 360000$$

$$\text{Proportional sum} = 7 + 2 + 1 = 10$$

$$\text{Profit of A} = 360000 \times \frac{7}{10} = \text{Rs. } 252000$$

$$\text{Profit of B} = 360000 \times \frac{2}{10} = \text{Rs. } 72000$$

Net profit of A after paying tax

$$= 252000 \times \frac{70}{100} = \text{Rs. } 176400$$

Net profit of B after paying tax

$$= 72000 \times \frac{80}{100} = \text{Rs. } 57600$$

Now we can see that, the net profit of A is more than net profit of B.

$$\text{Required\%} = \frac{176400 - 57600}{57600} \times 100\% = 206.25\%$$

So, option (e) is the correct answer.

248. A coin is tossed above the ground with a velocity of 9.8 m/s, then it will rise to the following height :

- (a) 9.8 m (b) 10 m
(c) 4.9 m (d) 49 m

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (c)

Let, the velocity of coin at the height of h meter = 0

$$\text{then } \because v^2 = u^2 - 2gh$$

(where u = initial velocity and v = final velocity)

$$0 = (9.8)^2 - 2(9.8)h$$

$$h = \frac{(9.8)^2}{2 \times (9.8)} = \frac{9.8}{2}$$

249. There are two drains, A and B in the bottom of a water tank. If only A is open, it takes 30 minutes to empty the full tank and if only B is open, it takes 20 minutes. If both drains are open for 10 minutes, then B is closed, how much time does take to empty a full tank?

- (a) 18 minutes (b) 15 minutes
(c) 17 minutes (d) 20 minutes
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

Part of tank, empty by drain A in 1 min = $\frac{1}{30}$

Part of tank, empty by drain B in 1 min = $\frac{1}{20}$

Part of tank, empty by both drains in 1 min

$$= \left(\frac{1}{30} + \frac{1}{20} \right) \text{Part}$$

Part of tank, empty by both drains A and B in 10 min.

$$= 10 \left(\frac{1}{30} + \frac{1}{20} \right) \text{Part}$$

$$= \frac{1}{3} + \frac{1}{2} \Rightarrow \frac{5}{6} \text{ Part}$$

Remaining, $1 - \frac{5}{6} \Rightarrow \frac{1}{6}$ Part

Time taken by drain A, to empty part of tank = 30 minutes

Time taken by drain A to empty remaining part of tank

$$= \frac{1}{6} \times 30 \Rightarrow 5 \text{ min.}$$

Total time taken to empty full tank = (10 + 5) min.
 = 15 min.

So, option (b) is correct answer.

250. If $x = -\frac{2}{3}$, then $9x^2 - 3x - 11$ is equal to-

- (a) -13 (b) 13
(c) -5 (d) -17
(e) 17

64th B.P.S.C. (Pre) 2018

Ans. (c)

If $x = -\frac{2}{3}$

$$\begin{aligned} \text{then } 9x^2 - 3x - 11 &= 9 \left(-\frac{2}{3} \right)^2 - 3 \left(-\frac{2}{3} \right) - 11 \\ &= 9 \times \frac{4}{9} + 2 - 11 \\ &= 4 + 2 - 11 \\ &= 6 - 11 = -5 \end{aligned}$$

251. In an examination, each candidate takes Hindi or History or both. 66% take Hindi and 59% take History. The total number of candidates was 3000. How many candidates took both Hindi and History?

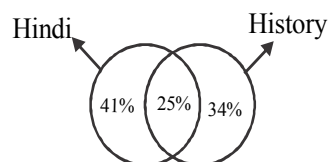
- (a) 500 (b) 750
(c) 542 (d) 738
(e) 830

64th B.P.S.C. (Pre) 2018

Ans. (b)

Percentage of candidates take both Hindi and History as subject-

$$\begin{aligned} &= \{(66+59) - 100\}\% \\ &= (125 - 100)\% \\ &= 25\% \end{aligned}$$



Number of candidates taking both the subjects-

$$\begin{aligned} &= 3000 \times \frac{25}{100} \\ &= 30 \times 25 = 750 \end{aligned}$$

252. If $x = \sqrt{\frac{0.00001225}{0.00005329}}$ then, the value of x is-

- (a) $\frac{35}{73}$ (b) $\frac{525}{933}$
(c) $\frac{205}{403}$ (d) $\frac{135}{233}$
(e) None of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

$$x = \sqrt{\frac{0.00001225}{0.00005329}}$$

$$= \sqrt{\frac{1225}{5329}}$$

$$= \sqrt{\frac{35 \times 35}{73 \times 73}}$$

$$= \frac{35}{73}$$

Therefore, option (a) is correct answer.

253. If $x + \frac{1}{y} = 1$ and $y + \frac{1}{z} = 1$, then the value of $z + \frac{1}{x} = 1$

is-

- (a) $x - y$ (b) 1
(c) Uncountable (d) 2
(e) None of the above

64th B.P.S.C. (Pre) 2018

Ans. (b)

Given that

$$x + \frac{1}{y} = 1$$

$$x = 1 - \frac{1}{y} = \frac{y-1}{y}$$

$$\therefore \frac{1}{x} = \frac{y}{y-1} \quad \dots\dots\dots (i)$$

and $y + \frac{1}{z} = 1$

$$\frac{1}{z} = 1 - y$$

$$\therefore z = \frac{1}{1-y} \quad \dots\dots\dots (ii)$$

Adding eqⁿ (i) and (ii)

$$\begin{aligned} z + \frac{1}{x} &= \frac{y}{y-1} + \frac{1}{1-y} \\ &= \frac{y}{y-1} - \frac{1}{y-1} \\ &= \frac{1}{y-1} (y-1) = 1 \end{aligned}$$

254. The average age of 4 sisters is 7 years. If we add the age of mother, the average increases by 6 years. Find the age of mother?

- (a) 46 years (b) 39 years
(c) 37 years (d) 47 years
(e) 57 years

64th B.P.S.C. (Pre) 2018

Ans. (c)

Average age of 4 sisters = 7 years

Total age of 4 sisters = $7 \times 4 = 28$ years

Average age of all members including mother = $7 + 6 = 13$ years

Total age of all members (Mother + 4 sisters) = $13 \times 5 = 65$ years

Mother age = $(65 - 28)$ years
= 37 years

255. If $3^{x+8} = 27^{2x+1}$ then the value of x is-

- (a) 9 (b) 1
(c) -1 (d) 10
(e) -10

64th B.P.S.C. (Pre) 2018

Ans. (b)

$$3^{x+8} = 27^{2x+1}$$

$$3^{x+8} = (3^3)^{2x+1}$$

$$3^{x+8} = 3^{6x+3}$$

Since, the bases of both the sides are equal, hence the power will also be equal.

$$x + 8 = 6x + 3$$

$$6x - x = 8 - 3$$

$$5x = 5$$

$$x = 1$$

256. If $S = \sum_{n=1}^{10} (2n + \frac{1}{2})$, then S is-

- (a) $55\frac{1}{2}$ (b) 56
(c) 111 (d) 115
(e) $110\frac{1}{2}$

64th B.P.S.C. (Pre) 2018

Ans. (d)

$$S = \sum_{n=1}^{10} (2n + \frac{1}{2})$$

$$n = 1, 2, 3, \dots\dots\dots, 10$$

$$S_1 = 2 \times 1 + \frac{1}{2} = \frac{5}{2}$$

$$S_2 = 2 \times 2 + \frac{1}{2} = \frac{9}{2}$$

$$S_3 = 2 \times 3 + \frac{1}{2} = \frac{13}{2}$$

$$S_{10} = 2 \times 10 + \frac{1}{2} = \frac{41}{2}$$

$$\therefore S = \frac{5}{2} + \frac{9}{2} + \frac{13}{2} + \dots\dots\dots + \frac{41}{2}$$

$$= \frac{1}{2} [5 + 9 + 13 + \dots\dots\dots, +41]$$

$$\begin{aligned}
&= \frac{1}{2} \left[\frac{10}{2} \{2 \times 5 + (10-1)4\} \right] \therefore S_n = \frac{n}{2} [2a + (n-1)d] \\
&= \frac{1}{2} \times 5 \times [10 + 9 \times 4] \\
&= \frac{1}{2} \times 5 \times 46 = 5 \times 23 \Rightarrow 115
\end{aligned}$$

257. The value of x is a consecutive whole number and the first four value of the expression $x^3 + 3y - 3$ are 7, 20, 45, 88, then the fifth value will be-

- (a) 137 (b) 155
(c) 158 (d) 143
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

Given that, the expression is $x^3 + 3y - 3$

where the value of x is a whole number

or $x = \{0, 1, 2, 3, 4, \dots\}$

putting $x=0$ in expression

$$(0)^3 + 3y - 3 = 7$$

$$y = \frac{10}{3}$$

putting, $x=1$ in expression

$$(1)^3 + 3y - 3 = 20$$

$$y = \frac{22}{3}$$

putting, $x=2$ in expression

$$(2)^3 + 3y - 3 = 45$$

$$y = \frac{40}{3}$$

On putting, $x=3$ in expression

$$(3)^3 + 3y - 3 = 88$$

$$y = \frac{64}{3}$$

So, the next value of x will be-

$$\frac{10}{3} + 4 = \frac{22}{3}, \frac{22}{3} + 6 = \frac{40}{3}$$

$$\frac{40}{3} + 8 = \frac{64}{3}, \frac{64}{3} + 10 = \frac{94}{3}$$

then the fifth value of expression will be-

$$(4)^3 + 3 \times \frac{94}{3} - 3$$

$$= 64 + 94 - 3$$

$$= 64 + 91$$

$$= 155$$

So, option (b) is correct answer-

258. Find the value of $x - [y - \{z - (x - \overline{y - z})\}]$

- (a) $x + y + z$ (b) $x - y - z$
(c) 1 (d) 0

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

$$\begin{aligned}
&x - [y - \{z - (x - \overline{y - z})\}] \\
&= x - [y - \{z - (x - y + z)\}] \\
&= x - [y - \{z - x + y - z\}] \\
&= x - [y - z + x - y + z] \\
&= x - y + z - x + y - z \\
&= 0
\end{aligned}$$

259. The divisor of the expression $x(x^2-1)(3x+2)$ for every integer x is-

- (a) 13 (b) 15
(c) 24 (d) 25

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (c)

Given that, the expression is $F(x) = x(x^2-1)(3x+2)$

For each integer-

On putting, $x = 1, 2, 3, 4, 5, \dots$

$$x = 1, F(1) = 0 = 24 \times 0$$

$$x = 2, F(2) = 2(2^2-1).(3 \times 2+2) = 2 \times 3 \times 8 = 24 \times 2$$

$$x = 3, F(3) = 3(3^2-1).(3 \times 3+2) = 3 \times 11 \times 8 = 24 \times 11$$

$$x = 4, F(4) = 4(4^2-1).(3 \times 4+2) = 4 \times 15 \times 14 = 24 \times 35$$

$$x = 5, F(5) = 5(5^2-1).(3 \times 5+2) = 5 \times 24 \times 17 = 24 \times 85$$

Thus, the divisor of the given expression $x.(x^2-1).(3x+2)$ for each integer is $x = 24$

260. The inverse of matrix $\begin{bmatrix} 5 & 2 \\ 7 & 3 \end{bmatrix}$ is-

- (a) $\begin{bmatrix} 3 & 2 \\ -7 & 5 \end{bmatrix}$
(b) $\begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}$
(c) $\begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$
(d) $\begin{bmatrix} -3 & 2 \\ 7 & -5 \end{bmatrix}$

53rd to 55th B.P.S.C. (Pre) 2011

$$\text{Let, } A = \begin{bmatrix} 5 & 2 \\ 7 & 3 \end{bmatrix}$$

$$\therefore \text{Adj } A = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$$

$$\text{and } |A| = 3 \times 5 - 7 \times 2 \\ = 15 - 14 \\ = 1$$

$$\therefore A^{-1} = \frac{\text{Adj } A}{|A|} = \frac{1}{1} \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix} \\ = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$$

261. The solution of the differential equation

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^{5x} \text{ is-}$$

$$(a) y = C_1 e^x + C_2 e^{2x} + \frac{1}{12} e^{5x}$$

$$(b) y = C_1 e^{-x} + C_2 e^{2x} + \frac{1}{12} e^{5x}$$

$$(c) y = C_1 e^x + C_2 e^{-2x} + \frac{1}{12} e^{5x}$$

$$(d) y = C_1 e^x + C_2 e^{2x} + \frac{1}{5} e^{5x}$$

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^{5x}$$

$$\text{Let, } \frac{d^2 y}{dx^2} = D^2$$

$$D^2 - 3D + 2 = 0$$

$$D^2 - 2D - D + 2 = 0$$

$$D(D-2) - 1(D-2) = 0$$

$$(D-2)(D-1) = 0$$

$$D = 2, 1$$

$$\text{then } y = C_1 e^x + C_2 e^{2x}$$

Specific solution

$$\text{P.I.} = \frac{1}{f(D)} e^{5x}$$

$$= \frac{1}{D^2 - 3D + 2} e^{5x}$$

$$= e^{5x} \cdot \frac{1}{25 - 15 + 2}$$

$$= \frac{1}{12} e^{5x}$$

then, General solution.

$$y = C_1 e^x + C_2 e^{2x} + \frac{1}{12} e^{5x}$$

where C_1 & C_2 are constant.

262. If function $f: I^+ \rightarrow R$, $f(x) = \log x$

then the value of $f(x) + f(y)$ is-

$$(a) f(xy)$$

$$(b) f(x+y)$$

$$(c) f\left(\frac{x}{y}\right)$$

$$(d) f\left(\frac{y}{x}\right)$$

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

$$\therefore f(x) = \log x \dots\dots\dots (i)$$

Substituting y in place of x

$$\therefore f(y) = \log y \dots\dots\dots (ii)$$

Substituting $x = v$

$$f(v) = \log v \dots\dots\dots (iii)$$

$$\therefore f(x) + f(y) = \log x + \log y$$

$$= \log xy (\because \log m + \log n = \log mn)$$

Let, $x, y = v$

$$\therefore f(x) + f(y) = \log v$$

$$\therefore f(x) + f(y) = f(v) \text{ (from eqⁿ (iii))}$$

$$\therefore f(x) + f(y) = f(xy) \quad (\because v = xy)$$

263. If triangle PQR has $\angle P = 120^\circ$ and $PQ = PR$, then $\angle Q$ and $\angle R$ will be respectively:

$$(a) 60^\circ, 30^\circ$$

$$(b) 30^\circ, 40^\circ$$

$$(c) 30^\circ, 30^\circ$$

$$(d) 20^\circ, 40^\circ$$

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

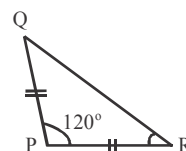
$$\therefore \text{ In } \triangle PQR \angle P = 120^\circ$$

and $PQ = PR$

$\therefore$ If two sides of a triangle are equal, then the angles opposites to it will also be.

$$\therefore \angle Q = \angle R \dots\dots\dots (i)$$

$$\therefore \angle P + \angle Q + \angle R = 180^\circ$$



$$\begin{aligned}\therefore 120 + \angle Q + \angle Q &= 180^\circ \\ (\because \text{From eq}^n \text{ (i) } \angle Q &= \angle R) \\ 2\angle Q &= 180^\circ - 120^\circ \\ 2\angle Q &= 60^\circ \\ \angle Q &= 30^\circ \\ \therefore \angle R &= 30^\circ\end{aligned}$$

264. The coordinates of the mid-point joining the point P (4, 6) and Q (-4, 8) will be-

- (a) (2, 7) (b) (7, 2)
(c) (7, 0) (d) (0, 7)

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

The coordinates of the mid-point joining the point

P (x_1, y_1) and Q (x_2, y_2)

$$= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$\therefore$ The coordinates of the mid-point joining the point (4, 6) and (-4, 8)

$$\begin{aligned}&= \left(\frac{4 - 4}{2}, \frac{6 + 8}{2} \right) \\ &= (0, 7)\end{aligned}$$

265. The triangle formed by the points (2, 7), (4, -1) and (-2, 6) will be -

- (a) Equilateral (b) Right angle
(c) Isosceles (d) None of these

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (b)

The three points of the triangle respectively.

$$A = (2, 7) \quad B = (4, -1), \quad C = (-2, 6)$$

Length of first side, AB

$$\begin{aligned}&= \sqrt{(4 - 2)^2 - (-1 - 7)^2} \\ &= \sqrt{2^2 + (-8)^2} = \sqrt{68}\end{aligned}$$

Length of second side, BC

$$\begin{aligned}&= \sqrt{(-2 - 4)^2 + (+6 - 1)^2} \\ &= \sqrt{(-6)^2 + 7^2} = \sqrt{85}\end{aligned}$$

Length of the third side, CA

$$\begin{aligned}&= \sqrt{(-2 - 2)^2 + (6 - 7)^2} \\ &= \sqrt{(-4)^2 + (-1)^2} = \sqrt{17}\end{aligned}$$

According to Pythagoras theorem-

$$BC^2 = AB^2 + CA^2$$

$$(\sqrt{85})^2 = (\sqrt{68})^2 + (\sqrt{17})^2$$

$$85 = 85$$

i.e. $\triangle ABC$ is a right angled triangle.

266. The length of the minute hand of a clock is 12 cm, then the area covered by minute in one minute will be-

- (a) 22.12 sq. cm
(b) 23.10 sq. cm
(c) 24.12 sq. cm
(d) None of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

The angle subtended by the min. hand of the clock in

One minutes = 6°

$$\therefore \text{Area covered by min. hand in one minute} = \frac{\pi r^2 \theta}{360^\circ}$$

$$= \frac{22}{7} \times \frac{12 \times 12 \times 6^\circ}{360^\circ} \text{ cm}^2$$

$$= \frac{22}{7 \times 60} \times 144 \text{ cm}^2$$

$$= \frac{11 \times 72}{7 \times 15} \Rightarrow 7.54 \text{ cm}^2$$

267. A player completes a run of 2200 meters by completing 7 rounds of circular path, then the radius of the circular

path is- $\left(\pi = \frac{22}{7} \right)$

- (a) 30 meters
(b) 40 meters
(c) 50 meters
(d) 60 meters

53rd to 55th B.P.S.C. (Pre) 2011

Let, the radius of circular path = r

$\therefore$ Distance covered by player in one round = $2\pi r$

$\therefore$ According to question

$$7 \times 2\pi r = 2200$$

$$r = \frac{2200 \times 7}{14 \times 22}$$

$$= \frac{100}{2} \Rightarrow 50 \text{ meters.}$$

268. The variance of 7, 7, 7, 7, 7 will be-

- (a) 7 (b) 0
(c) 2 (d) 7.5

56th to 59th B.P.S.C. (Pre) 2015

Ans. (b)

The variance of observations is represented by σ^2

$$\sigma^2 = \frac{(x - \bar{x})^2}{\text{no. of observation}}$$

Now,

$$\bar{x} = \frac{7+7+7+7+7}{5} = \frac{35}{5} = 7$$

then,

$$(x - \bar{x})^2 = 0 \text{ (because all observations are same)}$$

$$\text{So Variance, } (\sigma)^2 = \frac{0}{5} = 0$$

269. The median of 2, 4, 6, 8, 10, 12, 14, 16 is-

- (a) 8 (b) 9
(c) 10 (d) 11

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

Given that, the series is 2, 4, 6, 8, 10, 12, 14, 16

(Total 8 term i.e. the no- of terms is even)

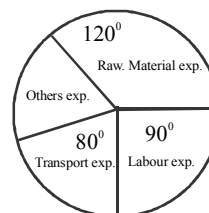
$$\therefore \text{ Median } = \frac{\frac{n}{2} \text{th term} + \left(\frac{n}{2} + 1\right) \text{th term}}{2}$$

$$= \frac{\frac{8}{2} \text{th term} + \left(\frac{8}{2} + 1\right) \text{th term}}{2}$$

$$= \frac{4 \text{th term} + 5 \text{th term}}{2}$$

$$= \frac{8+10}{2} \Rightarrow 9$$

270. In the following pie-chart, the various expenditures of an industry are represented. Total expenditure on raw material is Rs. 30 lakh, if labour expenditure is increased by 5%. How much expenditure on other expenses is reduced to maintain the profit?



- (a) 5.9% (b) 12.86%
(c) 6.43% (d) 6.21%
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

According to question,

Total expenditure on raw material = 30 lakh

$$120^\circ = 30 \text{ lakh}$$

$$1^\circ = \frac{30}{120} \text{ lakh}$$

$$= \frac{1}{4} \text{ lakh}$$

$$\text{Exp. on labour, } 90^\circ = \frac{90}{4} \text{ lakh}$$

$$\text{Increased expenditure on labour} = \frac{90}{4} \times \frac{5}{100}$$

$$= \frac{9}{8} \text{ lakh}$$

In order to keep the profit constant, the expenditure must be constant.

$$\text{Area of other exp.} = 360^\circ - (120^\circ + 80^\circ + 90^\circ)$$

$$= 70^\circ$$

$$= \frac{70}{4} \text{ lakh} \left(\because 1^\circ = \frac{1}{4} \text{ Lakh} \right)$$

$$\text{deduction in other expenses} = \frac{9}{8} \text{ lakh}$$

$$\frac{70}{4} \text{ lakh } x \% \text{ of } = \frac{9}{8} \text{ lakh}$$

$$x = \frac{9}{8} \times \frac{4}{70} \times 100\%$$

$$= 6.43\%$$

271. Find the value of –

$$\frac{0.05 \times 0.05 \times 0.05 - 0.04 \times 0.04 \times 0.04}{0.05 \times 0.05 + 0.05 \times 0.04 + 0.04 \times 0.04}$$

- (a) 1.01 (b) 0.1
(c) 0 (d) 0.01

40th B.P.S.C. (Pre) 1995

Ans. (d)

$$\frac{(0.05)^3 - (0.04)^3}{(0.05)^2 + (0.05)(0.04) + (0.04)^2}$$

$$= 0.05 - 0.04$$

$$= 0.01$$

Note $[a^3 - b^3 = (a - b)(a^2 + ab + b^2)]$

272. If $x^2 + \frac{1}{x^2} = 3^4$, then $x + \frac{1}{x}$ is equal to—

- (a) $\sqrt{43}$ (b) $\sqrt{37}$
(c) $\sqrt{46}$ (d) $\sqrt{83}$

44th B.P.S.C. (Pre) 2000-01

Ans. (d)

$$x^2 + \frac{1}{x^2} = 3^4$$

$$x^2 + \frac{1}{x^2} + 2 = 3^4 + 2$$

$$\left(x + \frac{1}{x}\right)^2 = 83$$

$$\therefore \left(x + \frac{1}{x}\right) = \sqrt{83}$$

273. Which of the following statements is not correct.

- (a) π is an irrational number
(b) e is a rational number
(c) $\sqrt{2}$ is an even number
(d) ∞ is a real number

43rd B.P.S.C. (Pre) 1999

Ans. (c)

$\sqrt{2}$ is not an even number.

274. Choose the false statements—

- (a) Mathematical work written by Laghu Bhaskarya Bhaskar
(b) Is also enumerated in the vachali inscription
(c) Decimal Method discovered by the Europeans
(d) Mathematical Tilak is a work of eleventh century

40th B.P.S.C. (Pre) 1995

Ans. (c)

Decimal Method discovered by the Indian.

275. Which of the following statements is not true?

- (a) The square of every negative number is positive.
(b) A negative number does not have a square root.
(c) $-\infty$ is a negative number.
(d) $-\pi$ is an algebraic number.

44th B.P.S.C. (Pre) 2000-01

Ans. (c)

- (i) The square of every negative number is positive, this statement is true.
(ii) A negative number root does not have a square root, its square root is imaginary.
(iv) π is an algebraic number. In fact, π its the ratio of circumference and diameter, It is not possible for the ratio to be negative.

276. Choose the difference of 5 miles and 8 km in yards—

- (a) 51.1 (b) 51
(c) 52.2 (d) 52

42nd B.P.S.C. (Pre) 1997-98

Ans. (a)

Scale -
1 miles = 1760 yards
1 km = 1093.61 yards
According to question,
5 miles = (5×1760) yards
= 8800 yards
and 8 km. = (8×1093.61) yards
= 8748.88 yards
 $\therefore$ Required difference = $(8800 - 8748.88)$ yards
= 51.12 yards
= 51.1 yards

277. The difference of two positive numbers is 72 and when one is divided by the other, the dividend is 4. Choose the numbers—

- (a) 40, 112 (b) 24, 96
(c) 32, 128 (d) 30, 102

42nd B.P.S.C. (Pre) 1997-98

Ans. (b)

Let, the two number be a and b respectively,
According to question,
 $a - b = 72$ (i)
 $\frac{a}{b} = 4$

$$a = 4b \dots\dots\dots (ii)$$

Substituting the value of eqⁿ (i) in equation (ii)

$$4b - b = 72$$

$$b = \frac{72}{3} = 24$$

$$\therefore a = 24 \times 4 = 96$$

So, the number will be 24 and 96.

278. How many Prime (twin prime) numbers are between 70 and 100 ?

- (a) 1 (b) 0
(c) 2 (d) 3

42nd B.P.S.C. (Pre) 1997-98

Ans. (a)

Twin Prime number are such prime numbers in which there is a difference of two e.g. (3, 5) and (5, 3)

According to question–

There are 6 prime number between 70 and 100

71, 73, 79, 83, 89, 97

Hence the number of twin prime between 70 and 100 is 1 (71 & 73).

279. Find the lowest number, which should be multiplied into 56,700, so that it becomes a perfect square-

- (a) 3 (b) 6
(c) 7 (d) 9

39th B.P.S.C. (Pre) 1994

Ans. (c)

$$56,700 = 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 5 \times 5 \times 7$$

It is clear that, on multiplying by 7, number 56700 will become a perfect square

280. The sum of three prime number is 100. One of the number is 36 more than the other numbers. The highest prime number is–

- (a) 31 (b) 67
(c) 73 (d) 79

41st B.P.S.C. (Pre) 1996

Ans. (b)

Let the first prime number x is

$\therefore$ Second prime number $= x + 36$

Third number $= 100 - (x + x + 36)$

$$= 100 - (2x + 36)$$

$$= 64 - 2x$$

x must be such a prime number that the result obtained by putting it in $(64 - 2x)$ should also prime number.

$\therefore$ On putting $x = 31$

Third Number $= 2$ (2 is a prime number)

So the highest prime number is $= 31 + 36 = 67$

281. If $7^{13} + 1$ is divisible by 6 then the remainder is–

- (a) 0 (b) 1
(c) 2 (d) 3

41st B.P.S.C. (Pre) 1996

Ans. (c)

On dividing $7^{13} + 1$, by 6

$$\text{Obtained remainder, } \frac{7^{13} + 1}{6} = \frac{(6+1)^{13} + 1}{6}$$

$$= \frac{(+1)^{13} + 1}{6} = \frac{1+1}{6}$$

$$= \frac{2}{6}$$

Thus, remainder $= 2$

282. In a class test, x and y obtained the following marks:

Subject \ Student	x	y	Total Mark
English	84	92	100
Hindi	80	79	100
Mathematics	90	88	100
History	69	60	100

Who got more marks and how many.

- (a) x got 1% more marks
(b) x got 2% more marks
(c) y got 1% more marks
(d) y got 2% more marks

40th B.P.S.C. (Pre) 1995

Ans. (a)

According to question total marks obtained by x

$$= 84 + 80 + 90 + 69 = 323$$

$$\text{marks percentage of } x = \frac{323 \times 100}{400} = 80.75\%$$

$$\text{marks percentage of } y = \frac{319 \times 100}{400} = 79.75\%$$

Percentage difference of x and $y = 80.75 - 79.75 = 1\%$
Hence x got 1% more marks.

283. If $33\frac{1}{3}\%$ of $x = 11$, the value of x will be?

- (a) 11 (b) 25
(c) 33 (d) 44

39th B.P.S.C. (Pre) 1994

Ans. (c)

$$33\frac{1}{3}\% \text{ of } x = 11$$

$$x \times \frac{100}{3} \% = 11$$

$$x \times \frac{100}{300} = 11$$

$$x = \frac{11 \times 300}{100} = 33$$

284. A man suffers a loss of 20% by selling a cycle for Rs. 450. By selling the cycle at 20% profits, its selling price will be—

- (a) Rs. 652.50 (b) Rs. 765
(c) Rs. 562.50 (d) Rs. 675

43rd B.P.S.C. (Pre) 1999

Ans. (d)

Selling price = Rs.450, the loss % = 20%

$$\text{Cost price} = \frac{100 \times \text{Selling price}}{100 - \text{Loss}\%}$$

$$\begin{aligned} \text{Cost price of cycle} &= \frac{100 \times 450}{80} \\ &= \text{Rs. } 562.50 \end{aligned}$$

Selling price of cycle of 20% profit =

$$\begin{aligned} &\frac{\text{Cost price} \times (100 + \text{profit}\%)}{100} \\ &= \frac{562.5 \times 120}{100} \end{aligned}$$

∴ The selling price of cycle at 20% profit = Rs. 675

- (a) Rs. 441 (b) Rs. 441

285. Choose the compound interest on Rs. 400 for 2 years at 5% per annum compounded annually.

(c) Rs. 410

(d) Rs. 401

43rd B.P.S.C. (Pre) 1999

Ans. (c)

Compound Interest = Amount - Principal

$$\begin{aligned} \text{Amount} &= P \left(1 + \frac{R}{100} \right)^{\text{Time}} \\ &= 4000 \left(1 + \frac{5}{100} \right)^2 \\ &= 4000 \left(\frac{21}{20} \right)^2 = \text{Rs. } 4410 \end{aligned}$$

Compound Interest = (4410 - 4000) Rs. = 410.

286. The market price of 1 share of Rs. 10 face value of a company is Rs. 25. The company declared a dividend of 10%. The dividend of 1500 shares of the company will be—

- (a) 3750 (b) 15000
(c) 7500 (d) 1500

47th B.P.S.C. (Pre) 2005

Ans. (d)

Market price = 10

No. of shares = 1500

Market price of 1500 share = 15000

dividend = 10%

$$\text{Now dividend} = 15000 \times \frac{10}{100} = 1500$$

287. The simple interest on a sum of money for 2 years is Rs. 160, while the compound interest is Rs. 164. Find the rate of interest?

- (a) 4% (b) 5%
(c) 6% (d) 10%

41st B.P.S.C. (Pre) 1996

Ans. (b)

Simple Interest for 2 years = Rs. 160

Compound Interest for 2 years = Rs. 164

Difference = Rs. 4

i.e. interest for one year's interest = Rs. 4

$$\text{Simple Interest of 1 year} = \frac{160}{2} = \text{Rs. } 80$$

$$\text{Rate of interest} = \frac{(\text{Compound Interest} - \text{Simple Interest}) \times 100}{\text{Simple Interest} \times \text{Time}}$$

$$= \frac{4 \times 100}{80 \times 1}$$

∴

288. A person buys 11 books for Rs. 10 and sells 10 books for Rs. 11. What percent profit will he make?

- (a) 12% (b) 12.5%
(c) 21.5% (d) 21%

42nd B.P.S.C. (Pre) 1997-98

Ans. (d)

$$\text{Cost price of 1 book} = \frac{10}{11} \text{ Rs.}$$

$$\text{Selling price of 1 book} = \frac{11}{10} \text{ Rs.}$$

$$\text{Profit} = \text{Selling Price} - \text{Cost price}$$

$$= \frac{11}{10} - \frac{10}{11}$$

$$= \frac{21}{110}$$

$$\text{Profit \%} = \frac{\text{Profit} \times 100\%}{\text{Cost Price}}$$

$$= \frac{\frac{21}{110} \times 100}{\frac{10}{11}}$$

$$= \frac{21}{110} \times \frac{11}{10} \times 100$$

$$= 21\%$$

289. In an examination, 20% students failed in English, 35% failed in Hindi. 27% of the students failed in both. If the number of passed student is 288, then the number of students who appeared in the examination is—

- (a) 600 (b) 500
(c) 400 (d) 300

39th B.P.S.C. (Pre) 1994

Ans. (c)

$$\text{Failed students in English} = n(E) = 20\%$$

$$\text{Failed students in Hindi} = n(H) = 35\%$$

$$\text{Students failed in both subjects} = n(E \cap H) = 27\%$$

$$\text{Total failed students} = n(E \cup H)$$

$$n(E \cup H) = n(E) + n(H) - n(E \cap H)$$

$$= (20 + 35 - 27)\%$$

$$n(E \cup H) = 28\%$$

$$\therefore \text{Total percentage of failed students} = 28\%$$

$$\text{Total percentage of passed students} = (100 - 28)\%$$

$$= 72\%$$

According to question,

$$72\% = 288$$

$$100\% = \frac{288 \times 100}{72}$$

$$= 400$$

290. Consider the following statements—

Assertion (A) : To increase a number by 40%, it is sufficient to multiply that by 1.4

Reason (R) : To reduce a number by 40%, it is sufficient to divide that number by 1.4,

- (a) A is true but R is false
(b) R is true but A is false
(c) Both A and R are true
(d) Both A and R are false

39th B.P.S.C. (Pre) 1994

Ans. (a)

Let the number is 1, if it is increased by 40% then its total value

$$= 1 + 40\% \text{ of } 1$$

$$= 1.4$$

But if it is decreased by 40%. then the number 1-40% of 1

$$= 0.6$$

Therefore, on reduction by 40% the number has to be multiplied by 0.6, dividing by 1.4 cannot yield 0.6 per unit.

291. If the population of a city increases at the rate of 4% per annum and there is an additional increase at the rate of 1% per annum due to refugees, then the increase population will after two years.

- (a) 10% (b) 10.25%
(c) 10.50% (d) 10.75%

40th B.P.S.C. (Pre) 1995

Ans. (b)

$$\text{Let, population} = 100$$

$$\text{Increased population after 1 years}$$

$$= 100 + 4 + 1$$

$$= 105$$

$$\text{Population after 2 years} = 105 + \frac{5}{100} \text{ of } 105$$

$$= 110.25$$

$$\text{Increase in population} = 10.25$$

$$\text{Percentage increased in population after 2 years} = 10.25\%$$

292. 80% of the voter participated in an election. There were only two candidates Ram and Shyam. Ram got 42% of total votes and defeated Shyam by 400 votes. Thus how

- (a) 4200 votes (b) 3800 votes
(c) 3,360 votes (d) 3000 votes

41st B.P.S.C. (Pre) 1996

Ans. (a)

∴ Ram got 42% votes
∴ Shyam got (80-42)% = 38% votes
Thus, the difference of their votes
(42-38)% = 400 (According to question)
4% = 400
1% = 100
Hence, Ram got votes = $42 \times 100 = 4200$

293. If $x\%$ of y is equal to $y\%$ of z , then z will be equal to-

- (a) x (b) y
(c) $x/100$ (d) $y/100$

40th B.P.S.C. (Pre) 1995

Ans. (a)

$$x\% \text{ of } y = y\% \text{ of } z$$

$$y \times \frac{x}{100} = z \times \frac{y}{100}$$

$$z = x$$

294. If $a : b :: c : d$ then-

- (a) $(a+b) : b :: (c+d) : d$
(b) $(a-b) : b :: (c-d) : d$
(c) $(a+b+c) : d$
(d) $(a-c) : b :: (d-a)$

40th B.P.S.C. (Pre) 1995

Ans. (b)

If $a : b :: c : d$
then, $(a+b) : b :: (c+d) : d$
or $(a-b) : b :: (c-d) : d$
or $a \times d = b \times c$
Hence, option (b) is the correct answer

295. While walking at $\frac{3}{4}$ of his normal rate, a man is delayed by 1.5 hr. His normal time is-

- (a) 1 hr (b) $4\frac{1}{2}$ hr
(c) $3\frac{1}{2}$ hr (d) $6\frac{1}{2}$ hr

41st B.P.S.C. (Pre) 1996

Ans. (b)

Let normal speed = x km/hr and
Time = y hr

distance = xy (∴ distance = Speed $\times$ Time)

Time taken with $\frac{3}{4}x$

$$\frac{xy}{\frac{3}{4}x} = y + \frac{3}{2} \quad \left(\text{Time} = \frac{\text{distance}}{\text{speed}} \right)$$

$$\frac{4y}{3} = \frac{2y+3}{2}$$

$$8y = 6y + 9$$

$$y = \frac{9}{2} \text{ hr} = 4\frac{1}{2} \text{ hr.}$$

$$= 4\frac{1}{2} \text{ h}$$

296. The distance between two stations A and B is 70 km. A train starts from A to B with a speed of 80 kmph and at the same time another train starts from B to A with a speed of 60 kmph, so in how much time they meet each other later?

- (a) 60 minutes (b) 45 minutes
(c) 30 minutes (d) 15 minutes

41st B.P.S.C. (Pre) 1996

Ans. (c)

Given that,

$(A \rightarrow B)$ Speed = 80 km/h

$(B \rightarrow A)$ = 60 km/h

(∴ Both trains are running in opposite direction)

Relative speed = 140 km/hr

∴ The optimum time taken to meet the two trains is the relative running distance between the two stations. to relative speed of both trains.

$$= \frac{70}{140}$$

$$= \frac{1}{2} \text{ hr} = 30 \text{ minutes}$$

297. Two trains from Delhi to Amritsar run at 10 A.M. and 11 A.M. on the same day at 60 km/hr and 75 km/hr. At what distance from Delhi will the trains meet?

- (a) 150 km (b) 200 km
(c) 250 km (d) 300 km

39th B.P.S.C. (Pre) 1994

The first train left 1 hr ago. So it covered a distance of 60 km.

∴ Relative speed of both trains = (75-60) km/hr
= 15 km/hr

(∴ Both trains are running in same direction)

∴ Required Time = $\frac{\text{distance}}{\text{speed}}$

$$= \frac{60}{15}$$

$$= 4 \text{ hr}$$

distance covered by another train in 4 hour

$$= (4 \times 75) = 300 \text{ km}$$

298. While climbing a tower of 180 steps, a man rests for 2 minutes after climbing every 30 steps how long did he rest till he reached the top of the tower?

- (a) 30 minutes (b) 12 minutes
(c) 10 minutes (d) 9 minutes

40th B.P.S.C. (Pre) 1995

Ans. (c)

Person took 2-2 minutes rest on 30th, 60th, 90th, 120th and 150th stairs. so he rested for a total of 10 minutes.

299. If 10 workers dig a well in 4 days then how many workers will be required to dig the same well in half a day?

- (a) 5 (b) 40
(c) 60 (d) 80

40th B.P.S.C. (Pre) 1995

Ans. (d)

∴ 10 workers dig a well in 4 days

∴ (4 × 10) workers can dig a well in 1/a day

∴ (4 × 10 × 2) workers can dig a well in 1/2 day
= 80 workers

Second Method-

$$M_1 D_1 = M_2 D_2$$

$$10 \times 4 = m_2 \times \frac{1}{2}$$

$$M_2 = (10 \times 4 \times 2) \text{ workers} \\ = 80 \text{ workers}$$

300. A cyclist covers a distance of 15 km, 20 km and 25 km at the speed of 15 km/hr, 25 km/hr and 15 km/hr respectively what will be his average speed?

- (a) 30 km/hr (b) 20 km/hr

(c) 22.5 km/hr

(d) 21.4 km/hr

47th B.P.S.C. (Pre) 2005

Ans. (*)

Average speed = $\frac{\text{Total distance covered}}{\text{total time taken}}$

$$= \frac{15 + 20 + 25}{\frac{15}{15} + \frac{20}{25} + \frac{25}{15}} \quad \left(\text{speed} = \frac{\text{distance}}{\text{time}} \right)$$

$$= \frac{60}{1 + \frac{4}{5} + \frac{5}{3}}$$

$$= \frac{60}{\frac{15 + 12 + 25}{15}}$$

$$= \frac{60 \times 15}{52} = \frac{225}{13}$$

$$= 17.3 \text{ km/hr}$$

301. The price of an article increased by 5% from 1995 to 1996, 8% from 1996 to 1997 and 77% from 1997 to 1998, then the average increase in price from 1995 to 1998 is-

- (a) 24% (b) 32%
(c) 26% (d) 30%

45th B.P.S.C. (Pre) 2002

Ans. (d)

$$\text{Average increase\%} = \left(\frac{5 + 8 + 77}{3} \right) \% = 30\%$$

302. Find the average of first ten prime numbers-

- (a) 12.7 (b) 13
(c) 12.9 (d) 12.8

Ans. (c)

42th B.P.S.C. (Pre) 1997-98

The first ten prime numbers are—2, 3, 5, 7, 11, 13, 17, 19, 23, 29

$$\text{Average} = \frac{2 + 3 + 5 + 7 + 11 + 13 + 17 + 19 + 23 + 29}{10}$$

$$= \frac{129}{10} = 12.9$$

303. The average temperature for Monday, Tuesday and Wednesday was 42°C. The average temp. for Tuesday, Wednesday and Thursday was 43°C. If the average temp. on Thursday is 44°C, then the temp on Monday was-

- (a) 41°C (b) 42°C
(c) 43°C (d) 44°C

39th B.P.S.C. (Pre) 1994

Ans. (a)

Let, the temp. on Monday, Tuesday, Wednesday, Thursday are a, b, c and d respectively.

According to question,

$$\frac{a+b+c}{3} = 42^\circ\text{C} \Rightarrow a+b+c = 126^\circ\text{C}$$

$$\frac{b+c+d}{3} = 43^\circ\text{C} \Rightarrow b+c+d = 129^\circ\text{C}$$

$d = 44^\circ\text{C}$ (According to question)

$$b+c = (129^\circ - 44^\circ)\text{C}$$

$$= 85^\circ\text{C}$$

$$a = (a+b+c) - (b+c)$$

$$= 126^\circ - 85^\circ = 41^\circ\text{C}$$

304. If the average annual income of 10 teachers is Rs. 25,000 and the annual income of two teachers is Rs. 20,000. then the annual income of the remaining teachers will be-

- (a) 26,250 Rs. (b) 25,500 Rs.
(c) 23,200 Rs. (d) 22,000 Rs.

40th B.P.S.C. (Pre) 1995

Ans. (a)

Total annual income of 10 teachers = 25000×10 Rs.

Total annual income of 2 teachers = 20000×2 Rs.

Total annual income of remaining 8 teachers
= $(250000 - 40000)$ Rs.

$$\text{Average income of 8 teachers} = \frac{250000 - 40000}{8} \\ = \text{Rs. } 26250$$

305. The sum of eleven consecutive numbers is 220, then the middle number is-

- (a) 25 (b) 20
(c) 15 (d) 10

41st B.P.S.C. (Pre) 1996

Ans. (b)

Let middle number = x

So the eleven consecutive numbers are—

$(x - 5), (x - 4), (x - 3), (x - 2), (x - 1), x,$
 $(x + 1), (x + 2), (x + 3), (x + 4), (x + 5),$

According to question,

$$(x - 5) + (x - 4) + (x - 3) + (x - 2) + (x - 1) + x + (x + 1) + (x + 2) \\ + (x + 3) + (x + 4) + (x + 5) = 220$$

$$11x = 220$$

$$x = 20$$

306. A man's age at the time of his death was twenty-ninth of the year of the year he was born. If the age at the time of death was in complete years, then what was his age in the year 1900?

- (a) 74 years (b) 45 years
(c) 44 years (d) 54 years

40th B.P.S.C. (Pre) 1995

Ans. (c)

Let the age of man is $1900 = x$ year.

It is clear that his birth year was $(1900 - x)$

According to q^n , $(1900 - x)$ is divisible by 29.

$$\frac{1900 - x}{29} \text{ is an integer.}$$

$$1900 - x = 1856 \text{ (1856 is also divisible by 29)}$$

$$\therefore x = 44 \text{ years}$$

307. Which of the following statements is false?

- (a) Two acute angles in any right triangle are complementary angle.
(b) The diagonal of a rhombus bisect each other at right angles.
(c) If the angle of a quadrilateral bisect each other, then the shape of a Parallelogram is formed.
(d) Every triangle have one acute angle.

44th B.P.S.C. (Pre) 2000-01

Ans. (d)

Each triangle must have almost one acute angle. This statement almost is wrong, because all angle of a triangle can be acute angle.

308. The perimeter of a square is $(4a+8)$ unit, the area of square is-

- (a) $(a^2 + 4a + 4)$ unit²
(b) $(b - a - 8)$ unit²
(c) $(a + 2 + 4)$ unit²
(d) $(4a^2 + a - 4)$ unit²

40th B.P.S.C. (Pre) 1995

$$\text{Perimeter of square} = 4a + 8$$

$$\text{Side of square} = \frac{4a+8}{4} = (a+2) \text{ Unit}$$

$$\text{here, side of square} = \frac{\text{Perimeter}}{4}$$

$$\begin{aligned} \text{Area of square, (side)}^2 &= (a+2)^2 \text{ Unit}^2 \\ &= (a^2 + 4a + 4) \text{ Unit}^2 \end{aligned}$$

309. If the sides of a triangle are 11m, 60m and 61m, then choose the height of triangle relative to longer side.

- (a) 10 m (b) 9 m
(c) 10.8 m (d) 9.8 m

42nd B.P.S.C. (Pre) 1997-98

Ans. (c)

It is clear that given triangle is right angled triangle Δ because $(11)^2 + (60)^2 = (61)^2$

$$\text{Area of triangle} = \frac{1}{2} \times 60 \times 11 = 330 \text{ sq. m.}$$

$$(\therefore \text{Area of right angle triangle} = \frac{1}{2} \times \text{base} \times \text{height})$$

Let height on hypotenuse = h

Now, let the hypotenuse as base and perpendicular as base then,

Area of triangle = area of ΔABC

$$330 = \frac{1}{2} \times h \times 61$$

$$h = 10.81 \text{ m}$$

So, height = 10.81 m

310. The diameter of a solid copper sphere is 6cm. The sphere is melted and shaped into a new wire whose diameter is 0.2 cm. Then the length of wire is-

- (a) 36 m (b) 360 m
(c) 24 m (d) 360 cm

42nd B.P.S.C. (Pre) 1997-98

Ans. (a)

$$\text{Radius of sphere} = \frac{6}{2} = 3 \text{ cm}$$

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \pi (3)^3$$

A solid is melted and shaped into a wire it is clear that the

new shape is cylindrical.

Let the length is l and given that diameter = 0.2cm

$$\therefore \text{radius of wire} = \frac{0.2}{2} = 0.1 \text{ cm}$$

Volume of sphere = Volume of cylinder

$$\frac{4}{3} \pi (3)^3 = \pi (0.1)^2 l$$

$$l = \frac{4}{3} \times \frac{27}{(0.1)^2}$$

$$l = 3600 \text{ cm, } l = 36 \text{ m}$$

311. Three cubes of sides 3cm, 4cm and 5cm are made of a metal, they are melted to form a new cube, the side of a new cube is-

- (a) 6 cm (b) $5\frac{1}{2}$ cm
(c) $4\frac{1}{2}$ cm (d) 4 cm

41st B.P.S.C. (Pre) 1996

Ans. (a)

$$\begin{aligned} (\text{Side of new cube})^3 &= \text{Total volume of all three cubes.} \\ &= 3^3 + 4^3 + 5^3 \\ &= 216 \\ \text{side of new cube} &= (216)^{1/3} \\ &= 6 \text{ cm} \end{aligned}$$

312. The sides of a wooden block is 5 cm $\times$ 10 cm $\times$ 20 cm. How many such complete blocks would be required to made solid density of wood of least measure.

- (a) 6 (b) 8
(c) 12 (d) 16

40th B.P.S.C. (Pre) 1995

Ans. (b)

$$\begin{aligned} \text{Given that, volume of wooden box} &= 5 \times 10 \times 20 = 1000 \text{ cm}^3 \\ \text{HCF of 5, 10, 20} &= 5 \text{ cm} \\ \text{Volume of cube of side 5 cm} &= 5 \times 5 \times 5 \text{ cm}^3 \\ \text{Total number of blocks} &= \frac{1000}{5 \times 5 \times 5} \\ &= 8 \end{aligned}$$

313. A engine whose wheel circumference is $7\frac{1}{2}$ m. takes 9 second to revolve 7 round, then the speed of that vehicle in km/hr will be?

(b) 21

(c) 30

(d) 35

40th B.P.S.C. (Pre) 1995

Ans. (b)

∴ A wheel covers a distance equal to its circumference in one revolution.

$$\therefore \text{distance covered by the wheel in one round} = 7 \frac{1}{2} \text{ m}$$

According to question,

Takes 9 second to revolve 7 rounds

$$\therefore \text{distance covered in 9 second} = 7 \times \frac{15}{2} \text{ m/s}$$

$$\therefore \text{distance covered in 1 second} = \frac{7 \times 15}{2 \times 9} \text{ m/s}$$

$$\therefore \text{speed} = \frac{7 \times 15}{18} \text{ m/s}$$

$$\begin{aligned} \therefore \text{speed of train} &= \frac{7 \times 15}{18} \times \frac{18}{5} \text{ km/hr} \\ &= 21 \text{ km/hr} \end{aligned}$$

314. Select of value of such that $2x + ay = 1$ and $3x - 5y = 7$ has no solution?

(a) $\frac{3}{10}$

(b) $-\frac{10}{3}$

(c) $\frac{10}{3}$

(d) $-\frac{3}{10}$

43rd B.P.S.C. (Pre) 1999

Ans. (b)

Given that,

$$2x + ay = 1$$

$$\text{and } 3x - 5y = 7$$

Condition for the equation which has no solution

$$ax + by + c = 0 \dots\dots\dots (i)$$

$$dx + ey + f = 0 \dots\dots\dots (ii)$$

$$\text{than } \frac{a}{d} = \frac{b}{e} + \frac{c}{f}$$

According to question, $2x + ay = 1$ and

$$3x - 5y = 7$$

$$\text{Now, } \frac{2}{3} = \frac{a}{-5} + \frac{1}{7}$$

$$a = \frac{-10}{3}$$

315. ${}^nC_1 + {}^nC_2 + \dots\dots\dots + {}^nC_n = 255$, then the value of n will be.-

(a) 6

(b) 8

(c) 4

(d) 10

47th B.P.S.C. (Pre) 2005

Ans. (b)

We know that— ${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots\dots\dots + {}^nC_n = 2^n$

$${}^nC_1 + {}^nC_2 + {}^nC_3 + \dots\dots\dots + {}^nC_n = 2^n - {}^nC_0$$

$$\text{Given that— } {}^nC_1 + {}^nC_2 + \dots\dots\dots + {}^nC_n = 255$$

$$2^n - {}^nC_0 = 255$$

$$2^n - 1 = 255$$

$$2^n = 256 = 2^8$$

Since, both sides have the same base so the powers will also same

$$n = 8$$

316. Series $\sum \frac{n \cdot 2^n}{n^n}$ is-

(a) Convergent

(b) Divergent

(c) Convergent and divergent

(d) None of these

46th B.P.S.C. (Pre) 2004

Ans. (a)

$$\sum \frac{n \cdot 2^n}{n^n}$$

putting $n = 1, 2, 3, \dots\dots\dots$

$$\sum \frac{n \cdot 2^n}{n^n} = 2, 2, \frac{24}{27}, \frac{1}{4}, \dots\dots\dots \text{ (which are in decreasing order)}$$

Hence the series is convergent.

317. If the roots of equation $x^2 + 11x + 50 = 0$ are α, β is,

$$\text{then } \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = ?$$

(a) 0.21

(b) 0.42

(c) 0.84

(d) 1.0

46th B.P.S.C. (Pre) 2004

Ans. (b)

$$\text{Equation } x^2 + 11x + 50 = 0$$

$$\alpha + \beta = \frac{-11}{1} \text{ (Sum of roots } = \frac{-b}{a} \text{)}$$

$$\alpha \cdot \beta = \frac{50}{1} \text{ (products of root } = \frac{c}{a} \text{)}$$

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$$

Putting the value of $\alpha + \beta$ and $\alpha \cdot \beta$

$$= \frac{(-11)^2 - 2(50)}{50}$$

$$= \frac{21}{50} = 0.42$$

318. The value of k, so that point (1, 2, 3), (k, 0, 4) and (-2, 4, 2) lie in a line—

- (a) 4 (b) 3
(c) 1 (d) 2

47th B.P.S.C. (Pre) 2005

Ans. (a)

The given points will be linear if

$$\begin{vmatrix} 1 & 2 & 3 \\ k & 0 & 4 \\ -2 & 4 & 2 \end{vmatrix} = 0$$

$$\Rightarrow 1 \begin{vmatrix} 0 & 4 \\ 4 & 2 \end{vmatrix} - 2 \begin{vmatrix} k & 4 \\ -2 & 2 \end{vmatrix} + 3 \begin{vmatrix} k & 0 \\ -2 & 4 \end{vmatrix} = 0$$

$$\Rightarrow 1(0-16) - 2(2k+8) + 3(4k) = 0$$

$$\Rightarrow -16 - 4k - 16 + 12k = 0$$

$$\Rightarrow 8k = 32$$

$$\Rightarrow k = 4$$

319. $x = \{a, b\}$, $y = \{2, 3\}$ and $z = \{3, 4\}$, then choose $x \times (y \cap z)$

- (a) $\{(a, 3), (b, 3)\}$
(b) $\{(3, a), (3, b)\}$
(c) $\{(a, 2), (a, 4), (b, 2), (b, 4)\}$
(d) $\{a, b, 2, 3, 4\}$

43rd B.P.S.C. (Pre) 1999

Ans. (a)

Given that $x = \{a, b\}$, $y = \{2, 3\}$ & $z = \{3, 4\}$

$$y \cap z = \{3\}$$

$$x \times (y \cap z) = (a, b) \times \{3\}$$

$$x \times (y \cap z) = (a, 3), (b, 3)$$

(b) 5

320. If $f: \{x, y, z\} \rightarrow \{a, b\}$, is defined by $f = \{(x-a), (y, b)\}$ then f —

- (a) is a unitary function
(b) is not a function
(c) is an interpolative function
(d) is an empty set

43rd B.P.S.C. (Pre) 1999

Ans. (c)

$f: \{x, y, z\} \rightarrow \{a, b\}$, $f = \{(x-a), (y, b)\}$, then f is an interpolative function

321. Set $(x : x = x + 1)$ is

- (a) Set of real number
(b) Set of complex number
(c) Set of prime number
(d) Empty set

44th B.P.S.C. (Pre) 2000-01

Ans. (a)

For any value of x , the value of $(x : x = x + 1)$ is real.

322. Choose co-suffix for (1, 6), (2, 9), (3, 6), (4, 7), (5, 8), (6, 5), (7, 12), (8, 3), (9, 17), (10, 1)–

- (a) 0.4 (b) 0.6
(c) 4 (d) 6

44th B.P.S.C. (Pre) 2000-01

Ans. (a)

$$\bar{x} = \frac{1+2+3+4+5+6+7+8+9+10}{10}$$

$$= 5.5$$

$$\bar{y} = \frac{6+9+6+7+8+5+12+3+17+1}{10}$$

$$= 7.4$$

$$n = 10$$

$$\sum (x - \bar{x})(y - \bar{y}) = 4.0$$

$$(\sigma)^2 = \frac{1}{n} \sum (x - \bar{x})(y - \bar{y}) = \frac{1}{10} \times 4$$

$$= 0.4$$

323. If A and B are two sets such that $A \cup B$ has 18, A has 8 and B has 15 members then choose the number of members in $A \cap B$ –

(c) 23

(d) 26

42th B.P.S.C. (Pre) 1997-98

Ans. (b)

We know that—

$$A \cup B = A + B - (A \cap B)$$

$$A \cap B = A + B - (A \cup B)$$

$$= 8 + 15 - 18 \text{ (According to question)}$$

$$= 5$$

324. If $(x - a)$ is factor of $x^3 - a^2x + x + 2$ then find the value of

a.

(a) 1

(b) 2

(c) -2

(d) -1

42th B.P.S.C. (Pre) 1997-98

Ans. (c)

$(x - a)$ is a factor of given equation

$$\therefore x - a = 0$$

$$x = a$$

Substituting the value in the given equation—

$$a^3 - a^3 + a + 2 = 0$$

$$\therefore a = -2$$

325. Let matrix $A = \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix}$, $B = \begin{vmatrix} 1 & \alpha \\ \beta & 1 \end{vmatrix}$ If $(A + B)(A - B) =$

$A^2 - B^2$ then

(a) $\alpha = \beta = -1$

(b) $\alpha = \beta = 0$

(c) $\alpha = \beta = 1$

(d) $\alpha = 1, \beta = -1$

46th B.P.S.C. (Pre) 2004

Ans. (b)

$$A^2 = \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix} \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix}$$

$$= \begin{vmatrix} -1 & -3 \\ 6 & 2 \end{vmatrix}$$

$$= -2 + 18$$

$$= 16$$

$$B^2 = \begin{vmatrix} 1 & \alpha \\ \beta & 1 \end{vmatrix} \begin{vmatrix} 1 & \alpha \\ \beta & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 + \alpha\beta & 2\alpha \\ 2\beta & \alpha\beta + 1 \end{vmatrix}$$

$$= (1 + \alpha\beta)^2 - 4\alpha\beta$$

cle by 1 unit = $r + 1$

$$= (1 - \alpha\beta)^2$$

$$A + B = \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix} + \begin{vmatrix} 1 & \alpha \\ \beta & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & \alpha - 1 \\ 2 + \beta & 3 \end{vmatrix}$$

$$A - B = \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix} - \begin{vmatrix} 1 & \alpha \\ \beta & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & -\alpha - 1 \\ 2 - \beta & 1 \end{vmatrix}$$

According to question, substituting the value in equation

$$(A + B)(A - B) = A^2 - B^2 \implies \alpha = \beta = 0$$

326. Let $f(x) = ax^7 + bx^3 - cx - 5$ where a, b, c are constant, if $f(-7) = 7$, then $f(7)$ is equal to—

(a) -17

(b) -7

(c) 14

(d) 17

45th B.P.S.C. (Pre) 2002

Ans. (a)

$$f(x) = ax^7 + bx^3 - cx - 5$$

$$f(-7) = a(-7)^7 + b(-7)^3 - c(-7) - 5$$

$$+ 7 = -a(7)^7 - b(7)^3 + 7c - 5 \quad (\because f(-7) = +7)$$

$$7 = -\{a(7)^7 + b(7)^3 - 7c\} - 5$$

$$12 = -\{a(7)^7 + b(7)^3 - 7c - 5 + 5\}$$

$$12 = -\{f(7) + 5\}$$

$$12 = -f(7) - 5$$

$$\therefore f(7) = -17$$

327. If the radius of a circle is increased by 1 unit then its area becomes double. Find the radius of circle?

(a) $2\sqrt{2}$

(b) $\sqrt{2}(\sqrt{2} - 1)$

(c) $\sqrt{2} - 1$

(d) $1 + \sqrt{2}$

46th B.P.S.C. (Pre) 2004

Ans. (d)

Let radius = r

Area of circle = πr^2

New area of circle = $\pi (r+1)^2$
 According to question.
 $2\pi r^2 = \pi (r+1)^2$
 $2r^2 = (r+1)^2$
 $r^2 - 2r - 1 = 0$
 $r = \frac{2 \pm 2\sqrt{2}}{2}$
 $r = \frac{2 + 2\sqrt{2}}{2}$ (r cannot be negative)
 $= 1 + \sqrt{2}$

328. A sector of a circle of radius 7 cm subtends an angle of 120° . Choose the area of the sector—

- (a) 51.31cm^2 (b) 53.13cm^2
 (c) 53.31cm^2 (d) 53.03cm^2

40th B.P.S.C. (Pre) 1995

Ans. (a)

Area of sector = $\frac{\text{Angle}}{360^\circ} \pi r^2$
 $= \frac{120^\circ}{360^\circ} \times \pi (7)^2$
 $= \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 = 51.31\text{cm}^2$

329. If the radius of a circle is doubled, then choose the percentage of its increased area.

- (a) 400 (b) 100
 (c) 300 (d) 200

43rd B.P.S.C. (Pre) 1999

Ans. (c)

Half of diameter means radius
 Let the radius of circle is r .
 $\therefore$ Area of circle = πr^2
 New radius after doubling the radius = $2r$
 Area of new circle = $\pi (2r)^2$
 $= 4\pi r^2$
 Percentage of increased area.
 $= \frac{4\pi r^2 - \pi r^2}{\pi r^2} \times 100$
 $= 300 \text{ percentage}$

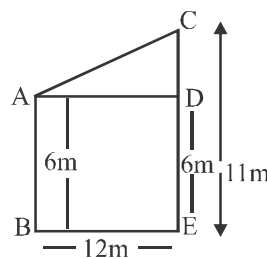
330. Two pillars are standing on flat ground, whose height are 6m and 11m. If the distance between their lower ends on the ground is 12 m. then. What will be the distance between their ends?

- (a) 16 m (b) 15 m
 (c) 13 m (d) 14 m

45th B.P.S.C. (Pre) 2001-02

Ans. (c)

AB and AC are pillars, whose heights are 6m and 11m.



$CD = CE - ED$
 $= 11 - 6$ ($ED = AB = 6\text{m}$)
 $= 5\text{m}$

In $\triangle ACD$
 $AD = 12\text{m}$ ($\therefore AD = BE$)
 $CD = 5\text{m}$

$AC^2 = AD^2 + CD^2$

$AC = \sqrt{12^2 + 5^2}$

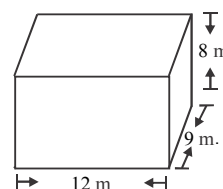
$AC = 13\text{m}$

331. The maximum length of pillar that can be placed in a room of length 12m, breadth 9 cm and height 8m is?

- (a) 864 m (b) 10 m
 (c) 17 m (d) 43 m
 (e) None of the above

64th B.P.S.C. (Pre) 2018

Ans. (c)



Maximum length of a room that can be placed in room =
 diagonal of room = $\sqrt{l^2 + b^2 + h^2}$

$$\begin{aligned}
 &= \sqrt{(12)^2 + (9)^2 + (8)^2} \\
 &= \sqrt{144 + 81 + 64} \\
 &= \sqrt{289} = 17 \text{ m}
 \end{aligned}$$

332. A train, whose length is 150m, travelling towards North at a speed of 144 km/hr can cross 250 m long bridge in-

- (a) In 20 seconds
- (b) In 100 seconds
- (c) In 45 seconds
- (d) In 10 seconds
- (e) In 28 seconds

64th B.P.S.C. (Pre) 2018

Ans. (d)

Speed of train = 144 km/hr

$$= \left(144 \times \frac{5}{18} \right) \text{ m/s}$$

= 40 m/s

distance covered by train to pass bridge = Length of train + length of bridge

$$= (150 + 250) \text{ m} = 400 \text{ m}$$

$$\begin{aligned}
 \text{Time taken by train to pass bridge} &= \frac{400}{40} \text{ seconds} \\
 &= 10 \text{ seconds}
 \end{aligned}$$

333. Driver of a car drives at an average speed of 40 km/hr between a hill station at a distance of 200 km from the plane. While travelling back he covers the same distance at an average speed of 20 km/hr. What will be the average speed of train in total 400 km?

- (a) 25 km/hr
- (b) 30 km/hr
- (c) 24.33 km/hr
- (d) 26.67 km/hr

45th B.P.S.C. (Pre) 2001-02

Ans. (d)

$$\begin{aligned}
 \text{Total time to cover 200 km with speed of 40 km/hr} &= \frac{200}{40} \\
 &= 5 \text{ hr}
 \end{aligned}$$

Time taken by train at a speed of 20 km/h

$$= \frac{200}{20} = 10 \text{ hr}$$

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

$$\begin{aligned}
 &= \frac{200 + 200}{10 + 5} \\
 &= 26.67 \text{ km/hr}
 \end{aligned}$$

334. If $y = |x|$, $x \neq 0$ then choose the value of $\frac{dy}{dx}$

- (a) $\frac{|x|}{x}$
- (b) 1
- (c) $\frac{x}{|x|}$
- (d) -1

44th B.P.S.C. (Pre) 2000-01

Ans. (*)

(i) If $x > 0$ then $|x| = x$

$$y = x$$

$$\frac{dy}{dx} = 1$$

(ii) If $x < 0$ then $|x| = -x$

$$y = -x$$

$$\frac{dy}{dx} = -1$$

$$\text{So, } \frac{dy}{dx} = \pm x$$

$\therefore$ None of the options is correct

335. If the function $f(x)$ is expressed as $f(x) = \frac{x - |x|}{x}$ then-

- (a) Function is continuous everywhere
- (b) Function is not continuous
- (c) Function is continuous where $x = 0$
- (d) Function is continuous for all x except zero

45th B.P.S.C. (Pre) 2002

Ans. (d)

$$\text{Given } f(x) = \frac{x - |x|}{x}$$

$$f(x) = \begin{cases} \frac{x+x}{x} & x < 0 \\ \frac{x-x}{x} & x > 0 \end{cases}$$

$$f(x) = \begin{cases} \frac{2x}{x} & x < 0 \\ 0 & x > 0 \end{cases}$$

$$f^{-1}(x) = \begin{cases} 0 & x < 0 \\ 0 & x > 0 \end{cases}$$

Now, $\lim_{x \rightarrow 0^+} f = \lim_{x \rightarrow 0^-} f = 0$

We got constant value so the function is continuous for all values of x except $x = 0$

336. Series $\sum \frac{x^2 - 2}{x^2 + 1} x^2, x < 0$

- (a) Convergent for $x \geq 1$
- (b) Convergent for $x = 1$
- (c) Divergent for $x \geq 1$
- (d) Divergent for $x \geq 1$

40th B.P.S.C. (Pre) 1995

Ans. (c)

From CR. Test

$$x^2 \neq 0$$

$$\lim_{x \rightarrow 0} \left(\frac{n^2 - 1}{n^2 + 1} x \right)^{\frac{1}{n}}$$

While $x > 0$

$$x \geq 1$$

So, it is divergent for $x \geq 1$

337. If function $f(x) = (x + 1)$ and $g(x) = x^2 - 3$ then

$\frac{gf\{(-1)\}}{f\{g(-1)\}}$ is equal to-

- (a) 0
- (b) 1
- (c) 2
- (d) 3

46th B.P.S.C. (Pre) 2004

Ans. (d)

$$f(-1) = -1 + 1 = 0$$

$$g(-1) = 1^2 - 3 = -2$$

$$\frac{gf\{(-1)\}}{f\{g(-1)\}} = \frac{g(0)}{f(-2)}$$

$$= \frac{0^2 - 3}{-2 + 1}$$

$$= \frac{-3}{-1} = 3$$

338. $f(x) = \frac{1}{1-x}$, the value of $f[f(f(x))]$ will be-

- (a) $\frac{1}{(1-x)^2}$
- (b) x^2
- (c) $(1-x)$
- (d) x

47th B.P.S.C. (Pre) 2005

Ans. (d)

$$f(x) = \frac{1}{1-x} \text{ (Given)}$$

$$f(f(x)) = \frac{1}{1 - \frac{1}{1-x}}$$

$$= \frac{1-x}{1-x-1}$$

$$= \frac{1-x}{-x}$$

$$= \frac{x-1}{x}$$

$$\begin{aligned} \text{Again } f[f(f(x))] &= \frac{\frac{1}{(x-1)} - 1}{\frac{1}{(1-x)}} \\ &= \frac{1-1+x}{(1-x)} \times \frac{(1-x)}{1} = x \end{aligned}$$

339. Curve $f(n) = x^3, x = 0$ to-

- (a) 26
- (b) 25
- (c) 2
- (d) 0

47th B.P.S.C. (Pre) 2005

Ans. (d)

$$y = x^3$$

$$\frac{dy}{dx} = 3x^2 \dots\dots\dots(i)$$

For the maximum and minimum value, $\frac{dy}{dx} = 0$

$$\frac{d^2y}{dx^2} = 6x$$

$$\frac{d^3y}{dx^3} = 6 \neq 0$$

Hence at: $x = 0$, the function is neither maximum nor minimum. This the point of change.

340. The demand function of a good is $x = \frac{1}{3}(24 - 2p)$, where x represents the number of units demanded and p represents per unit root. The value of p which will maximum the income?

- (a) 4 (b) 8
(c) 6 (d) 5

47th B.P.S.C. (Pre) 2005

Ans. (c)

$$x = \frac{1}{3}(24 - 2p) \dots\dots\dots(i)$$

$$x = 8 - \frac{2}{3}p$$

$$p = 12 - \frac{3}{2}x$$

$$\begin{aligned} \text{Revenue function } R(p) = px &= p \left(\frac{24 - 2p}{3} \right) \dots\dots[\text{by (i)}] \\ &= 8p - \frac{2}{3}p^2 \end{aligned}$$

For maximum income–

$$\frac{dR(p)}{dp} = \frac{d}{dp} \left(8p - \frac{2}{3}p^2 \right) = 0$$

$$8 - \frac{4}{3}p = 0$$

$$8 = \frac{4}{3}p$$

$$\boxed{p = 6}$$

341. The maximum value of $z = 50x + 15y$ under the condition $5x + y \leq 100$, and $x + y \leq 60$, $x \geq 0$, $y \geq 0$ will be.

- (a) 1500 (b) 1250
(c) 900 (d) 1000

47th B.P.S.C. (Pre) 2005

Ans. (b)

$$5x + y \leq 100 \dots\dots\dots(i)$$

$$x + y \leq 60 \dots\dots\dots(ii)$$

Subtracting eqⁿ (ii) from (i)

$$4x \leq 40$$

$$x \leq 10$$

$$y \leq 50$$

Given that $x \geq 0, y \geq 0$

$$\begin{pmatrix} 10 \geq x \geq 0 \\ 50 \geq y \geq 0 \end{pmatrix}$$

$$z = 50x + 15y$$

For the maximum value of z , the value of x and y should be maximum

$$\therefore x = 10$$

$$y = 50$$

$$z = 50 \times 10 + 15 \times 50$$

$$z = 1250$$

342. No. of solutions of equation $|x|^2 - 3|x| + 2 = 0$ is-

- (a) 2 (b) 3
(c) 4 (d) 5

46th B.P.S.C. (Pre) 2004

Ans. (c)

If $x > 0$ then $|x| = x$

$$|x|^2 - 3|x| + 2 = 0$$

$$x^2 - 3x + 2 = 0$$

$$x(x - 2) - 1(x - 2) = 0$$

$$(x - 2)(x - 1) = 0$$

$$x = 1 \text{ and } 2$$

If $x < 1$ then $|x| = -x$

$$|x|^2 - 3|x| + 2$$

$$(-x)^2 - 3(-x) + 2 = 0$$

$$x^2 + 3x + 2 = 0$$

$$x^2 + 2x + x + 2 = 0$$

$$x(x + 2) + 1(x + 2) = 0$$

$$(x + 2)(x + 1) = 0$$

$$\therefore x = -1 \text{ and } -2$$

It is clear that, x has four values are 1, -1, -2 and 2

343. $\lim_{x \rightarrow 0} \frac{xe^x - \log(1+x)}{x^2}$ is equal to–

- (a) 1/2 (b) 3/2
(c) 0 (d) 5/2

45th B.P.S.C. (Pre) 2002

Ans. (b)

$$\lim_{x \rightarrow 0} \frac{xe^x - \log(1+x)}{x^2} \left[\frac{0}{0} \right]$$

$$\frac{\frac{d}{dx}[xe^x - \log(1+x)]}{\frac{d}{dx}(x)^2}$$

$$\lim_{x \rightarrow 0} \frac{xe^x + e^x - \frac{1}{1+x} \left[\frac{0}{0} \right]}{2x}$$

Again by L-hospital rule

$$\lim_{x \rightarrow 0} \frac{\frac{d}{dx} \left(xe^x + e^x - \frac{1}{(1+x)} \right)}{\frac{d}{dx}(2x)}$$

$$\lim_{x \rightarrow 0} \frac{xe^x + e^x + e^x + \frac{1}{(1+x)^2}}{2}$$

$$= \frac{0+1+1+1}{2} = \frac{3}{2}$$

344. $f(x) = \frac{x-5}{x+5}$, $x \neq -5$, then the obtained of $f^{-1}(x)$ is-

- (a) R (b) R - {1}
(c) $(-\infty, 1)$ (d) $(1, \infty)$

46th B.P.S.C. (Pre) 2004

Ans. (b)

$$f(x) = y = \frac{x-5}{x+5}$$

Swap x and y -

$$x = \frac{y-5}{y+5}$$

$$x(y+5) = y-5$$

$$xy + 5x = y - 5$$

$$xy - y = -5x - 5$$

$$y(x-1) = -5x-5$$

$$y = \frac{5(x+1)}{1-x}$$

$$\text{so } f^{-1}(x) = \frac{5(x+1)}{(1-x)}$$

Clearly $x \neq 1$

So, obtained = $R = \{x \neq 1\}$

345. The value of $\lim_{x \rightarrow 0} \frac{1}{1-e^{-1/x}}$ is-

- (a) 1 (b) 0
(c) $-\infty$ (d) ∞

46th B.P.S.C. (Pre) 2004

Ans. (a)

$$\lim_{x \rightarrow 0} \frac{1}{1-e^{-1/x}}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{1}{1-\frac{1}{e^{1/x}}}$$

$$= 1$$

346. $x = a \cos \theta + b \sin \theta$

$y = a \sin \theta - b \cos \theta$

$x^2 + y^2$ is equal to-

- (a) $a^2 - b^2$ (b) $b^2 - a^2$
(c) $a^2 + b^2$ (d) $a^2 + 2ab$

46th B.P.S.C. (Pre) 2004

Ans. (c)

$$x = a \cos \theta + b \sin \theta$$

$$x^2 = (a \cos \theta + b \sin \theta)^2$$

$$y = (a \sin \theta - b \cos \theta)$$

$$y^2 = (a \sin \theta - b \cos \theta)^2$$

$$x^2 + y^2 = (a \cos \theta + b \sin \theta)^2 + (a \sin \theta - b \cos \theta)^2$$

$$= a^2 (\sin^2 \theta + \cos^2 \theta) + b^2 (\sin^2 \theta + \cos^2 \theta)$$

$$+ 2ab \cos \theta \sin \theta - 2ab \sin \theta \cos \theta$$

$$x^2 + y^2 = a^2 + b^2$$

347. Choose the value of $13^0 12'$ from the following-

- (a) $\frac{11}{150} \pi$ (b) $\frac{13}{150} \pi$
(c) $\frac{6}{125} \pi$ (d) $\frac{11}{25} \pi$

44th B.P.S.C. (Pre) 2000-01

Ans. (a)

$$13^0 12' = 13^0 \left(\frac{12}{60} \right)^0$$

$$= 13^0 + \frac{1^0}{5}$$

$$= \frac{66^0}{5}$$

$$\therefore 180^0 = \pi$$

$$\frac{66^0}{5} = \frac{\frac{66}{5} \times \pi}{180}$$

$$= \frac{11}{150} \pi$$

348. If $a = 13, b = 14, c = 15$ then find the value of $\tan \frac{c}{2}$ -

- (a) $1/2$ (b) $2/3$
(c) $4/7$ (d) $1/7$

44th B.P.S.C. (Pre) 2000-01

Ans. (b)

$$s = \frac{a+b+c}{2}$$

$$= \frac{13+14+15}{2}$$

$$s = 21$$

$$\tan \frac{c}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$$

$$= \sqrt{\frac{(21-13)(21-14)}{21(21-15)}} = \frac{2}{3}$$

349. Equation $x = 5 + 3 \cos \alpha$ and $y = 7 + 3 \sin \alpha$ is combinally a-

- (a) Parabola (b) Circle
(c) Hyperbola (d) Line pair

44th B.P.S.C. (Pre) 2000-01

Ans. (b)

$$x = 5 + 3 \cos \alpha \quad y = 7 + 3 \sin \alpha$$

$$\cos \alpha = \frac{x-5}{3} \quad \sin \alpha = \frac{y-7}{3}$$

On squaring both sides-

$$\cos^2 \alpha = \left(\frac{x-5}{3} \right)^2, \quad \sin^2 \alpha = \left(\frac{y-7}{3} \right)^2$$

$$\Rightarrow \therefore \sin^2 \alpha + \cos^2 \alpha = 1$$

$$\Rightarrow \therefore \left(\frac{x-5}{3} \right)^2 + \left(\frac{y-7}{3} \right)^2 = 1$$

$$\Rightarrow (x-5)^2 + (y-7)^2 = 3^2$$

$\Rightarrow$ It is an equation of circle whose radius is 3 and centre is (5,7).

Note— $x^2 + y^2 = a^2$
is an equation of circle

350. $x^2 y^2 = a^2 (x^2 + y^2)$ intangible parallel to the coordinates for the curve-

- (a) $x = \pm a, y = 0$
(b) $x = \pm a, y = \pm a$
(c) $x = a, y = 0$
(d) $x = -a, y = -a$

45th B.P.S.C. (Pre) 2002

Ans. (b)

$$x^2 y^2 = a^2 (x^2 + y^2)$$

$$y = \pm \frac{ax}{\sqrt{x^2 - a^2}}$$

y is imaginary when $-a < x < a$ and x is imaginary when $-a < y < a$. Hence, any part of curve is not between $x = \pm a$ and $y = \pm a$.

Hence $y = \pm a$ is tangent parallel to the axis.
and $x = \pm a$

351. The radius of the concentric circle touching the line $x - y = 0$ and the circle $x^2 + y^2 - 6x - 2y - 3 = 0$ is.

- (a) $\sqrt{2}$ (b) $\frac{1}{\sqrt{2}}$
(c) 1 (d) 2

46th B.P.S.C. (Pre) 2004

Ans. (a)

The radius of the concentric circle touching the straight line $x - y = 0$, centre (3, 1) of circle $x^2 + y^2 - 6x - 2y - 3 = 0$ equal to perpendicular dropped from the center.

$$\text{Radius of concentric circle} = \frac{3-1}{\sqrt{1^2 + 1^2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

352. The value of the subtle angle between the normal's of the planes $2x - y + z = 6$ and $x + y + 2z = 7$ will be-

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$

(c) $\frac{\pi}{3}$

(d) $\frac{\pi}{6}$

47th B.P.S.C. (Pre) 2005

Ans. (c)

$$2x - y + z = 6 \text{ here } a_1 = 2, b_1 = -1, c_1 = 1$$

$$x + y + 2z = 7 \text{ here } a_2 = 1, b_2 = 1, c_2 = 2$$

$$\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$= \frac{2 - 1 + 2}{\sqrt{6} \sqrt{6}}$$

$$= \frac{3}{6}$$

$$\cos \theta = \frac{1}{2} = \cos 60^\circ = \cos \frac{\pi}{3}$$

$$\therefore \theta = \frac{\pi}{3}$$

353. If in a code, DECIDE is written as 453945 then what will be written in place of ABIDE?

- (a) 94521 (b) 49521
(c) 12945 (d) 49251

40th B.P.S.C. (Pre) 1995

Ans. (c)

Clearly, It is a alphabetical number of english language.

D E C I D E
↓ ↓ ↓ ↓ ↓ ↓
4 5 3 9 4 5

So, ABIDE $\Rightarrow$ 1,2,9,4,5

354. In a code HORSE is written as DRONG then, in the some code, MONKEY is written as—

- (a) XDJMNL
(b) YEKNOM
(c) ESRIHD
(d) GNQRDM

39th B.P.S.C. (Pre) 1994

Ans. (*)

There is an error in this question due to which no correct answer can drawn If the question had HOPSE instead of HORSE, than the correct answer would be option (a).

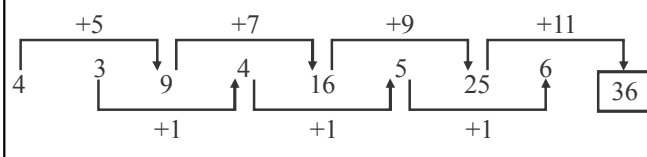
355. Which of the following numbers is in the blank space 2, 4, 3, 9, 4, 16, 5, 25, 6

- (a) 36 (b) 39
(c) 81 (d) 243

39th B.P.S.C. (Pre) 1994

Ans. (a)

The given series is increasing as follows—



356. x, y, z, u are in increasing order and v, y, w are in decreasing order. Which of the following has neither increasing nor decreasing order.

- (a) x, y, z are in increasing order (as per question)
(b) w, y, z are in increasing order (as per question)
(c) x, v, w are not in any order.
(d) v, y, w are in decreasing order (as per question)

39th B.P.S.C. (Pre) 1994

Ans. (c)

Option (c) is correct.

357. My clock is 3 minutes behind on Monday at 2 PM and 5 minutes ahead on Wednesday at 2PM. When did it show the right time?

- (a) Tuesday at 6 am
(b) Tuesday at 8 am
(c) Wednesday at 4 am
(d) Wednesday at 8 am

39th B.P.S.C. (Pre) 1994

Ans. (b)

$\therefore$ According to question, clock gained 8 min time interval in 48 hours

$\therefore$ The time interval per minutes is achieved in 6 hours. The clock was 3 minutes behind, hence just after completing 3 minutes time duration, clock will show right time.

Time taken by the clock in 3-minutes interval = $3 \times 6 = 18$